



DEPARTMENT OF PUBLIC HEALTH AND ENVIRONMENT

Air Quality Control Commission

REGULATION NUMBER 25

CONTROL OF EMISSIONS FROM SURFACE COATING, SOLVENTS, ASPHALT, GRAPHIC ARTS AND PRINTING, AND PHARMACEUTICALS

5 CCR 1001-29

[Editor's Notes follow the text of the rules at the end of this CCR Document.]

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Pursuant to Colorado Revised Statutes Section 24-4-103 (12.5), materials incorporated by reference are available for public inspection during normal business hours, or copies may be obtained at a reasonable cost from the Air Quality Control Commission (the Commission), 4300 Cherry Creek Drive South, Denver, Colorado 80246-1530. The material incorporated by reference is also available through the United States Government Printing Office, online at www.govinfo.gov. Materials incorporated by reference are those editions in existence as of the date indicated and do not include any later amendments.

PART A Applicability and General Provisions

I. Applicability

I.A.

I.A.1. The provisions of this regulation shall apply as follows:

I.A.1.a. All provisions of this regulation apply to the Denver 1-hour ozone attainment/maintenance area, to any nonattainment area for the 1-hour ozone standard, to the 8-hour Ozone Control Area, and to northern Weld County.

I.A.1.b. (State Only) All provisions of this regulation apply to any ozone nonattainment area, which includes areas designated nonattainment for either the 1-hour or 8-hour ozone standard, unless otherwise specified in Section I.A.1.c. Colorado's ozone nonattainment or attainment maintenance area maps and chronologies of attainment status are identified in Appendix A of this regulation.

I.A.1.c. The provisions marked by (State Only) are not federally enforceable, unless otherwise identified.

I.A.2. REPEALED

I.A.3. REPEALED

I.B. Sources

I.B.1. New Sources

I.B.1.a. New sources, defined as any sources located in the 1-hour ozone nonattainment area or attainment maintenance area, which either (1) submit a complete permit application on or after October 30, 1989, or (2) if no permit is required, commence operation on or after October 30, 1989, must comply with the provisions of this regulation upon commencement of operation.

I.B.1.b. (State Only) New sources are any sources which commenced construction on or after the date on which the area is first designated as being in nonattainment for ozone and are located in that area, or, if located in the 1-hour ozone nonattainment or attainment maintenance area, by October 30, 1989. New sources shall comply with the requirements of this regulation by whichever date comes later:

I.B.1.b.(i) (State Only) October 30, 1989, if they are located in what was previously designated as a 1-hour ozone nonattainment or attainment maintenance area;

I.B.1.b.(ii) (State Only) February 1, 2009, if they are located in an 8-Hour Ozone Control Area and outside of the 1-hour ozone nonattainment or attainment maintenance area; or

I.B.1.b.(iii) (State Only) Upon commencement of operation, if located within an ozone nonattainment or attainment maintenance area.

I.B.2. Existing Sources

I.B.2.a. Existing sources are

I.B.2.a.(i) (1) those sources for which a complete permit application was submitted prior to October 30, 1989, or (2) those sources, which commenced operation prior to October 30, 1989, if they are located in the 1-hour ozone nonattainment or attainment maintenance area.

I.B.2.a.(ii) Those sources that submitted a complete permit application or that commenced operation prior to February 1, 2009, if they are located in the 8-Hour Ozone Control Area and outside of the 1-hour ozone nonattainment or attainment maintenance area.

I.B.2.a.(iii) Those sources located in northern Weld County that submitted a complete application or that commenced operation prior to February 14, 2023.

I.B.2.b. (State Only) Existing sources located in the 1-hour ozone nonattainment or attainment maintenance area are those sources which commenced construction prior to the date on which the area is first designated as being in nonattainment for ozone and are located in that area, or, if located in the 1-hour ozone nonattainment or attainment maintenance area, by October 30, 1989.

I.B.2.c. Existing sources located in the 1-hour ozone nonattainment or attainment maintenance area shall not be required to comply with requirements of this regulation until on and after October 30, 1991. All existing sources shall comply with the requirements set forth in Exhibit A until October 30, 1991.

I.B.2.d. Existing sources shall be required to comply with requirements of this regulation by the following dates.

I.B.2.d.(i) (State Only) October 30, 1989, if they are located in what was previously designated as a 1-hour ozone nonattainment or attainment maintenance area;

I.B.2.d.(ii) (State Only) February 1, 2009, if they are located in an 8-hour Ozone Control Area and outside of the Denver 1-hour ozone nonattainment or attainment maintenance area;

I.B.2.d.(iii) February 14, 2023, if located in northern Weld County and subject to requirements in Part B, Sections III., IV.B.1., IV.B.2., or V.C.

I.B.2.d.(iv) December 31, 2024, if located in northern Weld County and not subject to requirements in Part B, Sections III., IV.B.1., IV.B.2., or V.C.; or

I.B.2.d.(v) (State Only) Three years after the date on which the area is first designated as being in nonattainment for ozone, if located within that ozone nonattainment or attainment maintenance area.

I.B.2.e. On and after October 30, 1991, all existing sources located in the 1-hour ozone nonattainment or attainment maintenance area shall comply with the requirements of this regulation, and Exhibit A shall no longer be applicable.

I.B.2.f. Repealed.

I.B.2.g. Repealed.

I.C. Once a source subject to this regulation exceeds an applicable threshold limit, the requirements of this regulation are irrevocably effective unless the source obtains a federally enforceable permit limiting emissions to levels below the threshold limit by restricting production capacity or hours of operation.

I.D. The owner or operator of a source not required to obtain a permit by provisions of law other than this section may apply for and shall be required to accept a permit as a condition of avoiding RACT requirements. Such permits shall contain only those conditions necessary to ensure the enforcement of the production capacity or hours of operation.

II. General Provisions

II.A. Definitions

II.A.1. “8-Hour Ozone Control Area” means the Counties of Adams, Arapahoe, Boulder (includes part of Rocky Mountain National Park), Douglas, and Jefferson; the Cities and Counties of Denver and Broomfield; and the following portions of the Counties of Larimer and Weld:

II.A.1.a. For Larimer County (includes part of Rocky Mountain National Park), that portion of the county that lies south of a line described as follows: Beginning at a point on Larimer County’s eastern boundary and Weld County’s western boundary intersected by 40 degrees, 42 minutes, and 47.1 seconds north latitude, proceed west to a point defined by the intersection of 40 degrees, 42 minutes, 47.1 seconds north latitude and 105 degrees, 29 minutes, and 40.0 seconds west longitude, thence proceed south on 105 degrees, 29 minutes, 40.0 seconds west longitude to the intersection with 40 degrees, 33 minutes and 17.4 seconds north latitude, thence proceed west on 40 degrees, 33 minutes, 17.4 seconds north latitude until this line intersects Larimer County’s western boundary and Grand County’s eastern boundary.

II.A.1.b. For Weld County, that portion of the county that lies south of a line described as follows: Beginning at a point on Weld County’s eastern boundary and Logan County’s western boundary intersected by 40 degrees, 42 minutes, 47.1 seconds north latitude, proceed west on 40 degrees, 42 minutes, 47.1 seconds north latitude until this line intersects Weld County’s western boundary and Larimer County’s eastern boundary.

II.A.2. “Denver 1-Hour Ozone Attainment/Maintenance Area” means the Counties of Jefferson and Douglas, the Cities and Counties of Denver and Broomfield, Boulder County (excluding Rocky Mountain National Park), Adams County west of Kiowa Creek, and Arapahoe County west of Kiowa Creek.

II.A.3. “Capture System” means the equipment used to contain, capture, or transport a pollutant to a control device.

II.A.4. “Capture System Efficiency (vapor gathering system efficiency)” means the percent by weight of VOC emitted by an operation subject to this

regulation, which is captured by the capture system and sent to the control device; i.e., $(\text{mass flow of VOC captured})/(\text{mass flow of VOC emitted by the operation}) \times 100\%$.

- II.A.5. "Carbon Adsorption System" means a device containing adsorbent material, an inlet and outlet for exhaust gases and a system to regenerate the saturated adsorbent.
- II.A.6. "Condenser" means any heat transfer device used to liquefy vapors by removing their latent heats of vaporization. Such devices include, but are not limited to, shell and tube, coil, surface, or contact condensers.
- II.A.7. "Control Device" means a carbon adsorber, refrigeration system, condenser, flare, firebox or other device, which will reduce the concentration of VOC in a gas stream by adsorption, combustion, condensation, or other means of removal.
- II.A.8. "Control Device Efficiency" means the percent removal by weight of VOC by a control device; i.e., $(\text{mass flow of VOC into control device} - \text{mass flow of VOC out of control device})/(\text{mass flow of VOC into control device}) \times 100\%$.
- II.A.9. "Gasoline" means a petroleum distillate having a Reid vapor pressure between 208 and 1040 torr (4-20 psi), which is used as fuel for internal combustion engines.
- II.A.10. "Highly Volatile Organic Compound" is defined as a Volatile Organic Compound or mixture of such compounds with a true vapor pressure in excess of 570 torr (11 psia) at 20 C.
- II.A.11. "Organic Material" means a chemical compound of carbon, excluding carbon monoxide, carbon dioxide, carbonic acid, metallic carbides or carbonates, and ammonium carbonate.
- II.A.12. (State Only) "Ozone Nonattainment Area" means any area designated as not in attainment with the ozone National Ambient Air Quality Standard as determined by the Environmental Protection Agency.
- II.A.13. "Northern Weld County" means the portion of the county that does not lie south of a line described as follows: Beginning at a point on Weld

County's eastern boundary and Logan County's western boundary intersected by 40 degrees, 42 minutes, 47.1 seconds north latitude, proceed west on 40 degrees, 42 minutes, 47.1 seconds north latitude until this line intersects Weld County's western boundary and Larimer County's eastern boundary.

- II.A.14. "Petroleum Refinery" means any facility engaged in producing gasoline, aromatics, kerosene, distillate fuel oils, residual fuel oils, lubricants, asphalt, or other products through distillation of petroleum or through redistillation, cracking, rearrangement or reforming of unfinished petroleum derivatives.
- II.A.15. "Reid Vapor Pressure" means the absolute vapor pressure of volatile crude oil and volatile nonviscous petroleum liquids except liquefied petroleum gases as determined by the American Society for Testing and Materials, Part 17, 1973, D-323-72 (Reapproved 1977).
- II.A.16. "True Vapor Pressure" means the equilibrium partial pressure exerted by petroleum (or other) liquid. This may be determined by the methods described in American Petroleum Institute Bulletin 2517, "Evaporation Loss from Floating Roof Tanks," 1962.
- II.A.17. "Vapor Recovery System" means a system that prevents release to the atmosphere of organic compounds emitted during the operation of any transfer, storage, or processing equipment.

- II.A.18. "Volatile Organic Compound (VOC)" means any compound of carbon, excluding carbon monoxide, carbon dioxide, carbonic acid,

metallic carbides or carbonates, and ammonium carbonate, which participates in atmospheric photochemical reactions, except those listed in Section II.B. as having negligible photochemical reactivity. VOC may be measured by a reference method, an equivalent method, an alternative method, or by procedures specified under 40 CFR Part 60 (July 1, 2025). A reference method, an equivalent method, or an alternative method, however, may also measure nonreactive organic compounds. In such cases, an owner or operator may exclude the compounds listed in Section II.B. when determining compliance with a standard if the amount of such compounds is accurately quantified, and such exclusion is approved by the Division. As a precondition to excluding such compounds as VOC, or at any time thereafter, the Division may require an owner or operator to provide monitoring or testing methods and results demonstrating, to the satisfaction of the Division, the amount of negligible-reactive compounds in the source's emissions.

II.B. Exemptions

Emissions of the organic compounds listed as having negligible photochemical reactivity in the common provisions definition of Negligibly Reactive Volatile Organic Compound are exempt from the provisions of this regulation.

II.C. General Emission Limitation

II.C.1. Existing Sources located in the 1-hour ozone nonattainment or attainment maintenance area or the 8-Hour Ozone Control Area

II.C.1.a. All existing sources shall comply with the requirements set forth in this regulation.

II.C.1.a.(i) Existing sources of VOC which are not subject to specific emission limitations set forth in this regulation, and which have the potential to emit 100 tons per year or more of VOC, shall utilize Reasonably Available Control Technology (RACT).

II.C.1.a.(ii) The potential to emit of such sources shall be based on design capacity or maximum production rate, whichever is greater, 8,760 hours/year operation, and before add-on controls.

II.C.1.a.(iii) Owners or operators of such sources with potential emissions of 100 tons per year or more, but with actual emissions less than 100 tons per year may obtain a federally

enforceable permit limiting emissions to actual rates by restricting production capacity or hours of operation, thus avoiding RACT requirements.

The owner or operator of a source not required to obtain a permit by provisions of law other than this section may apply for and shall be required to accept a permit as a condition of avoiding RACT requirements. Such permits shall contain only those conditions necessary to ensure the enforcement of the production capacity or hours of operation.

II.C.1.a.(iv) Such sources with potential emissions of 100 tons per year or more but with actual emissions of less than 50 tons per year, on a rolling 12-month total, may avoid RACT and permit requirements if the following requirements are met:

II.C.1.a.(iv)(A) The owner or operator shall submit revised Air Pollutant Emission Notices (APENs) by April 1 of each year, which demonstrate that the 50 tons per year threshold has not been exceeded.

II.C.1.a.(iv)(B) The owner or operator shall maintain records on site which include monthly VOC use and monthly VOC emissions. The records shall include calculation of total emissions for each rolling 12-month period. The records shall be made available to the Division for inspection upon request.

II.C.1.a.(v) (State Only) Existing sources that are modified – undergo any physical change, or changed in the method of operation of a stationary source which increase VOC or NO_x emissions – on or after March 30, 2008, shall utilize RACT control technologies pursuant to Regulation Number 7, Number 24, Number 25, and Number 26 and Regulation Number 3, Part B, Section III.D.2. upon recommencing operation.

II.C.1.b. Provided however, that no existing source of VOC emissions employing emission controls on or within the six-month period preceding the effective date of this regulation may reduce its level

of control of VOC emissions below that level of control actually achieved, even though such source may otherwise be subject to less stringent control requirements, except that no existing source shall be required to control emissions to an extent greater than that level of control which RACT would achieve.

II.C.1.c. (State Only) Existing sources with potential emissions equal to or greater than 100 tons per year of volatile organic compound emissions shall submit a permit modification application that includes a revised APEN (or APENs) and a RACT analysis, to the Division, as follows:

II.C.1.c.(i) (State Only) By October 30, 1991 if located in what was previously designated as the Denver 1-hour ozone nonattainment or attainment maintenance area; or

II.C.1.c.(ii) (State Only) By April 30, 2009 or within one year after the date on which the area is first designated as being in nonattainment for ozone, whichever comes later, if they are located in the 8-hour Ozone Control Area and outside of the Denver 1-hour ozone nonattainment or attainment maintenance area.

II.C.1.d. (State Only) Existing sources shall utilize RACT pursuant to Regulation Number 7, Number 24, Number 25, and Number 26 and Regulation Number 3, Part B, Section III.D.2. by whichever date comes later:

II.C.1.d.(i) (State Only) October 30, 1991, if they are located in what was previously designated as the Denver 1-hour ozone nonattainment or attainment maintenance area;

II.C.1.d.(ii) (State Only) November 21, 2011, if they are located in the 8-hour Ozone Control Area, and outside of the Denver 1-hour ozone nonattainment or attainment maintenance area;

II.C.1.d.(iii) (State Only) Three years after the date on which the area is first designated as being in nonattainment for ozone; or

II.C.1.d.(iv) (State Only) Two years after Division determination of case-by-case RACT pursuant to this Section II.C.1. The Division shall be deemed to have approved the RACT analysis for purposes of this Section II.C.1.d.(iv) if it does not

object after eighteen months from having received a complete permit application.

II.C.2. New Sources

All new sources shall utilize controls representing RACT, pursuant to applicable provisions in Regulation Number 7, Number 24, Number 25, and Number 26 and Regulation Number 3, Part B, Section III.D., upon commencement of operation.

II.D. Alternative Control Plans and Test Methods

II.D.1. Sources subject to specific requirements of this regulation shall submit for approval as a revision to the State Implementation Plan:

II.D.1.a. Any alternative emission control plan or compliance method other than control options specifically allowed in the applicable regulation. Such alternative control plans shall provide control equal to or greater than the emission control or reduction required by the regulation, unless the source contends that the control level required by the regulation does not represent RACT for their specific source.

II.D.1.b. Any alternative test method or procedure not specifically allowed in the applicable regulation.

II.D.2. No alternative submitted pursuant to this Section II.D. is effective until the alternative is approved as a revision to the State Implementation Plan.

II.E. REPEALED

II.F. Provisions for Specific Processes

II.F.1. The Gates Rubber Company Provision – REPEALED

Appendix A Colorado Ozone Nonattainment or Attainment Maintenance Areas

I. Chronology of Attainment Status

Denver Metropolitan Area Only

1978 Denver 1-hour Ozone Nonattainment Area designation first becomes effective in 7-county Denver Metropolitan Area

10/11/01 Denver 1-hour Ozone Attainment Maintenance Area designation replaces non-attainment designation and becomes effective in 7-county Denver Metropolitan Area

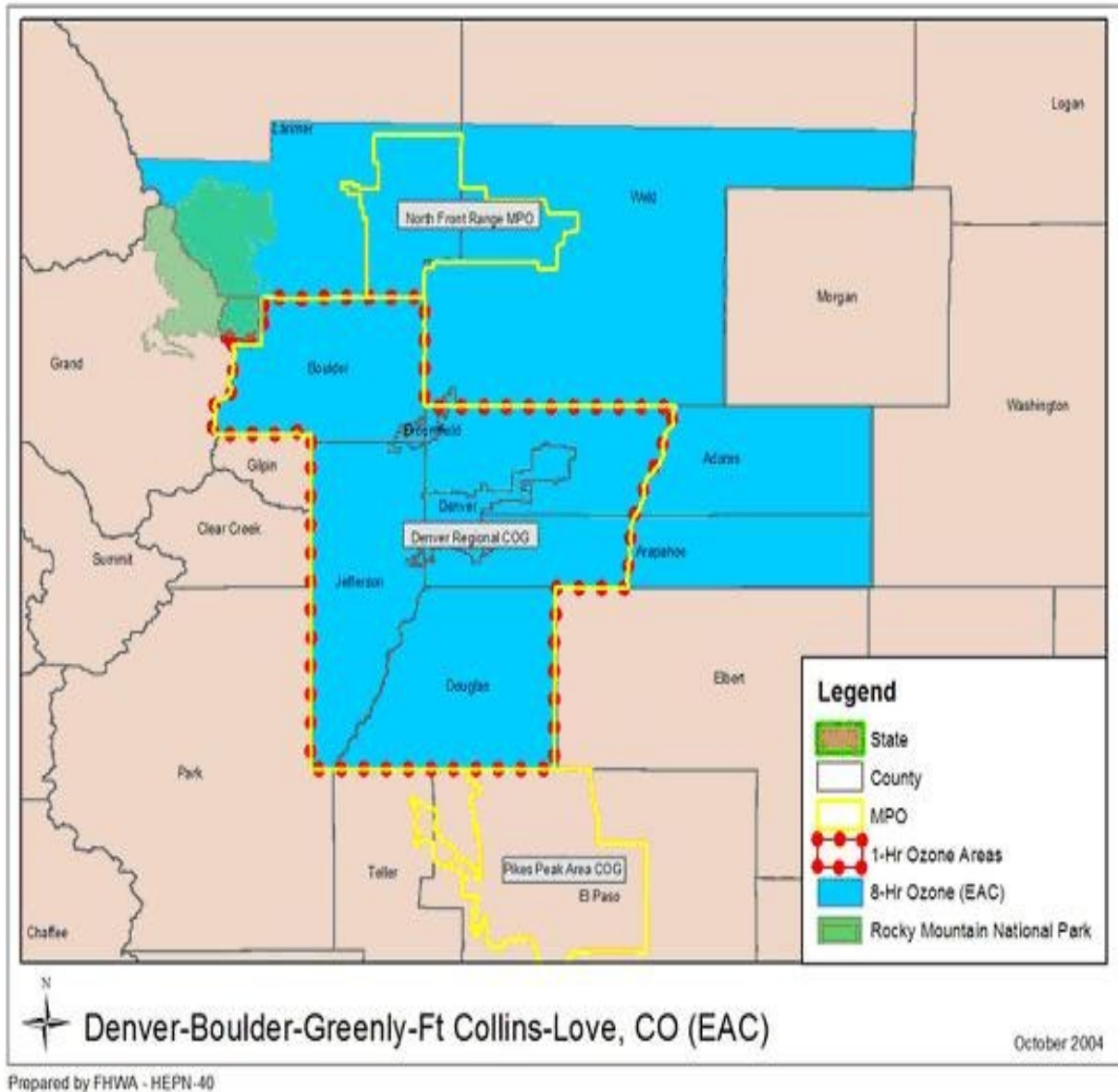
9/2/05 1-hour Ozone National Ambient Air Quality Standard is Revoked in
Colorado except for the Denver 1-hour Ozone Attainment Maintenance Area.

Denver Metropolitan Area and North Front Range

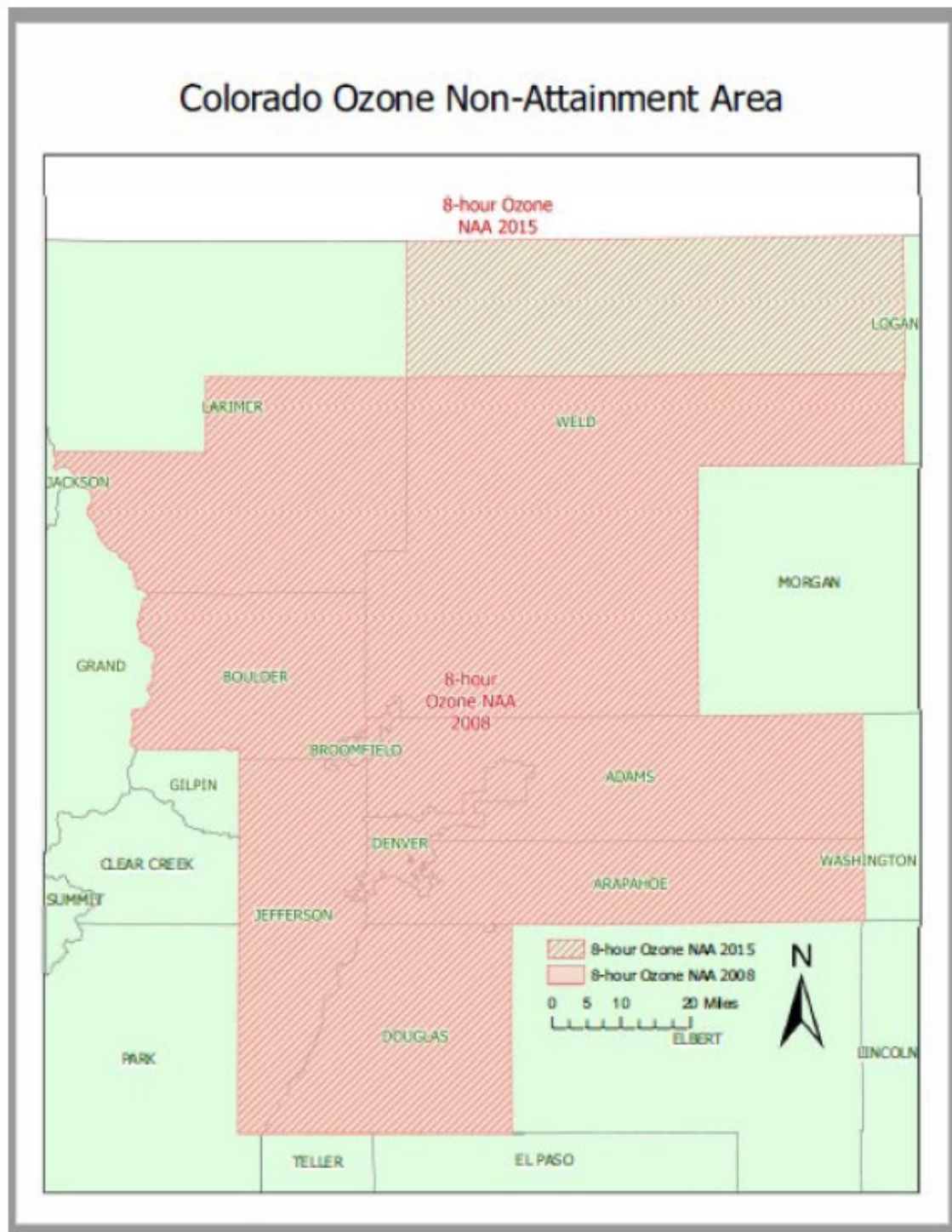
- 10/11/01 1-hour attainment maintenance area replaces non-attainment designation for the Denver Metro Area/North Front Range Area
- 4/15/04 EPA designates the Denver Metro Area/North Front Range region as an 8-hour ozone non-attainment area, designation deferred due to the implementation of the Early Action Compact
- 11/20/07 Denver 8-hour ozone non-attainment designation (1997 NAAQS) becomes effective in 9 county Denver Metropolitan Area
- 7/20/2012 Denver 8-hour ozone non-attainment designation (2008 NAAQS) becomes effective in 9 county Denver Metropolitan Area
- 8/3/2018 Denver 8-hour ozone nonattainment designation (2015 NAAQS) becomes effective in 9 county Denver Metropolitan Area
- 12/31/2021 EPA modification of the 9 county Denver Metropolitan Area 8-hour ozone nonattainment designation (2015 NAAQS) to include the portion of northern Weld County defined in Part A

II. Maps

Denver Metropolitan Area and North Front Range (2008 Ozone NAAQS)



**Denver Metropolitan Area and North Front Range and northern Weld County
(2015 ozone NAAQS)**



**PART B Surface Coating, Solvents, Asphalt, Graphic Arts and Printing, and
Pharmaceuticals**

I. Surface Coating Operations

I.A. General Provisions

I.A.1. Definitions, unless otherwise specified in Sections I.B. through I.Q.

I.A.1.a. "Coating" means a protective, functional or decorative film applied in a thin layer to a surface. This term often applies to paints such as lacquers or enamels, but is also used to refer to films applied to paper, plastics, or foils.

I.A.1.b. "Coating Applicator" means an apparatus used to apply a surface coating.

I.A.1.c. "Coating Line" means an operation which includes both (1) a coating applicator and (2) device(s) and/or area(s) to accomplish one or more of the following processes: flash-off, drying, curing, heat-setting and/or polymerization.

I.A.1.d. "Coating Solids" means that portion of a surface coating, which remains after volatile components have escaped.

I.A.1.e. "Final Repair Application" means that application of surface coating specifically intended to repair damage and imperfections in existing surface coats.

I.A.1.f. "Finished Coating Solids" means those coating-solids that remain on a coated substance after completion of all production processes.

I.A.1.g. "Flash-off Area" means the space between the application area and the oven.

I.A.1.h. "Prime Coat" (also termed "primer") means the first film of coating applied in a multiple-coat operation.

I.A.1.i. "Single Coat" means a single film of coating applied directly to the metal substrate, omitting the primer application.

I.A.1.j. "Surface Coating" means a liquid, liquefiable, or mastic composition which is converted to a solid (or semi-solid) protective, decorative, or adherent film or deposit after application as a thin layer or by impregnation. In a machine which has both coating and printing units, all units shall be considered as performing a printing operation. Such a machine is subject to the standards governing graphic arts, and thus is not covered by coating standards.

I.A.1.k. "Surface Coating Oven" means a chamber within which heat is used to bake, cure, polymerize, and/or dry a surface coating.

I.A.1.l. "Topcoat" means the final film of coating applied in a multiple-coat operation.

I.A.2. Abbreviations

I.A.2.a. Kg/lc shall be the abbreviation for: kilograms of solvent VOC per liter of coating (minus water and "exempt" solvents, as defined in Part A, Section II.B.).

I.A.2.b. Lb/gc shall be the abbreviation for: (avoirdupois) pounds of solvent VOC per gallon of coating (minus water and "exempt" solvents, as defined in Part A, Section II.B.).

I.A.3. Test Methods and Procedures

I.A.3.a. The owner or operator of any VOC source required to comply with this section shall, at their own expense, demonstrate compliance using EPA reference method 24 of 40 CFR Part 60 (September 14, 1989) for surface coatings, and reference method 25 and reference methods 1 through 4 (March 29, 2023) for add-on controls.

I.A.3.b. The test protocol should be in accordance with the requirements of the Air Pollution Control Division Compliance Test Manual and shall be submitted to the Division for review and approval at least thirty (30) days prior to testing. No test shall be conducted without prior approval from the Division.

I.A.3.c. The Division may use independent tests to verify test data submitted by the source operator or owner. The test methods shall be those listed in Section I.A.3.a. and the Division test results shall take precedence.

I.A.3.d. The Division may accept, instead of the testing required in this section, a certification by the manufacturer of the composition of the coatings if supported by actual batch formulation records. The owner or operator of the VOC source required to comply with this section shall obtain certification from the coating manufacturer(s) that the test method(s) used for determination of VOC content meet the requirements specified in Section I.A.3.a. The owner or operator shall have this certification readily available to Division personnel, in order to allow the results to be used in the daily compliance calculations specified in Section I.A.10.

I.A.3.e. The performance of add-on control device equipment shall be established with the required test methods of I.A.3.a. at equipment startup, and after major modification to the control equipment. Baseline operating parameters shall be established during the satisfactory (i.e. in-compliance) operation of the control equipment, including operation during all anticipated ranges of process throughput. During subsequent process operation, the owner or operator shall maintain the operating conditions of the add-on controls as close to these baseline conditions as possible. If serious operational problems with an add-on control system are evidenced from the daily monitoring required by Section I.A.8.b. (such problems may be indicated by changes from baseline conditions), repeat performance tests may be required by the Division, as necessary.

I.A.4. Sampling

To determine compliance with applicable surface coating standards, samples shall be taken from the coating as freshly delivered to the reservoir of the coating applicator.

I.A.5. Alternative compliance methods for processes and operations

For each process specified in Sections I.B. through I.N. the emission limits designated for that process shall be achieved by:

- I.A.5.a. Use of coatings with proportions of VOC less than or equal to the maximums specified by the applicable section of this regulation; or
- I.A.5.b. Use of the specified equipment and procedures prescribed by the applicable section of this regulation; or

- I.A.5.c. Use of an alternative means of control which satisfies the requirements of Section I.A.5.e., I.A.5.f., and Part A, Section II.D.; or
- I.A.5.d. Use of crossline averaging. The following requirements apply:
 - I.A.5.d.(i) The actual reduction shall be equivalent to the actual reduction that would be achieved on a line-by-line basis.
 - I.A.5.d.(ii) Credit shall not be received for downtime, however, credit is allowed for enforceable production limits.
 - I.A.5.d.(iii) Crossline averaging shall be used only across lines in the same control technique guidance group.
 - zxsdal.A.5.d.(iv) The crossline averaging shall be met on a daily weighted average.
 - I.A.5.d.(v) Sources subject to best available control technology (BACT) and lowest achievable emission rate (LAER) requirements shall not use cross line averaging.
 - I.A.5.d.(vi) VOC emissions shall be expressed as lbs/gallons solids to determine reduction over baseline (lb VOC/lb solids for graphic arts).
 - I.A.5.d.(vii) Organisol and plastisol coatings shall not be used to bubble emissions from vinyl surface or automobile topcoating operations.
 - I.A.5.d.(viii) Before crossline averaging may be used, the control methodology shall be approved as a revision to the State Implementation Plan.

I.A.5.e. The design, operation and efficiency of any capture system used in conjunction with any emission control system shall be certified in writing by the source owner or operator and approved by the Division. Unless the capture system meets the requirements for a total enclosure as specified in the New Source Performance Standard for the Magnetic Tape Manufacturing Industry, 53FR38892, October 3, 1988, or unless Division approved material balance techniques are used to adequately determine overall VOC capture and destruction/recovery efficiency, the efficiency of the capture system shall be determined by test methods approved as a revision to the State Implementation Plan. Testing for capture efficiency shall be performed on a case-by-case basis as required by the Division. The requirements of Sections I.A.3.e. and I.A.8.b. shall apply to the capture and control device system. When capture and control device efficiency must be independently determined, the overall VOC emission reduction rate equals the (percent capture efficiency X percent control device efficiency)/100.

I.A.5.f. Sources which use add-on controls, crossline averaging, or an approved alternative control strategy instead of low solvent technology to meet the applicable emission limit shall meet the equivalent VOC emission limit, on the basis of solids applied (lb VOC/gal solids applied, or lb VOC/lb solids applied, for graphic arts sources). Appendix E sets forth the procedure for converting emission limits and lists equivalent limits for various coating operations.

I.A.5.g. Owners or operators of sources which use a carbon adsorption system shall provide for the proper disposal or reuse of all VOC recovered.

I.A.6. Exemptions

I.A.6.a. The requirements of this Section I. do not apply to sources used exclusively for chemical or physical analysis or determination of product quality and commercial acceptance, provided;

I.A.6.a.(i) the operation of the source is not an integral part of the production process; and

I.A.6.a.(ii) the emissions from the source do not exceed 363 kilograms (800 lbs.) in any calendar month; and

I.A.6.a.(iii) the exemption is approved in writing by the Division.

I.A.6.b. The requirements of Sections I.C., D., E., F., G., H., I., L. and M. are not applicable to sources whose actual emissions, including fugitive emissions, before add-on controls, are less than 6.8 kilograms (15 lbs.) per day and less than 1.4 kilograms (3 lbs.) per hour. Emissions from all sources within the same control technique guidance group shall be totaled to determine actual emissions.

I.A.7. Fugitive emission control

I.A.7.a. Control techniques and work practices shall be implemented at all times to reduce VOC emissions from fugitive sources. Control techniques and work practices include, but are not limited to:

I.A.7.a.(i) tight-fitting covers for open tanks;

I.A.7.a.(ii) covered containers for solvent wiping cloths;

I.A.7.a.(iii) proper disposal of dirty cleanup solvent.

I.A.7.b. Emissions of organic material released during clean-up operations, disposal, and other fugitive emissions shall be included when determining total emissions, unless the source owner or operator documents that the VOCs are collected and disposed of in a manner that prevents evaporation to the atmosphere.

I.A.8. Recordkeeping, Reporting, and Monitoring

I.A.8.a. If add-on control equipment is used, continuous monitors of the following parameters shall be installed, calibrated, and operated at all times that the associated control equipment is operating:

I.A.8.a.(i) exhaust gas temperature of all incinerators;

I.A.8.a.(ii) temperature rise across a catalytic incineration bed;

I.A.8.a.(iii) breakthrough of VOC on a carbon adsorption unit;

I.A.8.a.(iv) any other monitoring and/or recording device, maintenance and/or control-media-replacement schedule(s) specified on a case-by-case basis by the Division.

I.A.8.b. If add-on control equipment is used, in addition to the requirements of Section I.A.8.a., the following information and any other necessary information, as determined applicable for each source by the Division, shall be monitored and recorded daily in order to assure continuous compliance. The substitution of continuous recordings for daily recording may be allowed by the Division.

I.A.8.b.(i) For the capture system: fan power use, duct flow, duct pressure.

I.A.8.b.(ii) For carbon adsorbers: bed temperature, bed vacuum pressure, pressure at the vacuum pump, accumulated time of operation, concentration of VOC in the outlet gas, solvent recovery.

I.A.8.b.(iii) For refrigeration systems: compressor discharge and suction pressures, condenser fluid temperature, solvent recovery.

I.A.8.b.(iv) For incinerator systems: exhaust gas temperature, temperature rise across a catalytic incinerator bed, flame temperature, accumulated time of incinerator.

I.A.8.c. Recordkeeping procedures shall follow the guidance in "Recordkeeping Guidance Document for Surface Coating Operations and the Graphic Arts Industry," July 1989, EPA 340/1-88-003.

I.A.9. Required and Prohibited Acts

I.A.9.a. No owner or operator of a source of VOCs subject to this section shall operate, cause, allow or permit the operation of the source, unless:

I.A.9.a.(i) For each category of surface coating as specified in Sections I.B. through I.M., the owner or operator of a surface coating line or facility subject to that section does not cause, allow or permit the discharge into the atmosphere of any VOCs in excess of the specified emission limit, calculated as delivered to the coating applicator or as applied to the substrate, whichever is greater.

I.A.9.a.(ii) The owner or operator of a surface coating operation maintains and operates surface coating operations in a manner consistent with good air pollution control practices for minimizing emissions, such as, but not limited to, coating application methods capable of achieving a transfer efficiency achieved by HVLP spraying.

I.A.10. Compliance Calculation Procedures

I.A.10.a. Compliance with this section shall be determined on a daily basis. Sources may request a revision to the State Implementation Plan for longer times for compliance determination.

I.A.10.b. Compliance calculation procedures shall follow the guidance in "Procedure for Certifying Quantity of Volatile Organic Compounds Emitted by Paint, Ink, and Other Coatings," December 1984, EPA-450/3-84/019. In addition, for add-on controls or other compliance alternatives, calculation procedures shall follow the guidance of Section I.A.5.f.

I.A.11. The requirements of Sections I.A.1. through I.A.10. apply to each category of surface coating as specified in Sections I.B. through I.M. The requirements of Sections I.A.7. through I.A.10. apply to the category in Section I.N. The requirements of Sections I.A.1. through I.A.9 apply to the category in Section I.O. The requirements of Sections I.A.1. through I.A.4., I.A.7, and I.A.8. apply to the categories in Sections I.P and I.Q.

I.A.12. The Division shall approve utilization of alternative compliance methods to the following sources pursuant to this Section I.

I.A.12.a. Lexmark International, Inc. shall be allowed to utilize the alternative compliance method of crossline averaging for processes and operations within the Manufactured Metal Parts and Metal products (Subgroup L) and within the Plastic Film Coating Operations (Subgroup J). The emission trading requirements of Regulation Number 3, Part A, Section V. shall be met, and utilization of the alternative compliance method shall be subject to the following generic conditions, which shall be written and specifically described as enforceable permit terms and conditions in its permits:

- I.A.12.a.(i) The alternative compliance method shall result in an actual reduction that is equivalent to the actual reduction that would otherwise be achieved on a line-by-line basis pursuant to this Regulation Number 25.
- I.A.12.a.(ii) Credit shall not be received for downtime; however, credit is allowed for emission reductions from enforceable production limits.
- I.A.12.a.(iii) Cross line averaging shall be used only across lines of the same control technique guidance group. Lexmark shall use cross line averaging between Metal Parts and Metal Products lines or between Plastic Film Coating lines. Lexmark shall not use cross line averaging where the emissions from Plastic film coating lines are averaged with Metal Parts and Metal Products lines.
- I.A.12.a.(iv) The cross line averaging shall be met on a daily weighted average.
- I.A.12.a.(v) Sources subject to Best Available Control Technology (BACT), and Lowest Achievable Emission Rate (LAER) shall not use cross line averaging.
- I.A.12.a.(vi) To determine reduction over baseline, VOC emissions shall be expressed according to Section I.A.5.f., as lbs/gallons solids.
- I.A.12.a.(vii) Monthly records shall be kept at the source to verify ongoing compliance with these conditions. The recordkeeping format shall be approved by the Division.
- I.A.12.a.(viii) An annual report demonstrating ongoing compliance with this regulation and all permit terms shall be filed with the Division. The report format shall be approved by the Division and specifically described in the permit.
- I.A.12.a.(ix) The Division shall issue a permit with federally enforceable terms and conditions to Lexmark limiting Lexmark's alternative compliance method emissions to those allowable under Section I.L. and Section I.J.

I.A.12.a.(x) Commercial and Product quality control laboratory equipment are exempt from APEN filing and construction permit requirements under Regulation Number 3, Part A, Section II.D.1.i., and Regulation Number 3, Part B, Section II.D.1.a.; and from construction permit requirements under Regulation Number 3, Part B, Section II.D.1.i. Qualifying sources shall be exempt from Regulation Number 25, Section I. A.6.

I.A.12.a.(xi) Nothing in the alternative compliance method is intended to relax any emissions limitation of this Regulation Number 25.

I.B. Automobile and Light-Duty Truck Assembly Plants

I.B.1. Definitions

I.B.1.a. "Application Area" means the area where the surface coating is applied by spraying, dipping or flow coating.

I.B.1.b. "Automobile" means a passenger motor-vehicle or a derivative of same, capable of seating twelve (12) or fewer passengers, and having at least two driven wheels.

I.B.1.c. "Automobile Assembly Facility" means a facility where parts (including assembled or partially assembled components) of automobiles are received, and finished automobiles are produced, partially or wholly by an assembly line.

I.B.1.d. "Light-Duty Truck" means any motor vehicle rated at 8,500 pounds (3,855 kilograms) gross vehicle weight or less, and having at least two driven wheels, which is designed primarily for purposes of transportation of property or is a derivative of such vehicles. It includes, but is not limited to, pickup trucks, vans, and window vans rated at 8,500 pounds' gross vehicular weight or less.

I.B.1.e. "Light-Duty Truck Assembly Facility" means a facility where parts (including assembled or partially assembled components) of light-duty trucks are received, and finished light-duty trucks are produced, partially or wholly by an assembly line.

I.B.2. Applicability

This section applies to all assembly and subassembly lines in an automobile or light-duty truck assembly facility, including those for frames, small parts, wheels, and main body parts. This section applies only to the manufacture of new vehicles.

I.B.3. Emission Limitations

	Kg/lc	Lb/gc
Prime application, flashoff area, and oven	0.23	1.9
Topcoat application area, flashoff area, and oven	0.34	2.8
Final repair application, flashoff area and oven	0.58	4.8

I.B.4. Coatings other than primer, surfacer (guidecoat), topcoat and final repair shall be considered under the miscellaneous metal parts Section I.L.

I.B.5. For topcoat application, if a complying coating is not used to meet the emission limit of Section I.B.3, then:

I.B.5.a. the alternate method shall meet an emission limit of 15.1 lb VOC/gal. solids deposited on the coated part; and

I.B.5.b. compliance shall be determined on a daily weighted average basis.

I.B.6. Topcoat operation shall include all spray booths, flash-off areas and ovens in which topcoat is applied, dried and cured, except for final offline repair.

I.C. Can Coating Operations

I.C.1. Definitions

I.C.1.a. "Can Coatings" means any coatings containing organic materials and applied -- or intended for application -- by spray, roller, or other means onto the inside and/or outside surfaces of formed cans and components of cans.

I.C.1.b. "End Sealing Compound" means a substance which is coated onto can ends and which functions as a seal when the end is assembled onto the can.

- I.C.1.c. "Exterior Base Coat" means a coating applied to the exterior of a can to provide protection to the metal and/or to provide background for any lithographic or printing operation.
- I.C.1.d. "Interior Base Coat" means the initial coating applied to the interior surface of a can by roller coater or spray.
- I.C.1.e. "Interior Body Spray" means a coating sprayed onto the interior surface of the can body to provide a protective film between the can and its contents.
- I.C.1.f. "Overvarnish" means a coating applied directly over ink to reduce the coefficient of friction, provide gloss, protect against abrasion, enhance product quality, and protect against corrosion.
- I.C.1.g. "Three-Piece Can Side Seam Spray" means a coating sprayed onto the interior and/or exterior of a can body seam on a three-piece can to protect the exposed metal.
- I.C.1.h. "Two-Piece Can Exterior End Coat" means a coating applied to the exterior of the bottom end of a two-piece can.

I.C.2. Applicability

This section applies to coating applicator(s), and oven(s) of sheet can or end coating lines involved in sheet basecoat (exterior and interior) and over varnish, two-and three-piece can interior body spray, two-piece can exterior end (spray or roll coat), three-piece can side-seam spray, and end sealing compound operations.

I.C.3. Emission Limitations

Can Coating	Kg/lc	Lb/gc
Sheet base coat (exterior and interior) and overvarnish two-piece can exterior (base coat and overvarnish)	0.34	2.8
Two and three-piece can interior body spray, two-piece can exterior end (spray or roll coat)	0.51	4.2
Three-piece can side-seam spray	0.66	5.5
End sealing compound	0.44	3.7
Any additional coats	0.51	4.2

I.D. Coil Coating Operations

I.D.1. Definitions

I.D.1.a. “Coil Coating” means any surface coating applied by spray, roller, or other means onto one or both surfaces of flat metal sheets or strips that come in rolls or coils.

I.D.1.b. “Quench Area” means a chamber where the hot metal exiting the oven is cooled by either a spray of water or a blast of air followed by water cooling.

I.D.2. Applicability

This section applies to the coating applicator(s), oven(s), and quench area(s) of coil coating operations involved in primer, intermediate, top-coat or single-coat operations.

I.D.3. Emission Limitations:

Coil Coating	Kg/lc	Lb/gc
Any coat (primer, intermediate coat, topcoat, single coat)	0.31	2.6

I.E. Fabric Coating Operations

I.E.1. Definitions

- I.E.1.a. "Fabric Coating" means the process of coating or impregnating the full, usable surface of a fabric web or sheet to impart properties that are not initially present such as strength, stability, water or acid repellency, or appearance. "Fabric Coating" excludes those processes normally included under fabric finishing (e.g. dyeing, treating for stain and wrinkle resistance, etc.).

I.E.2. Applicability

This section applies to fabric coating lines which includes, but is not limited to, coaters and drying ovens.

I.E.3. Emission Limitations

	Kg/lc	Lb/gc
Fabric Coating Line	0.35	2.9

I.F. Large Appliance Coating Operations

I.F.1. Definition

- I.F.1.a. "Large Appliances" includes doors, cases, lids, panels, interior support parts, and any other large (greater than one square decimeter (15.5 square inches)) coated surfaces of residential and commercial washers, dryers, ovens, ranges, refrigerators, freezers, water heaters, dishwashers, trash compactors, air conditioners, and all other products under SIC Code 363 according to the "Standard Industrial Classification Manual", Executive Office of the President, Office of Management and Budget, designated by convention of the industry as large appliances.

I.F.2. Applicability

This section applies to all large appliance coating lines.

I.F.3. Emission Limitations

	Kg/lc	Lb/gc
Large Appliance Coating Line; prime, single or topcoat application area, flashoff area, and oven	0.34	2.8

I.G. Magnet Wire Coating Operations

I.G.1. Definition

I.G.1.a. "Magnet Wire Coating" means those operations which apply a coating of electrically insulating varnish or enamel (or similar substance) to wire which is known as "magnet wire." Magnet wire is usually copper or aluminum, and is used for electric motors, generators, transformers, magnets, and related products.

I.G.2. Applicability

This section applies to, but is not limited to, coaters and drying ovens of magnet wire coating operations.

I.G.3. Emission Limitations

	Kg/lc	Lb/gc
Magnetic wire coating operation	0.20	1.7

I.H. Metal Furniture Coating Operations

I.H.1. Definitions

I.H.1.a. "Metal Furniture" means furnishings commonly considered furniture, for domestic, business, and/or institutional use, which have one or more essential, major components made of metal. "Metal furniture" includes, but is not limited to, tables, chairs, wastebaskets, beds, desks, lockers, shelving, cabinets, room dividers, clothing racks, chests of drawers, and sofas.

I.H.1.b. "Metal Furniture Coating" means applying a "surface coating" to "metal furniture" as defined. It excludes coating of non-metal components.

I.H.2. Applicability

This section applies to all metal furniture coating lines.

I.H.3. Emission Limitations

	Kg/lc	Lb/gc
Metal Furniture Coating Line: All coats (including prime, single, and topcoat)	0.36	3.0

I.I. Paper Coating Operations

I.I.1. Definition

"Paper Coating" means impregnating or applying a uniform layer of "surface coating" to paper. It includes, but is not limited to, the production of: coated, glazed, decorated, and varnished paper; carbon and pressure-sensitive copy papers; paper adhesive-labels and tapes; blue-print; photographic and copier paper. It also includes coating of metal foil such as gift wrap and packaging. Paper coating does not include impregnation using a batch dipping process.

I.I.2. Applicability

This section applies to paper coating lines, which includes, but is not limited to, coaters and drying ovens.

I.I.3. Emission Limitations

	Kg/lc	Lb/gc
Paper Coating Line	0.35	2.9

I.J. Plastic-Film Coating Operations

I.J.1. Definition

I.J.1.a. "Plastic-Film Coating" means applying a uniform layer of "surface coating" to a flexible web or sheet of thin plastic substance, excluding all rubbers and vinyl's* (polyvinyl chloride) except for the following two categories of vinyl products: (1) vinyl tapes and (2) vinyl's coated with an adhesive or pressure-sensitive coating. It includes, but is not limited to: plastic typewriter ribbons, photographic film, adhesive tapes, and magnetic recording tapes. (*see Section I.K.)

I.J.2. Applicability

This section applies to, but is not limited to, coaters and drying ovens of plastic-film coating lines.

I.J.3. Emission Limitations

	Kg/lc	Lb/gc
Plastic-Film Coating Line	0.35	2.9

I.K. Vinyl Coating Operations

I.K.1. Definition

"Vinyl Coating" means applying a uniform layer, decorative or protective topcoat to a vinyl (polyvinyl chloride) coated fabric or vinyl sheet. It includes printing of same. Excluded are*: (1) the coating of same with adhesive or pressure-sensitive coatings and (2) vinyl tapes. (*see Section I.J.)

I.K.2. Application

This section applies to vinyl coating lines which includes, but is not limited to, coaters and drying ovens.

I.K.3. Emission Limitations

	Kg/lc	Lb/gc
Vinyl Coating Line	0.45	3.8

I.L. Manufactured Metal Parts and Metal Products

I.L.1. General Provisions

I.L.1.a. Applicability

Before January 1, 2022, this section applies to the application area(s), flashoff area(s), oven(s), and drying areas including (but not limited to) air and forced air drier(s) used in the surface coating of the metal parts and products listed below where actual emissions, including fugitive emissions, before add-on controls, are greater than or equal to 6.8 kilograms (15 lbs.) per day and 1.4 kilograms (3 lbs.) per hour. This section applies to prime coat, top coat, and single coat operations.

On or after January 1, 2022, this section applies to each metal parts and product surface coating unit at a facility where the total actual VOC emissions from all metal parts and product surface coating operations, including related cleaning activities, at that facility are equal to or greater than 2.7 tons per 12-month rolling period, before consideration of controls.

This section is applicable to surface coating of manufactured metal parts and metal products which include:

- I.L.1.a.(i) Large farm machinery (harvesting, fertilizing, and planting machines, tractors, combines, etc.);

- I.L.1.a.(ii) Small-farm, lawn and garden machinery (lawn and garden tractors, lawn mowers, rototillers, etc.);
- I.L.1.a.(iii) Small appliances (fans, mixers, blenders, crock pots, dehumidifiers, vacuum cleaners, etc.);
- I.L.1.a.(iv) Commercial machinery (office equipment, computers and auxiliary equipment, typewriters, calculators, vending machines, etc.);
- I.L.1.a.(v) Industrial machinery (pumps, compressors, conveyor components, fans, blowers, transformers, etc.);
- I.L.1.a.(vi) Fabricated metal products (metal covered doors, frames, etc.);
- I.L.1.a.(vii) Furniture hardware made of metal for use with non-metal furniture; and
- I.L.1.a.(viii) Any other industrial category which coats metal parts or products under the standard industrial classification code of major group 33 (primary metal industries), major group 34 (fabricated metal products), major group 35 (non-electric machinery), major group 36 (electrical machinery), major group 37 (transportation equipment), major group 38 (miscellaneous instruments), and major group 39 (miscellaneous manufacturing industries), according to the "Standard Industrial Classification Manual" Executive Office of the President, Office of Management and Budget.

I.L.1.b. Exemptions

- I.L.1.b.(i) Section I.L. does not apply to the surface coating of metal parts and products inasmuch as these are covered in Sections I.B., I.C., I.D., I.F., I.G., and I.H.
- I.L.1.b.(ii) Section I.L. is not applicable to the following special purpose coatings:
 - I.L.1.b.(ii)(A) Division-approved exemptions for high performance coatings on a case-by-case basis.
 - I.L.1.b.(ii)(B) Full exterior repainting of automobiles and light-duty trucks if fewer than 18 vehicles are painted per day.

- I.L.1.b.(iii) Beginning January 1, 2022, Section I.L. does not apply to aerosol coatings, architectural coatings, and automobile refinish coatings regulated by 40 CFR Part 59 Subparts B (National Volatile Organic Compound Emission Standards for Automobile Refinish Coatings (August 23, 2019)), Subpart D (National Volatile Organic Compound Emission Standards for Architectural Coatings (August 23, 2019)), or Subpart E (National Volatile Organic Compound Emission Standards for Aerosol Coatings (August 23, 2019)).
- I.L.1.b.(iv) Beginning January 1, 2022, Section I.L. does not apply to architectural and industrial maintenance coatings regulated by Regulation Number 21.
- I.L.1.b.(v) Beginning January 1, 2022, Section I.L. does not apply to the coating of aerospace vehicles or components including, but not limited to, any fabricated part, processed part, assembly of parts, or completed unit of any aircraft including, but not limited to, airplanes, helicopters, missiles, rockets, and space vehicles at aerospace manufacturing and rework operations.
- I.L.1.b.(vi) Beginning January 1, 2022, Sections I.L.2 and I.L.3. do not apply to:
 - I.L.1.b.(vi)(A) Aerosol coating products.
 - I.L.1.b.(vi)(B) Powder coatings.
 - I.L.1.b.(vi)(C) Stencil coatings.
 - I.L.1.b.(vi)(D) Safety-indicating coatings.
 - I.L.1.b.(vi)(E) Solid-film lubricants.
 - I.L.1.b.(vi)(F) Electric-insulating and thermal-conducting coatings.
 - I.L.1.b.(vi)(G) Magnetic data storage disk coatings.
 - I.L.1.b.(vi)(H) Plastic extruded onto metal parts to form a coating.

I.L.1.b.(vii) Beginning January 1, 2022, Section I.L.3. does not apply to touch-up coatings, repair coatings, and textured finishes.

I.L.1.c. Definitions

For the purpose of this section, the following definitions apply:

I.L.1.c.(i) "Air Dried Coating" means coatings which are dried by the use of air or forced warm air at temperatures up to 90°C (194°F);

I.L.1.c.(ii) "Baked Coating" means a coating that is cured at a temperature at or above 90°C (194°F).

I.L.1.c.(iii) "Camouflage Coating" means a coating used, principally by the military, to conceal equipment from detection.

I.L.1.c.(iv) "Clear Coat" means a coating, which lacks color and opacity or a coating which is transparent.

I.L.1.c.(v) "Coating Application System" means all operations and equipment which apply, convey, and dry a surface coating, including, but not limited to, spray booths, flow coaters, flashoff areas, air dryers and ovens.

I.L.1.c.(vi) "Coating Unit" means a series of one or more coating applicators and any associated drying area and/or oven wherein a coating is applied, dried, and/or cured. A coating unit ends at the point where the coating is dried or cured, or prior to any subsequent application of a different coating. It is not necessary for a coating unit to have an oven or flash-off area.

I.L.1.c.(vii) "Drum" means any cylindrical metal shipping container larger than 12 gallons' capacity but no larger than 110 gallons' capacity.

I.L.1.c.(viii) "Electric-Insulating Varnish" means a non-convertible-type coating applied to electric motors, components of electric motors, or power transformers, to provide electrical, mechanical, and environmental protection or resistance.

- I.L.1.c.(ix) "Etching Filler" means a coating that contains less than 23 percent solids by weight and at least 1/2-percent acid by weight, and is used instead of applying a pretreatment coating followed by a primer.
- I.L.1.c.(xi) "Extreme Environmental Conditions" means exposure to any of the following: temperatures consistently above 95°C, detergents, abrasive and scouring agents, solvents, and corrosive environments.
- I.L.1.c.(xii) "Extreme High-Gloss Coating" means a coating which, when tested by the American Society for Testing Material Test Method D-523 adopted in 1980, shows a reflectance of 75 or more on a 60° meter.
- I.L.1.c.(xiii) "Extreme Performance Coatings" means coatings designed for extreme environmental conditions (i.e., chronic exposure to corrosive, caustic or acidic agents, chemicals, chemical fumes, chemical mixtures or solutions; repeated exposure to temperatures in excess of 250°F; or repeated heavy abrasion, including mechanical wear and repeated scrubbing with industrial grade solvents, cleaners, or scouring agents).
- I.L.1.c.(xiv) "Heat-Resistant Coating" means a coating that must withstand a temperature of at least 400°F during normal use.
- I.L.1.c.(xv) "High-Performance Architectural Coating" means a coating used to protect architectural subsections and which meets the requirements of the Architectural Aluminum Manufacturer Association's publication number AAMA 2604-05 (Voluntary Specification, Performance Requirements and Test Procedures for High Performance Organic Coatings on Aluminum Extrusions and Panels) or 2605-05 (Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing Organic Coatings on Aluminum Extrusions and Panels).
- I.L.1.c.(xv) "Metallic Coating" means a coating which contains more than 5 grams of metal particles per liter of coating as applied. "Metal particles" are pieces of a pure elemental metal or a combination of elemental metals.

- I.L.1.c.(xvi) "Military Specification Coating" means a coating which has a formulation approved by a United States Military Agency for use on military equipment.
- I.L.1.c.(xvii) "Mold-Seal Coating" means the initial coating applied to a new mold or a repaired mold to provide a smooth surface which, when coated with a mold release coating, prevents products from sticking to the mold.
- I.L.1.c.(xviii) "Multi-Component Coating" means a coating requiring the addition of a separate reactive resin, commonly known as a catalyst or hardener, before application to form an acceptable dry film.
- I.L.1.c.(xix) "One-Component Coating" means a coating that is ready for application as it comes out of its container to form an acceptable dry film. A thinner, necessary to reduce the viscosity, is not considered a component.
- I.L.1.c.(xx) "Pan Backing" means a coating applied to the surface of pots, pans, or other cooking implements that are exposed directly to a flame or other heating elements.
- I.L.1.c.(xxi) "Prefabricated Architectural Coating" means coatings applied to metal parts and products which are to be used as an architectural structure.
- I.L.1.c.(xxii) "Repair Coating" means a coating used to re-coat portions of a previously coated product which has sustained mechanical damage to the coating following normal coating operations.
- I.L.1.c.(xxiii) "Silicone Release Coating" means any coating which contains silicone resin and is intended to prevent food from sticking to metal surfaces such as baking pans.
- I.L.1.c.(xxiv) "Solar-Absorbent Coating" means a coating which has as its prime purpose the absorption of solar radiation.
- I.L.1.c.(xxv) "Touch-Up Coating" means a coating used to cover minor coating imperfections appearing after the main coating operation.

I.L.1.c.(xxvi) “Vacuum-Metalizing Coating” means the undercoat applied to the substrate on which the metal is deposited or the overcoat applied directly to the metal film. Vacuum metalizing/physical vapor deposition (PVD) is the process whereby metal is vaporized and deposited on a substrate in a vacuum chamber.

I.L.2. Provisions for Specific Processes

I.L.2.a. Until January 1, 2022, no owner or operator of a facility or operation engaging in the surface coating of manufactured metal parts or metal products may operate a coating application system subject to this regulation that emits VOC in excess of:

I.L.2.a.(i) Clear coatings: 0.52 kg/lc (4.3 lb/gc)

I.L.2.a.(ii) Extreme Performance Coatings: 0.42 kg/lc (3.5 lb/gc)

I.L.2.a.(iii) Air-Dried Coatings: 0.42 kg/lc (3.5 lb/gc)

I.L.2.a.(iv) Other coatings and systems: 0.36 kg/lc (3.0 lb/gc) delivered to a coating applicator for all other coatings and coating application systems.

I.L.2.b. Beginning January 1, 2022, the owner or operator of a facility or operation engaging in the surface coating of manufactured metal parts or metal products must:

I.L.2.b.(i) Reduce VOC emissions from coatings and thinners with an emission control system having a control efficiency of 90% or greater;

I.L.2.b.(ii) Use products that comply with the following VOC content limits (VOC per volume solid) in Table 1; or

Table 1 – Metal parts and products VOC content limits (kg VOC/l solids)				
Coating category	Air dried		Baked	
	kg VOC/l solids	lb VOC/gal solids	kg VOC/l solids	lb VOC/gal solids
General one component	0.54	4.52	0.40	3.35
General multi-component	0.54	4.52	0.40	3.35
Camouflage	0.80	6.67	0.80	6.67

Table 1 – Metal parts and products VOC content limits (kg VOC/l solids)				
Coating category	Air dried		Baked	
	kg VOC/l solids	lb VOC/gal solids	kg VOC/l solids	lb VOC/gal solids
Electric-insulating varnish	0.80	6.67	0.80	6.67
Etching filler	0.80	6.67	0.80	6.67
Extreme high-gloss	0.80	6.67	0.61	5.06
Extreme performance	0.80	6.67	0.61	5.06
Heat-resistant	0.80	6.67	0.61	5.06
High performance architectural	4.56	38.0	4.56	38.0
High temperature	0.80	6.67	0.80	6.67
Metallic	0.80	6.67	0.80	6.67
Military specification	0.54	4.52	0.40	3.35
Mold-seal	0.80	6.67	0.80	6.67
Pan backing	0.80	6.67	0.80	6.67
Prefabricated architectural multi-component	0.80	6.67	0.40	3.35
Prefabricated architectural one-component	0.80	6.67	0.40	3.35
Pretreatment coatings	0.80	6.67	0.80	6.67
Silicone release	0.80	6.67	0.80	5.06
Solar-absorbent	0.80	6.67	0.61	5.06
Vacuum-metalizing	0.80	6.67	0.80	6.67
Drum coating, new, exterior	0.54	4.52	0.54	4.52
Drum coating, new, interior	0.80	6.67	0.80	6.67
Drum coating, reconditioned, exterior	0.80	6.67	0.80	6.67

Table 1 – Metal parts and products VOC content limits (kg VOC/l solids)				
Coating category	Air dried		Baked	
	kg VOC/l solids	lb VOC/gal solids	kg VOC/l solids	lb VOC/gal solids
Drum coating, reconditioned, interior	1.17	9.78	1.17	9.78

I.L.2.b.(iii) Use products that comply with the following VOC content limits (excluding water and exempt compounds, as applied) in Table 2.

Table 2 – Metal parts and products VOC content limits (lb VOC/gallon coating)				
Coating category	Air dried		Baked	
	kg VOC/l coating	lb VOC/gal coating	kg VOC/l coating	lb VOC/gal coating
General one component	0.34	2.8	0.28	2.3
General multi-component	0.34	2.8	0.28	2.3
Camouflage	0.42	3.5	0.42	3.5
Electric-insulating varnish	0.42	3.5	0.42	3.5
Etching filler	0.42	3.5	0.42	3.5
Extreme high-gloss	0.42	3.5	0.36	3.0
Extreme performance	0.42	3.5	0.36	3.0
Heat-resistant	0.42	3.5	0.36	3.0
High performance architectural	0.74	6.2	0.74	6.2
High temperature	0.42	3.5	0.42	3.5
Metallic	0.42	3.5	0.42	3.5
Military specification	0.34	2.8	0.28	2.3
Mold-seal	0.42	3.5	0.42	3.5
Pan backing	0.42	3.5	0.42	3.5

Table 2 – Metal parts and products VOC content limits (lb VOC/gallon coating)				
Coating category	Air dried		Baked	
	kg VOC/l coating	lb VOC/gal coating	kg VOC/l coating	lb VOC/gal coating
Prefabricated architectural multi-component	0.42	3.5	0.28	2.3
Prefabricated architectural one-component	0.42	3.5	0.28	2.3
Pretreatment coatings	0.42	3.5	0.42	3.5
Repair and touch-up	0.42	3.5	0.36	3.0
Silicone release	0.42	3.5	0.42	3.5
Solar-absorbent	0.42	3.5	0.36	3.0
Vacuum-metalizing	0.42	3.5	0.42	3.5
Drum coating, new, exterior	0.34	2.8	0.34	2.8
Drum coating, new, interior	0.42	3.5	0.42	3.5
Drum coating, reconditioned, exterior	0.42	3.5	0.42	3.5
Drum coating, reconditioned, interior	0.50	4.2	0.50	4.2

I.L.2.c. If more than one emission limitation in Sections I.L.2.a., I.L.2.b.(ii), or I.L.2.b.(iii) applies to a specific coating, then the least stringent emission limitation applies.

I.L.2.d. Pioneer Metal Finishing, Inc., a surface coating operation, is authorized pursuant to Regulation Number 3, Part A, Section V.E.2.e. and Regulation Number 25, Part A, Section II.D.1.a. to use up to twenty (20) tons of certified emission reduction credits of volatile organic compounds (VOC) as an alternative compliance method to satisfy the surface coating emission limitations of Regulation Number 25 in accordance with and upon demonstration of the conditions set forth:

- I.L.2.d.(i) Certified emission reduction credits for VOCs (methanol) to be used in this transaction were formerly owned by the Coors Brewing Company, registered and issued in Emissions Reduction Credit Permit 91AR120R on July 25, 1994;
- I.L.2.d.(ii) Those emission reduction credits were originally obtained by Coors from Verticel, a company that produced honeycomb packaging material and was located within five miles of the PMF facility;
- I.L.2.d.(iii) The use of these VOC emission reduction credits identified shall be used to satisfy VOC limitations of certain specified surface coatings in excess of Control Technique Guidance as specified in Regulation Number 25, Section I.L.2.a. and Section I.A.6.b., and applicable to the Pioneer Metal finishing operations;
- I.L.2.d.(iv) Such emission reduction credits identified will be used by PMF to achieve compliance with Regulation Number 25 to compensate for ozone precursor emission of VOCs from non-compliant coatings as authorized by Regulation Number 3, Part A, Section V.E.2.e. In order to satisfy the photochemical reactivity equivalency requirement of VOC trades, the methanol VOC ERCs will be reduced on a ratio of 1.1:1 for VOCs of toluene, ethylbenzene, xylene and ketones emitted from non-compliant coatings. All other VOCs involved in this transaction are considered to be of the same degree of photochemical reactivity;
- I.L.2.d.(v) This transaction is only valid within the Denver/Boulder nonattainment area as described at 40 CFR 81, Subchapter C - Air Programs, Subpart C, Section 107 - Attainment Status Designations, Section 81.306 (February 16, 1995);
- I.L.2.d.(vi) This transaction shall be calculated upon a pound for pound basis and averaged over a maximum 24-hour period.
- I.L.2.d.(vii) This transaction shall be effective upon approval by the U.S. Environmental Protection Agency as a revision to the Colorado State Implementation Plan and after issuance

of a State Construction Permit incorporating, but not limited to, the conditions and requirements of the Section;

- I.L.2.d.(viii) This transaction may not be used to satisfy any current or future requirements of NSPS, BACT, LAER, or MACT requirements of HAPs which may apply to PMF, except that this transaction may be used to satisfy control technique guidance or RACT requirements contained in Regulation Number 25 which are applicable to PMF;
- I.L.2.d.(ix) This transaction shall not interfere with any applicable requirement concerning attainment and reasonable further progress in the Colorado State Implementation Plan or any other applicable requirements of the Clean Air Act;
- I.L.2.d.(x) This transaction shall be registered and enforced through a State Construction Permit issued to Pioneer Metal Finishing, Inc. containing, but not limited to the conditions and limitations set forth in this Section;
- I.L.2.d.(xi) Such state Construction Permit issued to Pioneer Metal Finishing, Inc. shall specify, among other, things the necessary monitory, recordkeeping and reporting requirements to insure that the emission reduction credits are applied in accordance with the conditions and requirements of this Section;
- I.L.2.d.(xii) The state Construction Permit shall allow a daily maximum limitation of 160 lbs. of VOC emissions from non-compliant surface coatings and an annual limitation of 40,000 lbs. of non-compliant VOC emissions. The annual limitation shall be calculated on a 12-month rolling total calculated on the first day of each month using the previous 12 months.
- I.L.2.d.(xiii) The state Construction Permit shall limit the VOC-HAP emissions to less than ten (10) per year of any one HAP or twenty-five (25) tons per year of any combination of HAP emissions; and
- I.L.2.d.(xiv) PMF will maintain records of daily and monthly totals of non-compliant surface coatings used in its operation and report such usages on an annual basis to the Division or as otherwise requested.

- I.L.3. Application methods. Beginning January 1, 2022, unless the owner or operator is reducing VOC emissions by complying with Section I.L.2.b.(i), the owner or operator must use one or more of the following coating application methods
 - I.L.3.a. Electrostatic application.
 - I.L.3.b. High volume low pressure (HVLP) spray.
 - I.L.3.c. Flow coat.
 - I.L.3.d. Roller coat.
 - I.L.3.e. Dip coat, including electrodeposition.
 - I.L.3.f. Airless spray,
 - I.L.3.g. Air-assisted airless spray.
 - I.L.3.h. Other coating application method capable of achieving a transfer efficiency equivalent to or better than that achieved by HVLP spraying.
- I.L.4. Work practices, beginning January 1, 2022, owners or operator's must
 - I.L.4.a. Store all VOC containing coatings, thinners, coating-related waste materials, cleaning materials, and used shop towels in closed containers.
 - I.L.4.b. Keep mixing and storage containers used for VOC containing coatings, thinners, coating-related waste materials, cleaning materials, and used shop towels closed at all times except when depositing or removing materials.
 - I.L.4.c. Minimize spills of VOC containing coatings, thinners, coating-related waste materials cleaning materials, and used shop towels.
 - I.L.4.d. Convey VOC containing coatings, thinners, coating-related waste materials, cleaning materials, and used shop towels in closed containers.
 - I.L.4.e. Minimize VOC emissions from cleaning of application, storage, mixing, and conveying equipment by cleaning equipment without atomizing the cleaning solvent and capturing spent solvent in closed containers.

I.L.5. Recordkeeping. Beginning January 1, 2022, owners or operators must maintain the following records for five (5) years and make them available to the Division upon request

I.L.5.a. Records of the calendar year VOC emission estimates demonstrating whether the coating operation meets or exceeds the applicability threshold in Section I.L.1.a.

I.L.5.b. If applicable, monthly records such as safety data sheets or other analytical data from the coating manufacturer showing the VOC type and VOC content used to comply with the control requirements in Sections I.L.2.b. and I.L.2.c.

I.L.5.c. If applicable, records demonstrating that a listed exemption to this Section I.L. applies.

I.L.5.d. If applicable, monthly records sufficient to demonstrate compliance with the control requirement in Section I.L.2.a.

I.L.6. Reporting.

I.L.6.a. By April 30 2026, and each year thereafter, the owner or operator of a metal parts and product surface coating facility where the total actual VOC emissions from all metal parts and product surface coating operations, including related cleaning activities, at that facility are equal to or greater than 2.7 tons per 12-month rolling period must submit a single annual report using the Division-approved format that includes a demonstration, such as safety data sheets or other analytical data from the coating manufacturer, that products used during the calendar year (e.g., calendar year 2025 for the April 30, 2026, report) comply with the VOC content limits (VOC per volume solid) in Table 1.

I.L.6.b. Beginning in 2025, the owner or operator of a metal parts and product surface coating facility where the total actual VOC emissions from all metal parts and product surface coating operations, including related cleaning activities, at that facility are equal to or greater than 25 tons per year must submit with the semi-annual report required by the facility's operating permit documentation that the products used during the previous semi-annual reporting period comply with the VOC content limits (VOC per volume solid) in Table 1.

I.M. Flat Wood Paneling Coating.

I.M.1. Definitions

- I.M.1.a. "Class II Hardboard Paneling Finish" means finishes which meet the specifications of Voluntary Product Standard PS-59-73 as approved by the American National Standards Institute.
- I.M.1.b. "Coating Application System" means all operations and equipment which apply, convey, and dry a surface coating, including, but not limited to, spray booths, flow coaters, conveyers, flashoff areas, air dryers and ovens.
- I.M.1.c. "Hardboard" is a panel manufactured primarily from inter-felted ligno-cellulosic fibers which are consolidated under heat and pressure in a hot press.
- I.M.1.d. "Hardboard Plywood" is plywood whose surface layer is a veneer of hardwood.
- I.M.1.e. "Natural Finish Hardwood Plywood Panels" means panels whose original grain pattern is enhanced by essentially transparent finishes frequently supplemented by fillers and toners.
- I.M.1.f. "Printed Interior Panels" means panels whose grain or natural surface is obscured by fillers and basecoats upon which a simulated grain or decorative pattern is printed.
- I.M.1.g. "Thin Particleboard" is a manufactured board 1/4 inch or less in thickness made of individual wood particles which have been coated with a binder and formed into flat sheets by pressure.
- I.M.1.h. "Tileboard" means paneling that has a colored waterproof surface coating.

I.M.2. Applicability

This section applies to all flat wood manufacturing and surface finishing facilities that manufacture printed interior panels made of hardwood plywood and thin particle board; natural finish hardwood plywood panels, or hardboard paneling with Class II finishes. This section does not apply to the manufacture of exterior siding, tileboard, or particleboard used as a furniture component.

I.M.3. Emission Limitations

- I.M.3.a. 2.9 kg per 100 square meters of coated finished product (6.0 lb/1,000 sq. ft.) from printed interior panels, regardless of the number of coats applied;

I.M.3.b. 5.8 kg per 100 square meters of coated finished product (12.0 lb/1,000 sq. ft.) from natural finish hardwood plywood panels, regardless of the number of coats applied; and

I.M.3.c. 4.8 kg per 100 square meters of coated finished product (10.0 lb/1,000 sq. ft.) from Class II finishes on hardboard panels, regardless of the number of coats applied.

I.N. Manufacture of Pneumatic Rubber Tires

I.N.1. Definitions

I.N.1.a. "Bead Dipping" means the dipping of an assembled tire bead into a solvent-based cement.

I.N.1.b. "Green Tires" means assembled tires before holding and curing have occurred.

I.N.1.c. "Green Tire Spraying" means the spraying of green tires, both inside and outside, with release compounds which help remove air from the tire during molding and prevent the tire from sticking to the mold after curing.

I.N.1.d. "Pneumatic Rubber Tire Manufacture" means the production of pneumatic rubber, passenger type tires on a mass production basis.

I.N.1.e. "Passenger Type Tire" means agricultural, airplane, industrial, mobile home, light and medium duty truck, and passenger vehicle tires with a bead diameter up to 20.0 inches and cross section dimension up to 12.8 inches.

I.N.1.f. "Tread End Cementing" means the application of a solvent-based cement to the tire tread ends.

I.N.1.g. "Undertread Cementing" means the application of a solvent-based cement to the underside of a tire tread.

I.N.1.h. "Water Based Sprays" means release compounds, sprayed on the inside and outside of green tires, in which solids, water, and emulsifiers have been substituted for organic solvents.

I.N.2. Applicability

This section applies to VOC emissions from the following operations in all pneumatic rubber tire facilities: undertread cementing, tread end cementing, bead dipping, and green tire spraying.

The provisions of this section do not apply to the production of specialty tires for antique or other vehicles when produced on an irregular basis or with short production runs. This exemption applies only to tires produced on equipment separate from normal production lines for passenger type tires.

I.N.3. Provisions for Specific Processes

I.N.3.a. The owner or operator of an undertread cementing, tread end cementing, or bead dipping operation subject to this regulation shall:

I.N.3.a.(i) Install and operate a capture system, designed to achieve maximum reasonable capture, up to 85 percent by weight of VOC emitted, from all undertread cementing, tread end cementing and bead dipping operations. Maximum reasonable capture shall be consistent with the following documents:

I.N.3.a.(i)(A) Industrial Ventilation, A Manual of Recommended Practices, 17th Edition, American Federation of Industrial Hygienists, 1982.

I.N.3.a.(i)(B) Recommended Industrial Ventilation Guidelines, U.S. Department of Health, Education and Welfare, National Institute of Occupational Safety and Health, January 1976.

I.N.3.a.(ii) Install and operate a control device that meets the requirements of one of the following:

I.N.3.a.(ii)(A) A carbon adsorption system designed and operated in a manner such that there is at least a 95.0 percent removal of VOC by weight from the gases ducted to the control device; or,

I.N.3.a.(ii)(B) An incineration system that oxidizes at least 90.0 percent of the nonmethane volatile organic compounds (VOC measured as total combustible carbon) which enter the incinerator to carbon dioxide and water.

I.N.4. The owner or operator of a green tire spraying operation subject to this regulation must implement one of the following means of reducing volatile organic compound emissions:

I.N.4.a. Substitute water-based sprays for the normal solvent-based mold release compound; or,

I.N.4.a.(i) Install a capture system designed and operated in a manner that will capture and transfer at least 90.0 percent of the VOC emitted by the green tire spraying operation to a control device; and,

I.N.4.a.(ii) In addition to Section I.N.4.a.(i), install and operate a control device that meets the requirements of one of the following:

I.N.4.a.(ii)(A) a carbon adsorption system designed and operated in a manner such that there is at least 95.0 percent removal of VOC by weight from the gases ducted to the control device; or,

I.N.4.a.(ii)(B) an incineration system that oxidizes at least 90 percent of the nonmethane volatile organic compounds (VOC measured as total combustible carbon) to carbon dioxide and water.

I.N.5. Testing of capture system efficiency shall meet the requirements of Section I.A.5.e.

I.N.6. Control devices shall meet the applicable requirements, including recordkeeping, of Sections I.A.3.a., b., c., and e., and I.A.8.a. and b.

I.N.7. The applicable EPA reference methods 1 through 4, and 25, of 40 CFR Part 60 (March 29, 2023), shall be used to determine the efficiency of control devices.

I.O. Wood Products Coating

I.O.1. Definitions

- I.O.1.a. “As Applied” means the VOC and solids content of the finishing material that is actually used for coating the substrate. It includes the contribution of materials used for in-house dilution of the finishing material.
- I.O.1.b. “Cleaning Operation” means operations in which organic solvent is used to remove coating materials from equipment used in wood furniture manufacturing operations.
- I.O.1.c. “Conventional Air Spray” means a spray coating method in which the coating is atomized by mixing it with compressed air at an air pressure greater than 10 pounds per square inch (gauge) at the point of atomization. Airless and air assisted spray technologies are not conventional air spray because the coating is not atomized by mixing it with compressed air. Electrostatic spray technology is also not considered conventional air spray because an electrostatic charge is employed to attract the coating to the workplace.
- I.O.1.d. “Equipment Leak” means emissions of VOCs from pumps, valves, flanges, or other equipment used to transfer or apply finishing materials or organic solvents.
- I.O.1.e. “Finishing Material” means a coating used in the wood furniture industry including, but not limited to, basecoats, stains, washcoats, sealers, and topcoats.
- I.O.1.f. “Finishing Operation” means those activities in which a finishing material, including, but not limited to, basecoats, stains, washcoats, sealers, and topcoats, is applied to a substrate and is subsequently air-dried, cured in an oven, or cured by radiation.
- I.O.1.g. “Organic Solvent” means a liquid containing VOCs that is used for dissolving or dispersing constituents in a coating, adjusting the viscosity of a coating, cleaning, or washoff. When used in a coating, the organic solvent evaporates during drying and does not become part of the dried film.
- I.O.1.h. “Sealer” means a finishing material used to seal the pores of a wood substrate before additional coats of finishing material are applied. Washcoats, which are used in some finishing systems to optimize aesthetics, are not sealers.

- I.O.1.i. “Strippable Booth Coating” means a coating that is applied to a booth wall to provide a protective film to receive overspray during finishing operations that is subsequently peeled off and disposed, and reduces or eliminates the need to use organic solvents to clean booth walls.
- I.O.1.j. “Topcoat” means the last film-building finishing material applied in a finishing system. Non-permanent final finishes are not topcoats.
- I.O.1.k. “Washcoat” means a transparent special purpose coating that has a solids content by weight of 12 percent or less. Washcoats are applied over initial stains to protect and control color and to stiffen the wood fibers in order to aid sanding.
- I.O.1.l. “Washoff Operation” means those operations in which organic solvent is used to remove coating from a substrate.
- I.O.1.m. “Wood Furniture” means any product made of wood, a wood product such as rattan or wicker, or an engineered wood product such as particleboard.
- I.O.1.n. “Wood Furniture Component” means any part that is used in the manufacture of wood furniture including, but not limited to, drawer sides, cabinet doors, seat cushions, and laminated tops.
- I.O.1.o. “Wood Furniture Manufacturing Operation” means the finishing, cleaning, and washoff operations associated with the production of wood furniture or wood furniture components.

I.O.2. Applicability

This section applies to wood furniture manufacturing operations with uncontrolled actual VOC emissions greater than or equal to 25 tons per calendar year.

Beginning July 1, 2021, this section applies to other wood products coating operations with uncontrolled actual VOC emissions greater than or equal to 50 tons per year (as of January 27, 2020, located in the 8-Hour Ozone Control Area).

Beginning May 1, 2023, this section applies to other wood products coating operations with uncontrolled actual VOC emissions greater than or equal to 25 tons per year (as of November 7, 2022, located in the 8-Hour Ozone Control Area).

I.O.3. Emission Limitations

I.O.3.a. The owner or operator of a wood furniture manufacturing or other wood products coating operation must limit VOC emissions from finishing operations by:

I.O.3.a.(i) Using topcoats with a VOC content equal to or less than 0.8 lb VOC/lb solids (0.8 kg VOC/kg solids); or

I.O.3.a.(ii) Using a finishing system of:

I.O.3.a.(ii)(A) Sealers with a VOC content equal to or less than 1.9 lb VOC/lb solids (1.9 kg VOC/kg solids), as applied; and

I.O.3.a.(ii)(B) Topcoats with a VOC content equal to or less than 1.8 lb VOC/lb solids (1.8 kg VOC/kg solids), as applied; or

I.O.3.a.(iii) For sources using acid-cured alkyd amino vinyl sealers or acid-cured alkyd amino conversion varnish topcoats:

I.O.3.a.(iii)(A) Use acid-cured alkyd amino vinyl sealers with a VOC content equal to or less than 2.3 lb VOC/lb solids (2.3 kg VOC/kg solids), as applied, and an acid-cured alkyd amino conversion varnish topcoat with a VOC content equal to or less than 2.0 lb VOC/lb solids (2.0 kg VOC/kg solids), as applied; or

I.O.3.a.(iii)(B) Use acid-cured alkyd amino conversion varnish topcoat with a VOC content equal to or less than 2.0 lb VOC/lb solids (2.0 kg VOC/kg solids), as applied, and sealers with a VOC content equal to or less than 1.9 lb VOC/lb solids (1.9 kg VOC/kg solids); or

I.O.3.a.(iii)(C) Use acid-cured alkyd amino vinyl sealers with a VOC content equal to or less than 2.3 lb VOC/lb solids (2.3 kg VOC/kg solids), as applied, and topcoats with a VOC content equal to or less than 1.8 lb VOC/lb solids (1.8 kg VOC/kg solids), as applied.

I.O.3.b. The owner or operator of a wood furniture manufacturing or other wood products coating operation must use strippable booth coatings with a VOC content equal to or less than 0.8 lb VOC/lb solids (0.8 kg VOC/kg solids), as applied.

I.O.3.c. The owner or operator of a wood furniture manufacturing or other wood products coating operation must use compounds containing equal to or less than 8.0 percent by weight of VOC for cleaning spray booth components other than conveyors, continuous coaters and their enclosures, and/or metal filters, unless the spray booth is being refurbished. If the spray booth is refurbished (i.e., spray booth coating or other material used to cover the booth is being replaced), the owner or operator must use equal to or less than 1.0 gallon of organic solvent to prepare the booth prior to applying the booth coating.

I.O.4. Work Practices

I.O.4.a. In addition to complying with Sections I.A.7. and I.A.9., the owner or operator of a wood furniture manufacturing or other wood products coating operation must:

I.O.4.a.(i) Develop an operator training program that includes, at a minimum, appropriate application techniques, appropriate cleaning and washoff procedures, appropriate equipment setup and adjustment to minimize finishing material usage and overspray, and appropriate management of cleanup wastes;

I.O.4.a.(ii) Conduct monthly visual inspections of all equipment used to transfer or apply finishing materials or organic solvents for equipment leaks and repair equipment leaks within 15 working days, or within 3 months if the leaking equipment must be replaced by a new purchase;

I.O.4.a.(iii) Collect cleaning and washoff solvents into closed containers;

I.O.4.a.(iv) Use conventional air spray guns only to:

I.O.4.a.(iv)(A) Apply finishing materials with a VOC content equal to or less than 1.0 lb VOC/lb solids (1.0 kg VOC/kg solids), as applied;

I.O.4.a.(iv)(B) Touch-up and repair after completion of the finishing operation, after stain and before other finishing material, or to apply stain on a part for which it is technically or economically infeasible to use any other spray application technology.

I.O.5. Recordkeeping

I.O.5.a. The owner or operator of a wood furniture manufacturing or other wood products coating operation must keep the following records for five (5) years and make them available for inspection by the Division upon request:

I.O.5.a.(i) Records of calendar year VOC emission estimates demonstrating whether the wood furniture manufacturing operation means or exceeds the applicability threshold in Section I.O.2.;

I.O.5.a.(ii) Records of the operator training program;

I.O.5.a.(iii) Records of the date and results of the monthly equipment inspections and any repairs that were made;

I.O.5.a.(iv) Records such as, but not limited to, data sheets documenting how the as applied values were determined and safety data sheets or other analytical data from the manufacturer showing the VOC content of each sealer, topcoat, strippable booth coating, or cleaning booth compound subject to the emission limits in Section I.O.3.; and

I.O.5.a.(v) Monthly records of the quantity and type of organic cleaning and washoff solvent used.

I.O.6. Reporting. Beginning in 2025, the owner or operator must submit with the semi-annual report required by the facility's operating permit documentation that the products used during the previous semi-annual reporting period comply with the VOC content limits in Section I.O.3.

I.P. Motor Vehicle Materials

I.P.1. Applicability

I.P.1.a. Beginning May 1, 2023, this section applies to automotive coating facilities where the total actual VOC emissions from motor vehicle materials and automotive coatings, including related cleaning activities, at that facility are equal to or greater than 2.7 tons per 12-month rolling period, before consideration of controls.

I.P.1.b. Beginning May 1, 2026, (i.e., after the effective date of a finding by EPA of a failure by Colorado to attain by the serious ozone attainment date or meet an applicable milestone under the 2008 ozone National Ambient Air Quality Standard), this section applies to any person who manufactures, distributes, and/or sells automotive coatings that are used to coat any part or component of motor vehicles (such as cars, buses, and golf carts) or mobile equipment (such as railcars and tractors).

I.P.1.c. This section does not apply to

I.P.1.c.(i) Automotive coatings or associated solvents offered for sale, sold, or manufactured for use outside of Colorado or for shipment to other manufacturers for reformulation or repackaging.

I.P.1.c.(ii) Aerosol coating products.

I.P.1.c.(iii) Automotive coatings sold, supplied, or offered for sale in 0.5 fluid ounce or smaller containers intended to be used by the general public to repair tiny surface imperfections.

I.P.1.c.(iv) Automotive coating facilities subject to Section I.B.

I.P.2. Definitions

I.P.2.a. “Adhesion Promoter” means a coating that is labeled and formulated to be applied to uncoated plastic surfaces to facilitate bonding of subsequent coatings and on which a subsequent coating is applied.

I.P.2.b. “Associated Parts and Components” means structures, devices, pieces, modules, sections, assemblies, subassemblies, or elements of motor vehicles or mobile equipment that are designed to be a part of motor vehicles or mobile equipment but which are not attached to motor vehicles or mobile equipment at the time of coating the structure, device, piece, module, section, assembly, subassembly. Associated parts and components does not include circuit boards.

- I.P.2.c. “Automotive Coating Facility” means any shop, business, location, or parcel of land where motor vehicles or mobile equipment or their associated parts and components are coated, including autobody collision repair shops. Automotive coating facility does not include the original equipment manufacturing plant where the motor vehicle or mobile equipment is completely assembled.
- I.P.2.d. “Clear Coating” means any coating that contains no pigments and is labeled and formulated for application over a color coating or clear coating.
- I.P.2.e. “Color Coating” means any pigmented coating applied over a primer, adhesion promotor, or color coating that requires a subsequent clear coating. Color coatings include metallic/iridescent color coatings. Color coatings does not include adhesion promoters, primers, and multi-color coatings.
- I.P.2.f. “Metallic/Iridescent Color Coating” means any coating that contains more than 0.042 pounds per gallon (5 grams per liter) of metal or iridescent particles as applied, where such particles are visible in the dried film.
- I.P.2.g. “Mobile Equipment” means any device that may be drawn and/or driven on rails or a roadway including, but not limited to, trains, railcars, truck trailers, mobile cranes, bulldozers, street cleaners, and implements of husbandry or agriculture.
- I.P.2.h. “Motor Vehicle” means any self-propelled vehicle including, but not limited to, cars, trucks, buses, golf carts, vans, motorcycles, tanks, and armored personnel carriers.
- I.P.2.i. “Motor Vehicle Bedliner” means a multi-component coating applied to a cargo bed after the application of topcoat to provide additional durability and chip resistance.
- I.P.2.j. “Motor Vehicle Cavity Wax” means a coating applied into the cavities of the vehicle primarily for the purpose of enhancing corrosion protection.
- I.P.2.k. “Motor Vehicle Deadener” means a coating applied to selected vehicle surfaces primarily for the purpose of reducing the sound of road noise in the passenger compartment.

- I.P.2.l. “Motor Vehicle Gasket/Sealing Material” means a fluid applied to coat a gasket or replace and perform the same function as a gasket and includes room temperature vulcanization (RTV) seal material.
- I.P.2.m. “Motor Vehicle Lubricating Wax/Compound” means a protective lubricating material applied to vehicle hubs and hinges.
- I.P.2.n. “Motor Vehicle Sealer” means a high viscosity material generally, but not always, applied in the paint shop after the body has received an electrodeposition primer coating and before the application of subsequent coatings (e.g., primer-surfacer) with the primary purpose to fill body joints completely so that there is no intrusion of water, gases, or corrosive materials into the passenger area of the body compartment. The material may also be referred to as sealant, sealant primer, or caulk.
- I.P.2.o. “Motor Vehicle Trunk Interior Coating” means a coating applied to the trunk interior to provide chip protection.
- I.P.2.p. “Motor Vehicle Underbody Coating” means a coating applied to the undercarriage or firewall to prevent corrosion and/or provide chip protection.
- I.P.2.q. “Multi-color Coating” means any coating that exhibits more than one color in the dried film after a single application and is packaged in a single container.
- I.P.2.r. “Pretreatment Coating” means any coating that contains a minimum of one-half percent acid by weight and not more than 16 percent solids by weight necessary to provide surface etching and is labeled and formulated for application directly to bare metal surfaces to provide corrosion resistance and adhesion.
- I.P.2.s. “Primer” means any coating labeled and formulated for application to a substrate to provide a bond between the substrate and subsequent coats, corrosion resistance, a smooth substrate surface, or resistance to penetration of subsequent coats and on which a subsequent coating is applied. Primers may be pigmented.
- I.P.2.t. “Primer Sealer” means any coating labeled and formulated for application prior to the application of a color coating for the purpose of color uniformity or to promote the ability of the underlying coating to resist penetration by the color coating.

- I.P.2.u. “Single-stage Coating” means any pigmented coating labeled and formulated for application without a subsequent clear coat. Single-stage coatings include single-stage metallic/iridescent coatings. Single-stage coating does not include primers and multi-color coatings.
- I.P.2.v. “Temporary Protective Coating” means any coating which is labeled and formulated for the purpose of temporarily protecting areas from overspray or mechanical damage.
- I.P.2.w. “Truck Bedliner Coating” means any coating labeled and formulated for application to a truck bed to protect it from surface abrasion. Truck bedliner coating does not include clear, color, multi-color, and single-stage coatings.
- I.P.2.x. “Underbody Coating” means any coating labeled and formulated for application to wheel wells, the inside of door panels or fenders, the underside of a trunk or hood, or the underside of the motor vehicle.
- I.P.2.y. “Uniform Finish Coating” means any coating labeled and formulated for application to the area around a spot repair for the purpose of blending a repaired area’s color or clear coat to match the appearance of an adjacent area’s existing coating.
- I.P.3. Beginning May 1, 2026, (i.e., after the effective date of a finding by EPA of a failure by Colorado to attain by the serious ozone attainment date or meet an applicable milestone under the 2008 ozone National Ambient Air Quality Standard), no person can
 - I.P.3.a. Manufacture for sale in Colorado any automotive coating with a VOC content in excess of the VOC limit specified in Table 4.
 - I.P.3.b. Sell, supply, offer for sale, or distribute for sale in Colorado any automotive coating that is manufactured on or after the date in Section I.P.3. with a VOC content in excess of the VOC limit specified in Table 4.
- I.P.4. Except as specified by the timing in Section I.P.4.b., the owner or operator of automotive coating facilities must
 - I.P.4.a. Use products that comply with the following VOC content limits in Table 3 or reduce VOC emissions from motor vehicle materials with an emission control system having a control efficiency of 90% or greater.

- I.P.4.b. Beginning May 1, 2026, (i.e., after the effective date of a finding by EPA of a failure by Colorado to attain by the serious ozone attainment date or meet an applicable milestone under the 2008 ozone National Ambient Air Quality Standard), use products manufactured on or after that effective date that comply with the following VOC content limits in Table 4 or reduce VOC emissions from automotive coatings with an emission control system having a control efficiency of 90% or greater.
- I.P.4.c. Use solvents with a VOC content less than or equal to 25 grams per liter.
- I.P.4.d. If more than one emission limitation applies to a specific material, then the most stringent emission limitation applies.

Table 3 – Motor vehicle materials VOC content limits		
Coating Category	kg VOC/liter coating	lbs VOC/gal coating
Motor vehicle cavity wax	0.65	5.4
Motor vehicle sealer	0.65	5.4
Motor vehicle deadener	0.65	5.4
Motor vehicle gasket/gasket sealing material	0.20	1.7
Motor vehicle underbody coating	0.65	5.4
Motor vehicle trunk interior coating	0.65	5.4
Motor vehicle bedliner	0.20	1.7
Motor vehicle lubricating wax/compound	0.70	5.8

Table 4 – Automotive coatings VOC content limits		
Coating Category	grams/liter	lbs/gallon (minus water and exempt compounds)
Adhesion promoter	540	4.5
Clear coating	250	2.1
Color coating	420	3.5
Multi-color coating	680	5.7
Pretreatment coating	660	5.5
Primer	250	2.1
Single-stage coating	340	2.8
Temporary protective coating	60	0.5
Truck bedliner coating	310	2.6
Underbody coating	430	3.6
Uniform finish coating	540	4.5
Any other coating type	250	2.1

I.P.5. Application methods. Unless the owner or operator of the automotive coating facility is reducing VOC emissions by complying with Section I.P.4.a. and except for the application of underbody coatings, truck bedliner coatings, and coatings of less than one fluid ounce, the owner or operator must use one or more of the following coating application methods

- I.P.5.a. Electrostatic application.
- I.P.5.b. High volume low pressure (HVLP) spray.
- I.P.5.c. Flow coat.
- I.P.5.d. Roller coat.

- I.P.5.e. Dip coat, including electrodeposition.
- I.P.5.f. Airless spray,
- I.P.5.g. Air-assisted airless spray.
- I.P.5.h. Other coating application method capable of achieving a transfer efficiency equivalent to or better than that achieved by HVLP spraying, as approved by EPA.
- I.P.6. Work practices. Owners or operators of automotive coating facilities must
 - I.P.6.a. Store all VOC containing coatings, thinners, coating-related waste materials, cleaning materials, and used shop towels in closed containers.
 - I.P.6.b. Keep mixing and storage containers used for VOC containing coatings, thinners, coating-related waste materials, cleaning materials, and used shop towels closed at all times except when depositing or removing materials.
 - I.P.6.c. Minimize spills of VOC containing coatings, thinners, coating-related waste materials cleaning materials, and used shop towels.
 - I.P.6.d. Convey VOC containing coatings, thinners, coating related waste materials, cleaning materials, and used shop towels in closed containers.
 - I.P.6.e. Minimize VOC emissions from cleaning of application, storage, mixing, and conveying equipment by cleaning equipment without atomizing the cleaning solvent and capturing spent solvent in closed containers.
- I.P.7. Recordkeeping.
 - I.P.7.a. Except as specified by the timing in Section I.P.7.a.(vi)., owners or operators of automotive coating facilities must maintain the following records for five (5) years and make them available to the Division upon request.
 - I.P.7.a.(i) Records of the calendar year VOC emission estimates demonstrating whether the coating operation meets or exceeds the applicability threshold in Section I.P.1.a.

- I.P.7.a.(ii) If applicable, monthly records such as safety data sheets or other analytical data from the product manufacturer showing the VOC type and VOC content and demonstrating compliance with the VOC content limits in Table 3.
- I.P.7.a.(iii) If applicable, records demonstrating that a listed exemption to this Section I.P. applies.
- I.P.7.a.(iv) If applicable, monthly records sufficient to demonstrate compliance with the control requirement in Section I.P.4.a.
- I.P.7.a.(v) If applicable, records sufficient to demonstrate that a coating application method used pursuant to Section I.P.5.h. is capable of achieving a transfer efficiency equivalent to or better than that achieved by HVLP spraying.
- I.P.7.a.(vi) Beginning May 1, 2026, (i.e., after the effective date of a finding by EPA of a failure by Colorado to attain by the serious ozone attainment date or meet an applicable milestone under the 2008 ozone National Ambient Air Quality Standard), maintain records such as safety data sheets or other analytical data from the product manufacturer showing the VOC type and VOC content and demonstrating compliance with the VOC content limits in Table 4.
- I.P.7.b. Beginning May 1, 2026, (i.e., after the effective date of a finding by EPA of a failure by Colorado to attain by the serious ozone attainment date or meet an applicable milestone under the 2008 ozone National Ambient Air Quality Standard), manufacturers, sellers, and distributors must maintain the following records for five (5) years and make them available to the Division upon request.
 - I.P.7.b.(i) Records demonstrating that the automotive coating meets the applicable VOC content limit specified in Table 4.
 - I.P.7.b.(ii) Records demonstrating that the automotive coating label includes the applicable use category(ies) and the VOC content as supplied.

I.P.8. Reporting.

I.P.8.a. By April 30 2026, and each year thereafter, the owner or operator of an automotive coating facility where the total actual VOC emissions from all automotive coating operations, including related cleaning activities, at that facility are equal to or greater than 2.7 tons per 12-month rolling period must submit a single annual report using the Division- approved format that includes a demonstration, such as safety data sheets or other analytical data from the coating manufacturer, that products used during the calendar year (e.g., calendar year 2026 for the April 30, 2027, report) comply with the VOC content limits (VOC per volume solid) in Table 3.

I.P.8.b. Beginning in 2025, the owner or operator of an automotive coating facility where the total actual VOC emissions from all automotive coating operations, including related cleaning activities, at that facility are equal to or greater than 25 tons per year must submit with the semi-annual report required by the facility's operating permit documentation that the products used during the previous semi-annual reporting period comply with the VOC content limits (VOC per volume solid) in Table 3.

I.Q. Aerospace Vehicles or Components

I.Q.1. Applicability

I.Q.1.a. Beginning January 1, 2027, this section applies to facilities that manufacture or rework commercial, civil, or military aerospace vehicles or components (aerospace operations) located in the 8-hour Ozone Control Area where the total actual VOC at the aerospace operation are equal to or greater than 25 tons per year.

I.Q.1.b. This section does not apply to the following activities where cleaning and coating of aerospace components and vehicles may take place.

I.Q.1.b.(i) Research and development, quality control, laboratory testing, and electronic parts and assemblies (except for cleaning and coating of completed assemblies).

I.Q.1.b.(ii) The manufacturing or rework operations involving space vehicles or rework operations performed on antique aerospace vehicles or components.

I.Q.2. Definitions

- I.Q.2.a. “Ablative coating” means a coating that chars when exposed to open flame or extreme temperatures, as would occur during the failure of an engine casing or during aerodynamic heating. The ablative char surface serves as an insulative barrier, protecting adjacent components from the heat or open flame.
- I.Q.2.b. “Adhesion promoter” means a very thin coating applied to a substrate to promote wetting and form a chemical bond with the subsequently applied material.
- I.Q.2.c. “Adhesive bonding primer” means a primer applied in a thin film to aerospace components for the purpose of corrosion inhibition and increased adhesive bond strength by attachment. There are two categories of adhesive bonding primers: primers with a design cure at 250°F or below and primers with a design cure above 250°F.
- I.Q.2.d. “Aerosol coating” means a hand-held, pressurized, nonrefillable container that expels an adhesive or a coating in a finely divided spray when a valve on the container is depressed.
- I.Q.2.e. “Aerospace vehicle or component” means any fabricated part, processed part, assembly of parts, or completed unit, with the exception of electronic components, of any aircraft including but not limited to airplanes, helicopters, missiles, rockets, and space vehicles.
- I.Q.2.f. “Aircraft fluid systems” means those systems that handle hydraulic fluids, fuel, cooling fluids, or oils.
- I.Q.2.g. “Aircraft transparency” means the aircraft windshield, canopy, passenger windows, lenses, and other components that are constructed of transparent materials.
- I.Q.2.h. “Antichafe coating” means a coating applied to areas of moving aerospace components that may rub during normal operations or installation.
- I.Q.2.i. “Antique aerospace vehicle or component” means an aircraft or component thereof that was built at least 30 years ago. An antique aerospace vehicle would not routinely be in commercial or military service in the capacity for which it was designed.
- I.Q.2.j. “Aqueous cleaning solvent” means a solvent in which water is at least 80 percent of the solvent as applied.

- I.Q.2.k. “Bearing coating” means a coating applied to an antifriction bearing, a bearing housing, or the area adjacent to such a bearing in order to facilitate bearing function or to protect base material from excessive wear. A material is not a bearing coating if it can also be classified as a dry lubricative material or a solid film lubricant.
- I.Q.2.l. “Bonding maskant” means a temporary coating used to protect selected areas of aerospace parts from strong acid or alkaline solutions during processing for bonding.
- I.Q.2.m. “Caulking and smoothing compounds” means semi-solid materials that are applied by hand application methods and are used to aerodynamically smooth exterior vehicle surfaces or fill cavities such as bolt hole accesses. A material is not a caulking and smoothing compound if it can also be classified as a sealant.
- I.Q.2.n. “Chemical agent-resistant coating (CARC)” means an exterior topcoat designed to withstand exposure to chemical warfare agents or the decontaminants used on these agents.
- I.Q.2.o. “Chemical milling maskant” means a coating that is applied directly to aluminum components to protect surface areas when chemical milling the component with a Type I or II etchant. Type I chemical milling maskants are used with a Type I etchant and Type II chemical milling maskants are used with a Type II etchant. This definition does not include bonding maskants, critical use and line sealer maskants, and seal coat maskants. Additionally, maskants that must be used with a combination of Type I or II etchants and any of the above types of maskants (i.e., bonding, critical use and line sealer, and seal coat) are not included. Maskants that are defined as specialty coatings are not included under this definition.
- I.Q.2.p. “Cleaning operation” means collectively spray-gun, hand-wipe, and flush cleaning operations.
- I.Q.2.q. “Cleaning solvent” means a liquid material used for hand-wipe, spray gun, or flush cleaning. This definition does not include solutions that contain no VOC.

- I.Q.2.r. “Clear coating” means a transparent coating usually applied over a colored opaque coating, metallic substrate, or placard to give improved gloss and protection to the color coat. In some cases, a clearcoat refers to any transparent coating without regard to substrate.
- I.Q.2.s. “Closed-cycle depainting system” means a dust free, automated process that removes permanent coating in small sections at a time and maintains a continuous vacuum around the area(s) being depainted to capture emissions.
- I.Q.2.t. “Coating” means a material that is applied to the surface of an aerospace vehicle or component to form a decorative, protective, or functional solid film, or the solid film itself.
- I.Q.2.u. “Coating operation” means using a spray booth, tank, or other enclosure or any area, such as a hangar, for applying a single type of coating (e.g., primer). Using the same spray booth for applying another type of coating (e.g., topcoat) constitutes a separate coating operation for which compliance determinations are performed separately.
- I.Q.2.v. “Coating unit” means a series of one or more coating applicators and any associated drying area and/or oven wherein a coating is applied, dried, and/or cured. A coating unit ends at the point where the coating is dried or cured, or prior to any subsequent application of a different coating. It is not necessary to have an oven or flashoff area to be included in this definition.
- I.Q.2.w. “Commercial exterior aerodynamic structure primer” means a primer used on aerodynamic components and structures that protrude from the fuselage, such as wings and attached components, control surfaces, horizontal stabilizers, vertical fins, wing-to-body fairings, antennae, and landing gear and doors, for the purpose of extended corrosion protection and enhanced adhesion.
- I.Q.2.x. “Commercial interior adhesive” means materials used in the bonding of passenger cabin interior components. These components must meet the FAA fireworthiness requirements.

- I.Q.2.y. “Compatible substrate primer” means either compatible epoxy primer or adhesive primer. Compatible epoxy primer is primer that is compatible with the filled elastomeric coating and is epoxy based. The compatible substrate primer is an epoxypolyamide primer used to promote adhesion of elastomeric coatings such as impact-resistant coatings. Adhesive primer is a coating that: (1) inhibits corrosion and serves as a primer applied to bare metal surfaces or prior to adhesive application; or (2) is applied to surfaces that can be expected to contain fuel. Fuel tank coatings are excluded from this category.
- I.Q.2.z. “Confined space” means a space that is large enough and so configured that an employee can bodily enter and perform assigned work; has limited or restricted means for entry or exit (for example, fuel tanks, fuel vessels, and other spaces that have limited means of entry); and is not suitable for continuous employee occupancy.
- I.Q.2.aa. “Corrosion prevention system” means a coating system that provides corrosion protection by displacing water and penetrating mating surfaces, forming a protective barrier between the metal surface and moisture. Coatings containing oils or waxes are excluded from this category.
- I.Q.2.bb. “Critical use and line sealer maskant” means a temporary coating, not covered under other maskant categories, used to protect selected areas of aerospace parts from strong acid or alkaline solutions such as those used in anodizing, plating, chemical milling and processing of magnesium, titanium, or high-strength steel, high-precision aluminum chemical milling of deep cuts, and aluminum chemical milling of complex shapes. Materials used for repairs or to bridge gaps left by scribing operations (i.e., line sealer) are also included in this category.
- I.Q.2.cc. “Cryogenic flexible primer” means a primer designed to provide corrosion resistance, flexibility, and adhesion of subsequent coating systems when exposed to loads up to and surpassing the yield point of the substrate at cryogenic temperatures (-275°F and below).

- I.Q.2.dd. “Cryoprotective coating” means a coating that insulates cryogenic or subcooled surfaces to limit propellant boil-off, maintain structural integrity of metallic structures during ascent or re-entry, and prevent ice formation.
- I.Q.2.ee. “Cyanoacrylate adhesive” means a fast-setting, single component adhesive that cures at room temperature.
- I.Q.2.ff. “Dry lubricative material” means a coating consisting of lauric acid, cetyl alcohol, waxes, or other noncross linked or resin-bound materials that act as a dry lubricant.
- I.Q.2.gg. “Electric or radiation-effect coating” means a coating or coating system engineered to interact, through absorption or reflection, with specific regions of the electromagnetic energy spectrum, such as the ultraviolet, visible, infrared, or microwave regions. Uses include, but are not limited to, lightning strike protection, electromagnetic pulse (EMP) protection, and radar avoidance. Coatings that have been designated as “classified” by the Department of Defense are exempt.
- I.Q.2.hh. “Electrostatic discharge and electromagnetic interference (EMI) coating” means a coating applied to space vehicles, missiles, aircraft radomes, and helicopter blades to disperse static energy or reduce electromagnetic interference.
- I.Q.2.ii. “Elevated-temperature Skydrol-resistant commercial primer” means a primer applied primarily to commercial aircraft (or commercial aircraft adapted for military use) that must withstand immersion in phosphate-ester (PE) hydraulic fluid (Skydrol 500b or equivalent) at the elevated temperature of 150°F for 1,000 hours.
- I.Q.2.jj. “Epoxy polyamide topcoat” means a coating used where harder films are required or in some areas where engraving is accomplished in camouflage colors.
- I.Q.2.kk. “Exempt solvent” means a specified organic compound that has been determined by the EPA to have negligible photochemical reactivity and is listed in 40 CFR 51.100 (Nov. 7, 1986).

- I.Q.2.ll. “Fire-resistant (interior) coating” means for civilian aircraft, fire-resistant interior coatings are used on passenger cabin interior parts that are subject to the FAA fireworthiness requirements. For military aircraft, fire-resistant interior coatings are used on parts that are subject to the flammability requirements of MIL-STD-1630A and MIL-A-87721. For space applications, these coatings are used on parts that are subject to the flammability requirements of SE-R-0006 and SSP 30233.
- I.Q.2.mm. “Flexible primer” means a primer that meets flexibility requirements such as those needed for adhesive bond primed fastener heads or on surfaces expected to contain fuel. The flexible coating is required because it provides a compatible, flexible substrate over bonded sheet rubber and rubber-type coatings as well as a flexible bridge between the fasteners, skin, and skin-to-skin joints on outer aircraft skins. This flexible bridge allows more topcoat flexibility around fasteners and decreases the chance of the topcoat cracking around the fasteners. The result is better corrosion resistance.
- I.Q.2.nn. “Flight test coating” means a coating applied to aircraft other than missiles or single-use aircraft prior to flight testing to protect the aircraft from corrosion and to provide required marking during flight test evaluation.
- I.Q.2.oo. “Flush cleaning” means removal of contaminants such as dirt, grease, oil, and coatings from an aerospace vehicle or component or coating equipment by passing solvent over, into, or through the item being cleaned. The solvent may simply be poured into the item being cleaned and then drained, or assisted by air or hydraulic pressure, or by pumping. Hand-wipe cleaning operations where wiping, scrubbing, mopping, or other hand action are used are not included.
- I.Q.2.pp. “Fuel tank adhesive” means an adhesive used to bond components exposed to fuel and must be compatible with fuel tank coatings.
- I.Q.2.qq. “Fuel tank coating” means a coating applied to fuel tank components for the purpose of corrosion and/or bacterial growth inhibition and to assure sealant adhesion in extreme environmental conditions.

I.Q.2.rr. “Grams of VOC per liter of coating (less water and less exempt solvent)” means the weight of VOC per combined volume of total volatiles and coating solids, less water and exempt compounds, and can be calculated by the following equation:

Grams of VOC per liter of coating (less water and less exempt solvent) = $(Ws - Ww - Wes) / (Vs - Vw - Ves)$

Where:

Ws = weight of total volatiles in grams

Ww = weight of water in grams

Wes = weight of exempt compounds in grams

Vs = volume of coating in liters

Vw = volume of water in liters

Ves = volume of exempt compounds in liters

I.Q.2.ss. “Hand-wipe cleaning operation” means removing contaminants such as dirt, grease, oil, and coatings from an aerospace vehicle or component by physically rubbing it with a material such as a rag, paper, or cotton swab that has been moistened with a cleaning solvent.

I.Q.2.tt. “High temperature coating” means a coating designed to withstand temperatures of more than 350°F.

I.Q.2.uu. “High volume low pressure (HVLP) spray equipment” means spray equipment that is used to apply coating by means of a spray gun that operates at 10.0 pounds per square inch gauge (psig) of atomizing air pressure or less at the air cap.

I.Q.2.vv. “Insulation covering” means material that is applied to foam insulation to protect the insulation from mechanical or environmental damage.

I.Q.2.ww. “Intermediate release coating” means a thin coating applied beneath topcoats to assist in removing the topcoat in depainting operations and generally to allow the use of less hazardous depainting methods.

- I.Q.2.xx. “Lacquer” means a clear or pigmented coating formulated with a nitrocellulose or synthetic resin to dry by evaporation without a chemical reaction. Lacquers are resolvable in their original solvent.
- I.Q.2.yy. “Leak” means any visible leakage, including misting and clouding.
- I.Q.2.zz. “Limited access space” means internal surfaces or passages of an aerospace vehicle or component that cannot be reached without the aid of an airbrush or a spray gun extension for the application of coatings.
- I.Q.2.aaa. “Metalized epoxy coating” means a coating that contains relatively large quantities of metallic pigmentation for appearance and/or added protection.
- I.Q.2.bbb. “Mold release” means a coating applied to a mold surface to prevent the molded piece from sticking to the mold as it is removed.
- I.Q.2.ccc. “Nonstructural adhesive” means an adhesive that bonds nonload bearing aerospace components in noncritical applications and is not covered in any other specialty adhesive categories.
- I.Q.2.ddd. “Operating parameter value” means a minimum or maximum value established for a control equipment or process parameter that, if achieved by itself or in combination with one or more other operating parameter values, determines that an owner or operator has continued to comply with an applicable emission limitation.
- I.Q.2.eee. “Optical antireflection coating” means a coating with a low reflectance in the infrared and visible wavelength ranges that is used for antireflection on or near optical and laser hardware.
- I.Q.2.fff. “Part marking coating” means coatings or inks used to make identifying markings on materials, components, and/or assemblies. These markings may be either permanent or temporary.
- I.Q.2.ggg. “Pretreatment coating” means an organic coating that contains at least 0.5 percent acids by weight and is applied directly to metal or composite surfaces to provide surface etching, corrosion resistance, adhesion, and ease of stripping.

- I.Q.2.hhh. “Primer” means the first layer and any subsequent layers of identically formulated coating applied to the surface of an aerospace vehicle or component. Primers are typically used for corrosion prevention, protection from the environment, functional fluid resistance, and adhesion of subsequent coatings. Primers that are defined as specialty coatings are not included under this definition.
- I.Q.2.iii. “Radome” means the nonmetallic protective housing for electromagnetic transmitters and receivers (e.g., radar, electronic countermeasures, etc.).
- I.Q.2.jjj. “Rain erosion-resistant coating” means a coating or coating system used to protect the leading edges of parts such as flaps, stabilizers, radomes, engine inlet nacelles, etc. against erosion caused by rain impact during flight.
- I.Q.2.kkk. “Research and development” means an operation whose primary purpose is for research and development of new processes and products and that is conducted under the close supervision of technically trained personnel and is not involved in the manufacture of final or intermediate products for commercial purposes, except in a de minimis manner.
- I.Q.2.III. “Rocket motor bonding adhesive” means an adhesive used in rocket motor bonding applications.
- I.Q.2.mmm. “Rocket motor nozzle coating” means a catalyzed epoxy coating system used in elevated temperature applications on rocket motor nozzles.
- I.Q.2.nnn. “Rubber-based adhesive” means a quick setting contact cement that provide a strong, yet flexible bond between two mating surfaces that may be of dissimilar materials.
- I.Q.2.ooo. “Scale inhibitor” means a coating that is applied to the surface of a part prior to thermal processing to inhibit the formation of scale.
- I.Q.2.ppp. “Screen print ink” means an ink used in screen printing processes during fabrication of decorative laminates and decals.

- I.Q.2.qqq. “Sealant” means a material used to prevent the intrusion of water, fuel, air, or other liquids or solids from certain areas of aerospace vehicles or components. There are two categories of sealants: (1) extrudable/rollable/brushable sealants; and (2) sprayable sealants.
- I.Q.2.rrr. “Seal coat maskant” means an overcoat applied over a maskant to improve abrasion and chemical resistance during production operations.
- I.Q.2.sss. “Self-priming topcoat” means a topcoat that is applied directly to an uncoated aerospace vehicle or component for purposes of corrosion prevention, environmental protection, and functional fluid resistance. More than one layer of identical coating formulation may be applied to the vehicle or component.
- I.Q.2.ttt. “Semiaqueous cleaning solvent” means a solution in which water is a primary ingredient (> 60 percent of the solvent solution as applied must be water).
- I.Q.2.uuu. “Silicone insulation material” means an insulating material applied to exterior metal surfaces for protection from high temperatures caused by atmospheric friction or engine exhaust. These materials differ from ablative coatings in that they are not “sacrificial.”
- I.Q.2.vvv. “Solids” means the nonvolatile portion of the coating that after drying makes up the dry film.
- I.Q.2.www. “Solid film lubricant” means a very thin coating consisting of a binder system containing as its chief pigment material one or more of the following: molybdenum, graphite, polytetrafluoroethylene (PTFE), or other solids that act as a dry lubricant between faying (i.e., closely or tightly fitting) surfaces.
- I.Q.2.xxx. “Space vehicle” means a man-made device, either manned or unmanned, designed for operation beyond earth's atmosphere. This definition includes integral equipment such as models, mock-ups, prototypes, molds, jigs, tooling, hardware jackets, and test coupons. Also included is auxiliary equipment associated with test, transport, and storage that through contamination can compromise the space vehicle performance.

- I.Q.2.yyy. “Specialty coating” means a coating that, even though it meets the definition of a primer, topcoat, or self-priming topcoat, has additional performance criteria beyond those of primers, topcoats, and self-priming topcoats for specific applications. These performance criteria may include, but are not limited to, temperature or fire resistance, substrate compatibility, antireflection, temporary protection or marking, sealing, adhesively joining substrates, or enhanced corrosion protection.
- I.Q.2.zzz. “Specialized function coating” means a coating that fulfills extremely specific engineering requirements that are limited in application and are characterized by low volume usage. This category excludes coatings covered in other specialty coating categories.
- I.Q.2.aaaa. “Spray gun” means a device that atomizes a coating or other material and projects the particulates or other material onto a substrate.
- I.Q.2.bbbb. “Structural autoclavable adhesive” means an adhesive used to bond load-carrying aerospace components that is cured by heat and pressure in an autoclave.
- I.Q.2.cccc. “Structural nonautoclavable adhesive” means an adhesive cured under ambient conditions that is used to bond load-carrying aerospace components or other critical functions, such as nonstructural bonding in the proximity of engines.
- I.Q.2.dddd. “Surface preparation” means the removal of contaminants from the surface of an aerospace vehicle or component or the activation or reactivation of the surface in preparation for the application of a coating.
- I.Q.2.eeee. “Temporary protective coating” means a coating applied to provide scratch or corrosion protection during manufacturing, storage, or transportation. Two types include: (1) peelable protective coatings; and (2) alkaline removable coatings. These materials are not intended to protect against strong acid or alkaline solutions. Coatings that provide this type of protection from chemical processing are not included in this category.
- I.Q.2.ffff. “Thermal control coating” means a coating formulated with specific thermal conductive or radiative properties to permit temperature control of the substrate.

- I.Q.2.gggg. “Topcoat” means a coating that is applied over a primer on an aerospace vehicle or component for appearance, identification, camouflage, or protection. Topcoats that are defined as specialty coatings are not included under this definition.
- I.Q.2.hhhh. “Touch-up and repair coating” means a coating used to cover minor coating imperfections appearing after the main coating operation.
- I.Q.2.iiii. “Touch-up and repair operation” means that portion of the coating operation that is the incidental application of coating used to cover minor imperfections in the coating finish or to achieve complete coverage. This definition includes out-of-sequence or out-of-cycle coating.
- I.Q.2.jjjj. “VOC composite vapor pressure” means the sum of the partial pressures of the compounds defined as VOCs, and is determined by the following calculation.

$$PP_c = \sum_{i=1}^n \frac{\frac{W_i}{MW_i} \times VP_i}{\frac{W_w}{MW_w} + \frac{\sum_{i=1}^n W_e}{MW_e} + \sum_{i=1}^n \frac{W_i}{MW_i}}$$

Wi = weight of the “i”th VOC compound, grams

Ww = weight of water, grams

We = weight of nonwatery, non-VOC compound, grams

MWi = molecular weight of the “i”th VOC compound, g/g-mole

MWw = molecular weight of water, g/g-mole

Mwe = molecular weight of exempt compound, g/g-mole

PPc = VOC composite partial pressure at 20°C, mm Hg

VPi = vapor pressure of the “i”th VOC compound at 20°C, mm Hg

I.Q.2.kkkk. “Waterborne (water-reducible) coating” means a coating which contains more than 5 percent water by weight as applied in its volatile fraction.

I.Q.2.III. “Wet fastener installation coating” means a primer or sealant applied by dipping, brushing, or daubing to fasteners that are installed before the coating is cured.

I.Q.2.mmmm. “Wing coating” means a corrosion-resistant topcoat that is resilient enough to withstand the flexing of the wings.

I.Q.3. The owner or operator of an aerospace operation must

I.Q.3.a. Apply specialty coatings to aerospace vehicles or components that comply with the VOC limits in Table 5 as applied, or reduce VOC emissions from aerospace specialty coatings with an emission control system having a control efficiency of 90% or greater.

I.Q.3.a.(i) The following coating applications are exempt from the VOC content limits in Table 5.

I.Q.3.a.(i)(A) Touchup, aerosol, and Department of Defense (DOD) classified coatings.

I.Q.3.a.(i)(B) Coating of space vehicles.

I.Q.3.a.(i)(C) Aerospace operations that use separate formulations in volumes of less than 50 gallons per year, subject to a maximum exemption of 200 gallons total for such formulations applied annually.

I.Q.3.a.(i)(D) Primers, topcoats, and chemical milling maskants that are used by and regulated for major aerospace operations under 40 C.F.R. Section 63.749(a) (Aug. 3, 2016).

I.Q.3.a.(ii) If more than one emission limitation applies to a specific material, then the most stringent emission limitation applies.

Table 5 – Aerospace specialty coatings VOC content limits	
Coating Type	Grams VOC per liter of coating (g/L), less water and less exempt solvent
Ablative coating	600
Adhesion promoter	890
Adhesive bonding primer – cured at 250°F or below	850
Adhesive bonding primer – cured above 250°F	1030
Adhesive – commercial interior adhesive	760
Adhesive – cyanoacrylate adhesive	1020
Adhesive – fuel tank adhesive	620
Adhesive – nonstructural adhesive	360
Adhesive – rocket motor bonding adhesive	890
Adhesive – rubber-based adhesive	850
Adhesive – structural autoclavable adhesive	60
Adhesive – structural nonautoclavable adhesive	850
Antichafe coating	660
Bearing coating	620
Caulking and smoothing compounds	850
Chemical agent-resistant coating	550

Table 5 – Aerospace specialty coatings VOC content limits	
Coating Type	Grams VOC per liter of coating (g/L), less water and less exempt solvent
Clear coating	720
Commercial exterior aerodynamic structure primer	650
Compatible substrate primer	780
Corrosion prevention compound	710
Cryogenic flexible primer	645
Cryoprotective coating	600
Dry lubricative material	880
Electric or radiation-effect coating	800
Electrostatic discharge and electromagnetic interference (EMI) coating	800
Elevated-temperature skydrol-resistant commercial primer	740
Epoxy polyamide topcoat	660
Fire-resistant (interior) coating	800
Flexible primer	640
Flight test coatings – missile or single use aircraft	420
Flight-test coatings – all other	840
Fuel-tank coating	720
High-temperature coating	850
Insulation covering	740
Intermediate release coating	750

Table 5 – Aerospace specialty coatings VOC content limits	
Coating Type	Grams VOC per liter of coating (g/L), less water and less exempt solvent
Lacquer	830
Maskants – bonding maskant	1230
Maskants – critical use and line sealer maskant	1020
Maskants – seal coat maskant	1230
Metallized epoxy coating	740
Mold release	780
Optical anti-reflective coating	750
Park marking coating	850
Pretreatment coating	780
Rain erosion-resistant coating	850
Rocket motor nozzle coating	660
Scale inhibitor	880
Screen print ink	840
Sealants – extrudable/rollable/brushable sealant	280
Sealants – sprayable sealant	600
Silicone insulation material	850
Solid film lubricant	880
Specialized function coating	890
Temporary protective coating	320
Thermal control coating	800
Wet fastener installation coating	675

Table 5 – Aerospace specialty coatings VOC content limits	
Coating Type	Grams VOC per liter of coating (g/L), less water and less exempt solvent
Wing coating	850

I.Q.3.b. Comply with the following solvent cleaning requirements.

I.Q.3.b.(i) Use cleaning solvents in hand-wipe cleaning operations that

I.Q.3.b.(i)(A) Meet the definition of aqueous cleaning solvent or

I.Q.3.b.(ii)(B) Have a VOC composite vapor pressure less than or equal to 45 millimeters of mercury (mm Hg) at 20°C.

I.Q.3.b.(ii) Empty cleaning solvents used in the flush cleaning of parts, assemblies, and coating unit components into an enclosed container or collection system that is kept closed when not in use. This requirement does not apply to aqueous cleaning solvents.

I.Q.3.b.(iii) Clean spray guns using one or more of the following methods.

I.Q.3.b.(iii)(A) Enclosed spray gun cleaning system.

I.Q.3.b.(iii)(A)(1) The enclosed spray gun cleaning system must be kept closed when not in use.

I.Q.3.b.(iii)(A)(2) The enclosed spray gun cleaner must be visually inspected for leaks at least once per month while in operation.

- I.Q.3.b.(iii)(A)(3) Leaks must be repaired within 14 days from when the leak is first discovered. If the leak is not repaired by the 15th day after detection, the solvent must be removed, and the enclosed cleaner must be shut down until the leak is repaired or its use is permanently discontinued.
- I.Q.3.b.(iii)(B) Unatomized discharge of solvent into a waste container that is kept closed when not in use.
- I.Q.3.b.(iii)(C) Disassembly of the spray gun and cleaning in a vat that is kept closed when not in use.
- I.Q.3.b.(iii)(D) Atomized spray into a waste container that is fitted with a device designed to capture atomized solvent emissions.
- I.Q.3.b.(iv) The following solvent cleaning operations are exempt from Section I.Q.3.b.(i).
 - I.Q.3.b.(iv)(A) Cleaning during the manufacture, assembly, installation, maintenance, or testing of components of breathing oxygen systems that are exposed to the breathing oxygen.
 - I.Q.3.b.(iv)(B) Cleaning during the manufacture, assembly, installation, maintenance, or testing of parts, subassemblies, or assemblies that are exposed to strong oxidizers or reducers (e.g., nitrogen tetroxide, liquid oxygen, hydrazine).
 - I.Q.3.b.(iv)(C) Cleaning and surface activation prior to adhesive bonding.
 - I.Q.3.b.(iv)(D) Cleaning of electronics parts and assemblies containing electronics parts.
 - I.Q.3.b.(iv)(E) Cleaning of aircraft and ground support equipment fluid systems that are exposed to the fluid, including air-to-air heat exchangers and hydraulic fluid systems.
 - I.Q.3.b.(iv)(F) Cleaning of fuel cells, fuel tanks, and confined spaces.

- I.Q.3.b.(iv)(G) Surface cleaning of solar cells, coated optics, and thermal control surfaces.
- I.Q.3.b.(iv)(H) Cleaning during fabrication, assembly, installation, and maintenance of upholstery, curtains, carpet, and other textile materials used on the interior of the aircraft.
- I.Q.3.b.(iv)(I) Cleaning of metallic and nonmetallic materials used in honeycomb cores during the manufacture or maintenance of these cores, and cleaning of the completed cores used in the manufacture of aerospace vehicles or components.
- I.Q.3.b.(iv)(J) Cleaning of aircraft transparencies, polycarbonate, or glass substrates.
- I.Q.3.b.(iv)(K) Cleaning and solvent usage associated with research and development, quality control, or laboratory testing.
- I.Q.3.b.(iv)(L) Cleaning operations, using nonflammable liquids, conducted within 5 feet of energized electrical systems. Energized electrical systems means any AC or DC electrical circuit on an assembled aircraft once electrical power is connected, including interior passenger and cargo areas, wheel wells, and tail sections.
- I.Q.3.b.(iv)(M) Cleaning operations identified as essential uses under the Montreal Protocol for which the Administrator has allocated essential use allowances or exemptions in 40 C.F.R. § 82.4. (May 10, 1995).

I.Q.4. Application methods.

- I.Q.4.a. The aerospace operation owner or operator must use one or more of the following coating application methods in applying any primer or topcoat to aerospace vehicles or components.
 - I.Q.4.a.(i) Brush coating.
 - I.Q.4.a.(ii) Cotton-tipped swab application.
 - I.Q.4.a.(iii) Electrodeposition coating.

I.Q.4.a.(iv) Electrostatic spray.

I.Q.4.a.(v) High volume low pressure (HVLP) spray.

I.Q.4.a.(vi) Flow/curtain coat.

I.Q.4.a.(vii) Roll coating.

I.Q.4.a.(viii) Dip coat.

I.Q.4.a.(ix) Other coating application methods that achieve emission reductions equivalent to HVLP or electrostatic spray application methods.

I.Q.4.b. The following situations are exempt from the application equipment requirements in Section I.Q.4.a.

I.Q.4.b.(i) Any situation that normally requires the use of an airbrush or an extension on the spray gun to properly reach limited access spaces.

I.Q.4.b.(ii) The application of specialty coatings.

I.Q.4.b.(iii) The application of coatings that contain fillers that adversely affect atomization with HVLP spray guns and that cannot be applied by any of the other application methods specified in Section I.Q.4.a.

I.Q.4.b.(iv) The application of coatings that normally have a dried film thickness of less than 0.0013 centimeter (0.0005 in.) and cannot be applied by any of the other application methods specified in Section I.Q.4.a.

I.Q.4.b.(v) The use of airbrush application methods for stenciling, lettering, and other identification markings.

I.Q.4.b.(vi) The use of hand-held spray can application methods.

I.Q.4.b.(vii) Touch-up and repair operations.

I.Q.4.c. As an alternative to the application equipment requirements in Section I.Q.4.a., a major aerospace operation may comply with applicable application techniques under 40 C.F.R. Section 63.749(a) (Aug. 3, 2016).

I.Q.5. Work practices. Owners or operators of aerospace operations must

- I.Q.5.a. Store all VOC containing coatings, coating-related waste materials, cleaning materials, and cleaning-related waste materials in closed containers.
- I.Q.5.b. Keep mixing and storage containers used for VOC containing coatings, coating-related waste materials, cleaning materials, and cleaning-related waste materials closed at all times except when depositing or removing materials.
- I.Q.5.c. Minimize spills of VOC containing coatings, coating-related waste materials cleaning materials, and cleaning-related waste materials.
- I.Q.5.d. Convey VOC containing coatings, coating related waste materials, cleaning materials, and cleaning-related waste materials in closed containers.
- I.Q.5.e. Minimize VOC emissions from cleaning of application, storage, mixing, and conveying equipment by cleaning equipment without atomizing the cleaning solvent and capturing spent solvent in closed containers.
- I.Q.6. Recordkeeping.
 - I.Q.6.a. Owners or operators of aerospace operations must maintain the following records for five (5) years and make them available to the Division upon request.
 - I.Q.6.a.(i) Records of the calendar year VOC emission estimates demonstrating whether the aerospace operation meets or exceeds the applicability threshold in Section I.Q.1.a.
 - I.Q.6.a.(ii) If applicable, records demonstrating that a listed exemption to this Section I.Q. applies.
 - I.Q.6.a.(iii) If applicable, monthly records such as safety data sheets or other analytical data from the product manufacturer showing the VOC type and VOC content and demonstrating compliance with the VOC content limits in Table 5 of coatings as applied.

I.Q.6.a.(iv) If applicable, monthly records such as safety data sheets or other analytical data from the product manufacturer demonstrating compliance with the cleaning solvent requirements in Section I.Q.3.b.(i).

I.Q.6.a.(v) If applicable, monthly records sufficient to demonstrate compliance with the enclosed spray gun cleaning system inspection requirements in I.Q.3.b.(iii)(A).

I.Q.6.a.(vi) If applicable, monthly records sufficient to demonstrate compliance with the control requirement in Section I.Q.3.a.

I.Q.6.a.(vii) If applicable, records sufficient to demonstrate that a coating application method used pursuant to Section I.Q.4.a.(ix) is capable of achieving a transfer efficiency equivalent to or better than that achieved by HVLP spraying.

I.Q.7. Reporting.

I.Q.7.a. Beginning in 2027, the aerospace operation owner or operator must submit with the semi-annual report required by the facility's operating permit documentation that the products used during the previous semi-annual reporting period comply with the VOC content limits in Table 5 and in Section I.Q.3.b.(i).

II. Solvent Use

II.A. General Provisions

II.A.1. Applicability

The provisions of this section apply to cold cleaners, non-conveyorized vapor degreasers, conveyorized degreasers, industrial cleaning solvent operations, and other operations that use solvents. Open top vapor degreasers are a subset of non-conveyorized vapor degreasers. The owner or operator of a unit subject to this section shall ensure that no such unit is used unless the requirements of this section are satisfied. Section II.E. requirements are effective on January 1, 2017. Section II.F. requirements are effective on May 1, 2021.

II.A.2. Definitions

II.A.2.a. "Cold-Cleaner" means a container of non-aqueous liquid solvent held below its boiling point, which is designed, used, or intended for cleaning solid objects in a batch-loaded process. A "cold-cleaner" may have provisions for heating the solvent. It does not include vapor degreasers or continuously loaded conveyorized degreasers.

II.A.2.b. "Composite Partial Vapor Pressure" means the sum of the partial pressures of the compounds defined as VOCs. Composite partial vapor pressure is calculated as follows:

$$PP_c = \frac{\sum_{i=1}^n \frac{(W_i)(VP_i)}{MW_i}}{\frac{W_w}{MW_w} + \sum_{e=1}^n \frac{W_e}{MW_e} + \sum_{i=1}^n \frac{W_i}{MW_i}}$$

Where:

Wi = Weight of the "i"th VOC compound, in grams

Ww = Weight of water, in grams

We = Weight of exempt compound, in grams

MWi = Molecular weight of the "i"th VOC compound, in g/g-mole

MWw = Molecular weight of water, in g/g-mole

MWc = Molecular weight of exempt compound, in g/g-mole

PPc = VOC composite partial vapor pressure at 20°C (68°F), in mm Hg

VPi = Vapor pressure of the "i"th VOC compound at 20°C(68°F), in mm Hg

II.A.2.c. "Conveyorized Degreaser" means an apparatus that performs degreasing or other cleaning functions through the use of non-aqueous liquid solvent and/or solvent vapors within a container, and which has a conveyor mechanism allowing continuous loading of items conveyed into and out of the solvent.

- II.A.2.d. "Freeboard" in a vapor degreaser means the vertical distance from the top of the vapor zone (as established by normal operations within the specifications of the degreaser manufacturer) to the top of the degreaser. For cold-cleaners "freeboard" means the vertical distance from the surface of the solvent liquid to the top of the degreaser. If all sides are not even, the vertical distance to the top of the lowest side shall be used to make the determination of freeboard.
- II.A.2.e. "Freeboard Ratio" means the ratio of the freeboard to the width of the solvent surface.
- II.A.2.f. "Industrial Cleaning Solvent" means a VOC-containing liquid used to perform industrial cleaning solvent operations.
- II.A.2.g. "Industrial Cleaning Solvent Operation" means the use of an industrial cleaning solvent for cleaning industrial operations such as spray gun cleaning, spray booth cleaning, large manufactured parts cleaning, equipment cleaning, floor cleaning, line cleaning, parts cleaning, tank cleaning, and small manufactured parts cleaning. Residential and janitorial cleaning are not considered industrial cleaning solvent operations.
- II.A.2.h. "Non-Conveyorized Vapor Degreaser" means an apparatus, which uses non-aqueous solvent vapors within some type of container to degrease or otherwise clean solid objects in a batch-loaded process. It excludes continuously loaded conveyorized degreasers.
- II.A.2.i. "Residential and Janitorial Cleaning" means the cleaning of a building or building components including, but not limited to, floors, ceilings, wall, windows, doors, stairs, bathrooms, furnishings, and exterior surfaces of office equipment, excluding the cleaning of work areas where manufacturing or repair activity is performed.
- II.A.2.j. "Solvent Metal Cleaning" means the process of cleaning soils from metal surfaces by cold cleaning, conveyorized degreasing, or non-conveyorized vapor degreasing.

II.A.3. Transfer of waste solvent and used solvent

In any disposal or transfer of waste or used solvent, at least 80 percent by weight of the solvent/waste liquid shall be retained (i.e., no more than 20 percent of the liquid solvent/solute mixture shall evaporate or otherwise be lost during transfers).

II.A.4. Storage of waste solvent and used solvent

Waste or used solvent shall be stored in closed containers unless otherwise required by law.

II.A.5. Any control device shall meet the applicable requirements of Sections I.A.3.a., b., c., e., and I.A.8.a. and b.

II.B. Control of Solvent Cold-Cleaners

II.B.1. Control Equipment

II.B.1.a. Covers

II.B.1.a.(i) All cold-cleaners shall have a properly fitting cover.

II.B.1.a.(ii) Covers shall be designed to be easily operable with one hand under any of the following conditions:

II.B.1.a.(ii)(A) Solvent true vapor pressure is greater than 15 torr (0.3 psia) at 38°C (100°F).

II.B.1.a.(ii)(B) The solvent is agitated by an agitating mechanism.

II.B.1.a.(ii)(C) The solvent is heated.

II.B.1.b. Drainage Facility

II.B.1.b.(i) All cold-cleaners shall have a drainage facility that captures the drained liquid solvent from the cleaned parts.

II.B.1.b.(ii) For cold-cleaners using solvent which has a vapor pressure greater than 32 torr (0.62 psia) measured at 38°C (100°F) either:

II.B.1.b.(ii)(A) There shall be an internal drainage facility within the confines of the cold-cleaner, so that parts are enclosed under the (closed) cover to drain after cleaning, or if such a facility will not fit within;

II.B.1.b.(ii)(B) An enclosed, external drainage facility that captures the drained solvent liquid from the cleaned parts.

II.B.1.c. A permanent, clearly visible sign shall be mounted on or next to the cold-cleaner. The sign shall list the operating requirements.

II.B.1.d. Solvent spray apparatus shall not have a splashing, fine atomizing, or shower type action but rather should produce a solid, cohesive stream. Solvent spray shall be used at a pressure that does not cause excessive splashing.

For solvents with a true vapor pressure above 32 torr (0.62 psia) at 38°C (100°F), or, for solvents heated above 50°C (120°F), one of the following techniques shall be used:

II.B.1.d.(i) A freeboard ratio greater than or equal to 0.7.

II.B.1.d.(ii) A water or a non-volatile liquid cover. The cover liquid shall not be soluble in the solvent and shall not be denser than the solvent and the depth of the cover liquid shall be sufficient to prevent the escape of solvent vapors.

II.B.2. Operating requirements

II.B.2.a. The cold-cleaner cover shall be closed whenever parts are not being handled within the cleaner confines.

II.B.2.b. Cleaned parts shall be drained for at least 15 seconds and/or until dripping ceases. Any pools of solvent shall be tipped out on the clean part back into the tank.

II.C. Control of Non-Conveyorized Vapor Degreasers

II.C.1. Control Equipment

II.C.1.a. The non-conveyorized vapor degreaser shall have a cover which shall be designed and operated so that it can be easily opened and closed through the use of mechanical assists such as spring loading, counterweights, etc.; opening and closing the cover shall not disturb the vapor zone.

II.C.1.b. Safety Switches

The following two types of switches shall be installed on vapor degreasers:

II.C.1.b.(i) Condenser flow switch and thermostat - (shuts off sump heat if the condenser coolant is either not circulating or is too warm); and

II.C.1.b.(ii) Spray safety switch - (shuts off spray pump if the vapor level drops more than four (4) inches).

II.C.1.c. Control Device

II.C.1.c.(i) For non-conveyorized vapor degreasers with an open area (with the cover open) of one square meter (10.8 ft²) or less, either the freeboard ratio shall be greater than or equal to 0.75, or one of the control devices in II.C.1.c.(ii) shall be used.

II.C.1.c.(ii) For non-conveyorized vapor degreasers with an open area (with the cover open) greater than one (1) square meter, (10.8 ft²), at least one of the following control systems shall be used:

II.C.1.c.(ii)(A) Both a powered cover and a freeboard ratio greater than or equal to 0.75.

II.C.1.c.(ii)(B) A refrigerated chiller with a cooling capacity equivalent to or greater than the applicable specifications in Appendix C.

II.C.1.c.(ii)(C) An enclosed design: A system where the cover(s) or door(s) opens only when a dry part is entering or exiting the degreaser.

II.C.1.c.(ii)(D) A carbon adsorption system with ventilation greater than or equal to 15 cubic meters each minute per square meter (50 cfm/ft²) of air/vapor area (when the cover(s) is [are] open), exhausting less than 25 parts per million (by volume) of solvent averaged over one complete adsorption cycle.

II.C.1.d. A permanent, clearly visible sign shall be mounted on or next to the degreaser. The sign shall list the operating requirements.

II.C.2. Operating Requirements

II.C.2.a. Keep cover closed at all times except when processing work loads into or out of the degreaser.

- II.C.2.b. The following operations shall be performed to minimize solvent carry-out:
 - II.C.2.b.(i) Rack parts to allow full drainage.
 - II.C.2.b.(ii) Move parts as slowly as is practicable in and out of the degreaser. A maximum of one foot every five seconds by hand or a maximum of 5.5 cm/sec. (10.8ft/min) for a mechanically operated system.
 - II.C.2.b.(iii) Allow the workload to clean in the vapor zone at least 30 seconds or until condensation ceases.
 - II.C.2.b.(iv) Tip out any pools of solvent that remain on the cleaned parts before removal from the vapor zone.
 - II.C.2.b.(v) Allow parts to dry within the degreaser at least 15 seconds and/or until visually dry.
- II.C.2.c. Solvents shall not be used to clean porous or absorbent materials; for example, cloth, leather, wood, rope, etc.
- II.C.2.d. Workloads shall not occupy more than half of the degreaser's open top area.
- II.C.2.e. Spraying shall not be done above the vapor level.
- II.C.2.f. Solvent leaks shall be repaired immediately, or the degreaser shall be shut down.
- II.C.2.g. Exhaust ventilation shall not exceed twenty (20) cubic meters per minute per square meter (65.6 cfm per sq. ft.) of degreaser open area, unless greater exhaust rates are necessary to meet Occupational and Safety Health Act requirements. Ventilation fans shall not be used near the degreaser opening, unless necessary to meet Occupational and Safety Health Act requirements.
- II.C.2.h. The water separator shall function so that no visible water is present in the solvent exiting the separator.

II.D. Control of ConveyORIZED Degreasers

II.D.1. Control Equipment

II.D.1.a. Control Device

For all conveyORIZED degreasers with a solvent surface area greater than two (2) square meters (21.5 square feet), the degreasing shall be controlled by at least one of the following:

II.D.1.a.(i) Carbon adsorption system, with ventilation greater or equal to 15 cubic meters per minute per square meter (49.2 cfm/ft²) of air/vapor interface for vapor degreasers (of air/liquid interface for non-vapor types) when down-time covers are open, and exhausting less than 25 parts per million of solvent (by volume) averaged over a complete adsorption cycle.

II.D.1.a.(ii) For vapor degreasers only: a refrigerated chiller with a cooling capacity equivalent to or greater than the applicable specifications in Appendix D.

II.D.1.b. Prevention of Carry-out

A drying tunnel, tumbling basket(s), or other demonstrably effective method(s) shall be employed to prevent cleaned parts from carrying out solvent liquid or vapor.

II.D.1.c. Safety Switches

II.D.1.c.(i) The following two (2) switch-circuits (or equivalent) shall be installed.

II.D.1.c.(i)(A) A spray safety switch shall shut off the spray pump and/or the conveyor if the vapor level drops more than four (4) inches.

II.D.1.c.(i)(B) A vapor level control thermostat shall shut off sump heat when the vapor level rises too high.

II.D.1.c.(ii) All conveyORIZED degreasers shall have a condenser thermostat and flow-detector switch (or equivalent) which shuts off sump heat if coolant is too warm or is not circulating.

II.D.1.d. Minimized Openings: Degreaser entrance and exit openings shall silhouette workloads so that the average clearance between parts (or parts-and the edge of the degreaser opening) is either:

II.D.1.d.(i) less than 10 centimeters (4 inches) or;

II.D.1.d.(ii) less than 10 percent of the width of the opening

II.D.1.e. Covers shall be provided to close off all the entrance(s) and exit(s) when the conveyor is not in use.

II.D.1.f. A permanent, clearly visible sign shall be mounted on or next to the degreaser. The sign shall list the operating requirements.

II.D.2. Operating Requirements

II.D.2.a. Exhaust ventilation shall not exceed 20 m³/minute per square meter of degreaser opening (65.6 cf/m per square foot), unless necessary to meet OSHA requirements. Work place fans shall not be located near, nor directed at degreaser openings, unless necessary to meet OSHA requirements. Exhaust flow shall be measured by EPA reference methods 1 and 2 of 40 CFR Part 60 (March 29, 2023).

II.D.2.b. Carry-out emissions shall be minimized by:

II.D.2.b.(i) Racking parts in such a manner to achieve best drainage.

II.D.2.b.(ii) Maintaining the vertical component of conveyor speed at less than 3.3 meters per minute (10.8 feet per minute).

II.D.2.c. Repair solvent leaks immediately, or shut down the degreaser.

II.D.2.d. The water separator shall function with an efficiency sufficient to prevent water from being visible in the solvent exiting the separator.

II.D.2.e. Down-time cover(s) shall be placed over entrances and exits of conveyorized degreasers immediately after the conveyor and exhaust are shut down. Covers shall be retained in position until immediately before start-up.

II.E. Control of Industrial Cleaning Solvent Operations

II.E.1. Control Requirements

The owner or operator of an industrial cleaning solvent operation with total combined uncontrolled actual VOC emissions equal to or greater than three (3) tons per calendar year (excluding VOC emissions from solvents used for cleaning operations that are exempt under Section II.E.4.) must:

II.E.1.a. Limit the VOC content of cleaning solvents to less than or equal to 0.42 lb of VOC/gal (50 grams VOC/liter); or

II.E.1.b. Limit the composite partial vapor pressure of the cleaning solvent to 8 millimeters of mercury (mmHg) at 20 degrees Celsius (68 degrees Fahrenheit); or

II.E.1.c. Reduce VOC emissions with an emission control system having a control efficiency of 90% or greater.

II.E.2. Work Practice Requirements

The owner or operator of an industrial cleaning solvent operation must implement the following work practice requirements at all times to reduce VOC emissions from fugitive sources:

II.E.2.a. Cover open containers and used applicators in a manner that minimizes evaporation into the atmosphere;

II.E.2.b. Properly dispose of used solvent and shop towels; and

II.E.2.c. Implement good air pollution control practices that minimize emissions, including, but not limited to, using only volumes necessary for cleaning and maintaining cleaning equipment to be leak free.

II.E.3. Monitoring, Recordkeeping and Reporting Requirements

II.E.3.a. The owner or operator of an industrial cleaning solvent operation must keep the following records for two (2) years and make them available for inspection by the Division upon request:

II.E.3.a.(i) If applicable, records demonstrating that a listed exemption to this Section II.E. applies.

II.E.3.a.(ii) If applicable, monthly records such as safety data sheets or other analytical data from the industrial cleaning solvent manufacturer showing the VOC type and VOC content, or the composite partial vapor pressure at 20 degrees Celsius, and total amount of VOC-containing solvent used in solvent cleaning operations to demonstrate compliance with the control requirements in Sections II.E.1.a. and II.E.1.b.

II.E.3.a.(iii) If applicable, monthly records sufficient to demonstrate compliance with the control requirement in Section II.E.1.c.

II.E.3.a.(iv) Records of calendar year VOC emission estimates demonstrating whether the industrial cleaning solvent operation meets or exceeds the applicability threshold in Section II.E.1.

II.E.3.b. Compliance with the control requirements in Section II.E.1. must be demonstrated using one of the following methods as applicable:

II.E.3.b.(i) Safety data sheets or other analytical data from the industrial cleaning solvent manufacturer to demonstrate compliance with Sections II.E.1.a. and II.E.1.b.;

II.E.3.b.(ii) A manufacturer guarantee of the control equipment's emission control efficiency and operation and maintenance of control equipment according to manufacturer's specifications to demonstrate compliance with Section II.E.1.c.; or

II.E.3.b.(iii) A performance test conducted during representative operations using one of the following methods, as applicable:

II.E.3.b.(iii)(A) EPA Method 24 (40 CFR Part 60, Appendix A) (October 7, 2020) to determine VOC content;

II.E.3.b.(iii)(B) EPA Method 18, 25, or 25A (40 CFR Part 60, Appendix A) (March 29, 2023) to determine control efficiency of the emission control equipment.

II.E.4. Exemptions

II.E.4.a. Industrial cleaning solvent operations are not subject to Section II.E. if they are subject to a work practice or emission control requirement in another federally enforceable section of Regulation Number 7, Number 24, Number 25, and Number 26 that establishes RACT.

II.E.4.b. The VOC control requirements in Section II.E.1. do not apply to:

II.E.4.b.(i) Cleaning of electrical and electronic components;

II.E.4.b.(ii) Cleaning of precision optics;

- II.E.4.b.(iii) Cleaning of numismatic dies;
- II.E.4.b.(iv) Stripping of cured inks, coatings, and adhesives;
- II.E.4.b.(v) Cleaning of resin, coating, ink, and adhesive manufacturing, mixing, molding, and application equipment;
- II.E.4.b.(vi) Cleaning of research and development laboratories;
- II.E.4.b.(vii) Cleaning of medical device or pharmaceutical manufacturing equipment;
- II.E.4.b.(viii) Performance testing to determine coating, adhesive, ink or ink performance;
- II.E.4.b.(ix) Cleaning of equipment and materials used in testing for quality control or quality assurance purposes;
- II.E.4.b.(x) Cleaning of digital printing operations; and
- II.E.4.b.(xi) Cleaning of screen printing operations.

II.E.4.c. In lieu of compliance with Section II.E.1. and II.E.2., the owner or operator of an area source aerospace facility, as defined in 40 CFR Part 63, Section 63.742 (November 17, 2016), may implement the solvent cleaning provisions of the National Emission Standards for Hazardous Air Pollutants for Aerospace Manufacturing and Rework facilities contained in 40 CFR Part 63, Section 63.744 (November 17, 2016) along with the applicable definitions contained in 40 CFR Part 63, Section 63.742 (November 17, 2016), except that:

- II.E.4.c.(i) VOC-containing solvents which meet the definition of “non-HAP materials” in 40 CFR Part 63, Section 63.742 (November 17, 2016) are not excluded from the housekeeping measures contained in 40 CFR Part 63, Section 63.744(a) (November 17, 2016); and
- II.E.4.c.(ii) The baseline reduction compliance option contained in 40 CFR Part 63, Section 63.744(b)(3) (November 17, 2016) is not available for purposes of compliance with this VOC control rule.

II.F. General Solvent Use

II.F.1. Applicability

II.F.1.a. Within the 8-Hour Ozone Control Area: As of May 1, 2021, the requirements of Section II.F. apply to operations that use solvents with uncontrolled actual VOC emissions greater than or equal to two (2) tons per year that existed at major sources of VOC (greater than or equal to 50 tpy VOC) as of January 27, 2020.

II.F.1.b. (State Only) Outside the 8-Hour Ozone Control Area: As of May 1, 2021, the requirements of Section II.F. apply to operations that use solvents with uncontrolled actual VOC emissions greater than or equal to five (5) tons per year that existed at sources of VOC greater than or equal to 50 tpy VOC as of January 27, 2020.

II.F.1.c. Within the 8-Hour Ozone Control Area: As of May 1, 2023, the requirements of Section II.F. apply to operations that use solvents with uncontrolled actual VOC emissions greater than or equal to two (2) tons per year that existed at major sources of VOC (greater than or equal to 25 tpy VOC) as of November 7, 2022.

II.F.2. Exemptions

The requirements of this Section II.F. do not apply to:

II.F.2.a. Operations that are subject to a solvent work practice or emission control requirement in another federally enforceable section of Regulation Number 7, Number 24, Number 25, and Number 26 that constitutes RACT, or;

II.F.2.b. Solvent use where the solvent does not contain VOCs.

II.F.3. Work practice requirements

The owner or operator of operations that use solvents must implement the following work practice requirements at all times to reduce VOC emissions from fugitive sources:

II.F.3.a. Cover open containers and used applicators in a manner that minimizes evaporation into the atmosphere;

II.F.3.b. Properly dispose of used solvent and solvent contaminated waste (e.g. shop towels and carbon filtration or other control device media), and;

II.F.3.c. Implement good air pollution control practices that minimize emissions, including but not limited to:

II.F.3.c.(i) Using low or no-VOC solvents, if possible;

- II.F.3.c.(ii) Using only volumes of solvent necessary for operations;
- II.F.3.c.(iii) Using submerged fill pipes in storage tanks and containers;
- II.F.3.c.(iv) Using closed loop systems to minimize solvent loss during transfer and use of solvents;
- II.F.3.c.(v) Maintaining solvent storage, transfer, and use operations equipment in such a way that it minimizes evaporation loss and remains leak free, and;
- II.F.3.c.(vi) Owners or operators of sources that use a carbon adsorption system must provide for the proper disposal or reuse of all VOC recovered.

II.F.4. Control of general solvent use

The owner or operator of operations that use solvents with uncontrolled actual VOC emissions greater than or equal to twenty-five (25) tons per year on a calendar year basis, and that are located in the 8-Hour Ozone Control Area, must reduce solvent use VOC emissions by 90%.

II.F.5. Monitoring requirements

- II.F.5.a. The owner or operator of operations that use solvents that utilize a closed-loop system for emission control must inspect the control system using audio, visual, olfactory (AVO) on a monthly basis for perceptible emissions. First attempt to repair must be made upon detection if feasible, but no later than three (3) calendar days from detection.
- II.F.5.b The owner or operator of operations that use solvents that utilize a control device must operate and maintain the control device consistent with the manufacturer's specifications.
- II.F.5.c. The owner or operator of operations that use solvents that are subject to the 90% control requirement in Section II.F.4. must:
 - II.F.5.c.(i) Complete a performance test once every three (3) years during representative operations to verify compliance with Section II.F.4. using one of the following methods, as applicable:

II.F.5.c.(i)(A) EPA Method 24 (40 CFR Part 60, Appendix A) (October 7, 2020) to determine VOC content.

II.F.5.c.(i)(B) EPA Method 18, 25, or 25A (40 CFR Part 60, Appendix A) (March 29, 2023) to determine control efficiency of the emission control equipment

II.F.5.c.(ii) Conduct all performance tests in accordance with EPA test methods and a test protocol submitted to the Division for review at least thirty (30) days prior to testing and in accordance with AQCC Common Provisions Regulation Section II.C.

II.F.5.c.(iii) Comply with control device and monitoring system manufacturers' specifications for operation and maintenance for equipment used to demonstrate compliance with Section II.F.4.

II.F.6. Recordkeeping

II.F.6.a. Records of calendar year VOC emission estimates demonstrating whether the solvent operation meets or exceeds the applicability threshold in Section II.F.1.

II.F.6.b. If applicable, records demonstrating that an exemption to Section II.F.2. applies.

II.F.6.c. Monthly solvent losses based on beginning and ending inventories, solvent received, inventory adjustments, solvent destroyed in a control device, solvent recovered, and any volume of solvent normally retained in recovery equipment. Solvent losses must be totaled on a rolling 12-month basis.

II.F.6.d. Monthly records such as safety data sheets or other analytical data from the solvent manufacturer showing the VOC type and VOC content, or the composite partial vapor pressure at 20 degrees Celsius, and total amount of VOC-containing solvent used in solvent operations.

II.F.6.e. Records of negative pressure ranges, and other records necessary to demonstrate compliance with Section II.F.3.

II.F.6.f. Manufacturer guarantee of the control equipment's emission control efficiency to demonstrate compliance with Section II.F.4.

- II.F.6.g. If applicable, monthly records of operation and maintenance of control device and monitoring system according to manufacturer's specifications to demonstrate compliance with Sections II.F.4. and II.F.5.
- II.F.6.h. If applicable, Records of performance tests conducted to demonstrate compliance with Section II.F.5.
- II.F.6.i. If applicable, monthly records of AVO inspections including:
 - II.F.6.i.(i) The date, facility name, and facility AIRS ID or facility location if the facility does not have an AIRS ID for each inspection;
 - II.F.6.i.(ii) A list of the leaks requiring repair,
 - II.F.6.i.(iii) The date of first attempt to repair the leak and, if necessary, any additional attempt to repair;
 - II.F.6.i.(iv) The date the leak was repaired and type of repair method applied.
- II.F.6.j. Records must be maintained for two (2) years and made available for inspection by the Division upon request.

III. Use of Cutback Asphalt

III.A. Definitions

- III.A.1. "Asphalt or Asphalt Cement" The dark-brown to black cementitious material (solid, semi-solid, or liquid in consistency) of which the main constituents are bitumen's which occur naturally or as a residue of petroleum refining.
- III.A.2. "Asphalt Concrete" A waterproof and durable paving material composed of dried aggregate, which is evenly coated with hot asphalt cement.
- III.A.3. "Cutback Asphalt or Cutback Asphalt Cement" Any asphalt which has been liquefied by blending with a VOC, such as a petroleum solvent diluents or, in the case of some slow cure asphalts (Road Oils), which has been produced directly from the distillation of petroleum.
- III.A.4. "Emulsified Asphalt" Asphalt emulsions produced by combining asphalt and water with emulsifying agent. Emulsified Asphalt or any other coating or sealant, including but not limited to those produced from petroleum or coal, which contain more than five (5) percent of oil distillate as determined by ASTM Method D-244 is included in this definition.

III.A.5. “Penetrating Prime Coat” An application of low-viscosity liquid asphalt to an absorbent surface in order to prepare it for overlaying with a layer or layers of asphalt cement or asphalt emulsion and mineral aggregate paving materials.

III.B. Limitations

III.B.1. Applicability

The provisions of this Section III. apply to the use and storage of cutback asphalt for the paving and maintenance of all public roadways (including alleys), private roadways, parking lots, and driveways only within ozone nonattainment areas.

III.B.2. Storage

Stockpiles of aggregate mixed with cutback asphalt are permitted October 1 through February 28 (29). Such storage is not permitted March 1 through September 30 except where it can be demonstrated to the Division that such storage is necessary.

III.B.3. Use

Cutback asphalt may be used for any paving purpose October 1 through February 28 (29). No person shall use cutback asphalt or any coating included in the definition of cutback asphalt in Section III.A.3. March 1 through September 30 except as provided:

III.B.3.a. If used solely as a penetrating prime coat, or

III.B.3.b. If the user can demonstrate to the Division that under the conditions of its intended use, there will be no emissions of volatile organic compounds to the ambient air.

III.C. Recordkeeping

During the months of March through September, the person responsible for the use or storage of any cutback asphalt as permitted in Sections III.B.3.a., III.B.3.b., and Section III.B.2. shall keep records of same, including type and amount of solvent(s) used.

IV. Graphic Arts and Printing

IV.A. Packaging Rotogravure, Publication Rotogravure, and Flexographic Printing

IV.A.1. Definitions

For the purpose of this section, the following definitions apply:

IV.A.1.a. "Flexographic Printing" means the application of words, designs, and pictures to a substrate by means of a roll printing technique in which the pattern to be applied is raised above the printing roll and the image carrier is made of rubber or other elastomeric materials.

IV.A.1.b. "Packaging Rotogravure Printing" means rotogravure printing upon paper, paperboard, metal foil, plastic film, and other substrates, which are, in subsequent operations, formed into packaging products and labels for articles to be sold.

IV.A.1.c. "Publication Rotogravure Printing" means rotogravure printing upon paper, which is subsequently formed into books, magazines, catalogues, brochures, directories, newspaper supplements, and other types of printed materials.

IV.A.1.d. "Roll Printing" means the application of words, designs, and pictures to a substrate usually by means of a series of hard rubber or steel rolls each with only partial coverage.

IV.A.1.e. "Rotogravure Printing" means the application of words, designs, and pictures to a substrate by means of a roll printing technique, which involves an intaglio or recessed image areas in the form of cells.

IV.A.2. Applicability

IV.A.2.a. This section applies to all packaging rotogravure, publication rotogravure, and flexographic printing facilities whose potential emissions of volatile organic compounds before control (determined at design capacity and 8760 hrs/year, or at maximum production, and accounting for any capacity or production limitations in a federally-enforceable permit) are equal to or more than 90,000 Kg per year (100 tons/year). Potential emissions are to be estimated by extrapolating historical records of actual consumption of solvent and ink. (e.g., the historical use of 20 gallons of ink for 4,000 annual hours would be extrapolated to 43.8 gallons for 8760 hours.) The before-control volatile organic compound emissions calculations shall be the summation of all volatile organic compounds in the inks and solvents (including cleaning liquids) used.

IV.A.3. Provisions for Specific Processes

- IV.A.3.a. No owner or operator of a facility subject to this section and employing VOC-containing ink shall operate, cause, allow, or permit the operation of the facility unless:
- IV.A.3.a.(i) The volatile fraction of ink, as it is applied to the substrate, contains 25.0 percent or less (by volume) of VOC and 75.0 percent or more (by volume) of water; or
 - IV.A.3.a.(ii) The ink (minus water) as it is applied to the substrate, contains 60.0 percent or more (by volume) non-volatile material; or
 - IV.A.3.a.(iii) The owner or operator installs and operates a control device and capture system in accordance with Sections IV.A.3.b. and IV.A.3.c.; or
 - IV.A.3.a.(iv) A combination of solvent-borne inks and low solvent inks that achieve a 70% (volume) overall reduction of solvent usage (compared to an all solvent borne ink usage) is used; or
 - IV.A.3.a.(v) Flexographic and packaging rotogravure printing facilities limit emissions to 0.5 pounds of VOC per pound of solids in the ink. The limit includes all solvent added to the ink: solvent in the purchased ink, solvent added to cut the ink to achieve desired press viscosity, and solvent added to ink on the press to maintain viscosity during the press run. (Publication rotogravure facilities shall not use this option); or
 - IV.A.3.a.(vi) Crossline averaging is used. The requirements of Section I.A.5.d. apply.
- IV.A.3.b. A capture system shall be used in conjunction with the emission control system in Section IV.A.3.a. The design and operation of a capture system shall be consistent with good engineering practice, and in conjunction with control equipment shall be required to provide for an overall reduction in volatile organic compound emissions of at least:
- IV.A.3.b.(i) 75.0 percent where a publication rotogravure process is employed;
 - IV.A.3.b.(ii) 65.0 percent where a packaging rotogravure process is employed; or

IV.A.3.b.(iii) 60.0 percent where a flexographic printing process is employed.

IV.A.3.c. The design, operation, and efficiency of any capture system used in conjunction with any emission control system shall be certified in writing by the source owner or operator and approved by the Division. Testing of any capture system may be required by the Division on a case-by-case basis, in cases where a total enclosure is not used or when material balance results are questionable. Testing of capture system efficiency shall meet the requirements of Section I.A.5.e.

IV.A.3.d. The overall reduction in VOC emissions specified in Section IV.A.3.b. shall be calculated by material balance methods approved by the Division, or by determination of capture and control device efficiencies. The overall VOC emission reduction rate equals the (percent capture efficiency X percent control device efficiency)/100.

IV.A.4. Testing and Monitoring

The owner or operator of a source subject to the requirements of this section is also subject to the requirements of Part B, Sections I.A.3., I.A.7, I.A.9., and I.A.10. In Part B, Section I.A.3., EPA reference method 24A (October 7, 2020) shall be the test method used for publication rotogravure inks, while EPA Reference method 24 (October 7, 2020) data is acceptable for all other inks. Test methods as set forth in Appendix A, Part 60, Chapter I, Title 40, of the Code of Federal Regulations (CFR), in effect July 1, 2022.

IV.A.5. The owner or operator of a source subject to the requirements of this section is also subject to the requirements of Section I.A.8. "A Guideline for Graphic Arts Calculations" shall be used for compliance determination.

IV.B. Lithographic and Letterpress Printing

IV.B.1. General Provisions

IV.B.1.a. Definitions

IV.B.1.a.(i) "Alcohol" means any of the hydroxyl-containing organic compounds with a molecular weight equal to or less than 74.12, which includes methanol, ethanol, propanol, and butanol.

IV.B.1.a.(ii) "Alcohol substitute" means nonalcohol additives that contain VOCs and are used in the fountain solution to reduce the surface tension of water or prevent ink piling.

IV.B.1.a.(iii) "Cleaning material" means a VOC-containing material used to remove ink and debris from the printing press area, operating surfaces of the printing press and, printing press parts. Blanket wash is a type of cleaning material.

IV.B.1.a.(iv) "Composite partial vapor pressure" means the sum of the partial pressures of the compounds defined as VOCs. Composite partial vapor pressure is calculated as follows:

$$PP_c = \sum_{i=1}^n \frac{(W_i)(VP_i)/MW_i}{\frac{W_w}{MW_w} + \sum_{c=1}^n \frac{W_c}{MW_c} + \sum_{i=1}^n \frac{W_i}{MW_i}}$$

Where:

W_i = Weight of the "i"th VOC compound, in grams

W_w = Weight of water, in grams

W_e = Weight of exempt compound, in grams

MW_i = Molecular weight of the "i"th VOC compound, in g/g-mole

MW_w = Molecular weight of water, in g/g-mole

MW_c = Molecular weight of exempt compound, in g/g-mole

PP_c = VOC composite partial vapor pressure at 20°C (68°F), in mm Hg

VP_i = Vapor pressure of the "i"th VOC compound at 20°C(68°F), in mm Hg

IV.B.1.a.(v) "Fountain solution" means a mixture of water, nonvolatile printing chemicals, and a liquid additive that reduces the surface tension of the water so that it spreads easily across the printing plate surface. The fountain solution wets the non-image areas so that the ink is maintained within the image areas.

IV.B.1.a.(vi) "Heatset" means any lithographic or letterpress printing operation where printing inks are set by the evaporation of the ink oils in a heatset dryer.

- IV.B.1.a.(vii) "Heatset dryer" means a hot air dryer used in heatset lithography to heat the printed substrate and to promote the evaporation of ink oils.
- IV.B.1.a.(viii) "Lithographic printing" means a planographic printing process where the image and non-image areas are chemically differentiated (the image area is oil receptive and the non-image area is water receptive). This printing process differs from other conventional printing methods, where the image is a raised or recessed surface.
- IV.B.1.a.(ix) "Letterpress printing" means a printing process in which the image area is raised relative to the non-image area and the paste ink is transferred to the substrate directly from the image surface.
- IV.B.1.a.(x) "Non-heatset" means any printing operation where the printing inks are set without the use of heat. For the purpose of Section IV.B., ultraviolet-cured and electron beam-cured inks are considered non-heatset.
- IV.B.1.a.(xi) "Offset lithographic printing" means a printing process that transfers the ink film from the lithographic plate to an intermediary surface (blanket), which in turn transfers the ink film to the substrate.
- IV.B.1.a.(xii) "Press" means a printing production assembly composed of one or more print units used to produce a printed substrate including any associated coating, spray powder application, heatset web dryer, ultraviolet or electron beam curing units, or infrared heating units.
- IV.B.1.a.(xiii) "Sheet-fed printing" means a printing process where individual sheets of paper or substrate are fed into the printing press.
- IV.B.1.a.(xiv) "Web printing" means a printing process where continuous rolls of substrate material are fed to the press and rewound or cut to size after printing.
- IV.B.1.b. Applicability

IV.B.1.b.(i) The provisions of this Section IV.B. apply to fountain solutions, cleaning materials, inks (which include varnishes) and coatings used in lithographic and letterpress printing presses. These materials are not subject to the requirements of Sections I. and II.

IV.B.1.b.(ii) The work practice requirements in Section IV.B.1.c. apply to all lithographic and letterpress printing operations.

IV.B.1.b.(iii) The VOC content limit for inks in Section IV.B.1.d. applies to lithographic and letterpress printing operations where total combined uncontrolled actual VOC emissions from each printing operation, including related cleaning materials and fountain solutions, are equal to or greater than three (3) tons per calendar year.

IV.B.1.b.(iv) The cleaning material requirements in Section IV.B.2. apply to letterpress printing operations where total combined uncontrolled actual VOC emissions from each printing operation, including related cleaning materials and fountain solutions, are equal to or greater than three (3) tons per calendar year.

IV.B.1.b.(v) The cleaning material and fountain solution requirements in Sections IV.B.2. and IV.B.3. apply to offset lithographic printing operations where total combined uncontrolled actual VOC emissions from each printing operation, including related cleaning materials and fountain solutions, are equal to or greater than three (3) tons per calendar year.

IV.B.1.b.(vi) The control requirements in Section IV.B.4. apply to each heatset web offset lithographic and heatset web letterpress printing press with the potential to emit from the dryer, prior to controls, at least 25 tons per calendar year of VOC (petroleum ink oil) from heatset inks.

IV.B.1.c. Work Practice Requirements

Lithographic and letterpress printing operations must implement the following work practices at all times to reduce VOC emissions from fugitive sources:

IV.B.1.c.(i) Cover open containers and keep cleaning materials in closed containers when not in use;

IV.B.1.c.(ii) Properly dispose of used cleaning materials, fountain solutions, and used shop towels; and

IV.B.1.c.(iii) Implement good air pollution control practices that minimize emissions, including, but not limited to, using only volumes necessary for cleaning and maintain cleaning equipment to repair cleaning materials leaks.

IV.B.1.d. VOC Content Limit for Inks

IV.B.1.d.(i) Lithographic and letterpress printing operations, excluding heatset web offset and heatset web letterpress printing operations, must use low-VOC inks, which average less than 30% (by weight) VOC on a monthly basis.

IV.B.1.d.(ii) Heatset web offset lithographic and heatset web letterpress printing operations must use low-VOC inks, which average less than 40% (by weight) VOC on a monthly basis.

IV.B.2. Offset lithographic printing and letterpress printing operations must comply with the following cleaning materials requirements;

IV.B.2.a. All cleaning materials must contain less than 70% (by weight) VOC or have a VOC composite vapor pressure less than 10 mmHg at 20°C.

IV.B.2.b. Exemptions

The following materials and operations are exempt from the cleaning material requirements in Section IV.B.2.a.:

IV.B.2.b.(i) Cleaners used on electronic components of a press.

IV.B.2.b.(ii) Pre-press cleaning operations.

IV.B.2.b.(iii) Post-press cleaning operations.

IV.B.2.b.(iv) Floor cleaning supplies (other than those used to clean dried ink).

IV.B.2.b.(v) Cleaning performed in parts washers or cold cleaners that are subject to Section II.

IV.B.2.c. Use of non-compliant cleaning materials

Cleaning materials not meeting the limits in Section IV.B.2.a. are limited to less than or equal to 110 gallons per calendar year.

IV.B.3. Offset lithographic printing operations must comply with the following fountain solution requirements:

IV.B.3.a. Heatset web offset lithographic printing operations must:

IV.B.3.a.(i) Use a fountain solution containing 1.6% alcohol (by weight) or less as applied;

IV.B.3.a.(ii) Use a fountain solution containing 3% alcohol (by weight) or less as applied if the fountain solution is refrigerated to below 60°F (15.5°C); or

IV.B.3.a.(iii) Use a fountain solution containing 5% alcohol substitute (by weight) or less as applied and no alcohol.

IV.B.3.b. Sheet-fed printing operations must

IV.B.3.b.(i) Use a fountain solution containing 5% alcohol (by weight) or less as applied;

IV.B.3.b.(ii) Use a fountain solution containing 8.5% alcohol (by weight) or less as applied if the fountain solution is refrigerated to below 60°F (15.5°C); or

IV.B.3.b.(iii) Use a fountain solution containing 5% alcohol substitute (by weight) or less as applied and no alcohol.

IV.B.3.b.(iv) The following are exempt from the fountain solution requirements in Section IV.B.3.b.:

IV.B.3.b.(iv)(A) Fountain solution use associated with a sheet-fed printing press with maximum sheet size 11x17 inches or smaller.

IV.B.3.b.(iv)(B) Fountain solution use associated with a sheet-fed printing press having a total fountain solution reservoir less than one (1) gallon.

IV.B.3.c. Non-heatset web printing must use a fountain solution containing 5% alcohol substitute (by weight) or less and no alcohol.

IV.B.4. Heatset web offset lithographic and heatset web letterpress printing operations must comply with the following control requirements:

- IV.B.4.a. Heatset web offset lithographic and heatset web letterpress printing operations must reduce VOC emissions from heatset dryers with an emission control system having a control efficiency of 90% or greater.
- IV.B.4.b. If the control device was first installed on or after January 1, 2017, heatset web offset lithographic and heatset web letterpress printing operations must reduce VOC emissions from heatset dryers with an emission control system having a control efficiency of 95% or greater.
- IV.B.4.c. Where inlet VOC concentration is low and a 90 or 95% control efficiency is not achievable due to low inlet concentrations or measurable due to equipment configuration, heatset web offset lithographic and heatset web letterpress printing operations may reduce the control device outlet concentration to 20 ppmv (as hexane on a dry basis).
- IV.B.4.d. The following are exempt from the control requirements in Section IV.B.4.:
 - IV.B.4.d.(i) Heatset presses used for book printing.
 - IV.B.4.d.(ii) Heatset presses with maximum web width of 22 inches or less.
 - IV.B.4.d.(iii) Waterborne or radiation (ultra-violet or electron beam) cured materials that are not heatset.
- IV.B.5. Monitoring, Recordkeeping and Reporting
 - IV.B.5.a. The owner or operator of a heatset web offset lithographic or heatset web letterpress printing operation required to demonstrate compliance with Section IV.B.4. must install, calibrate, maintain, and operate a temperature monitoring device, according to the manufacturer's specifications.
 - IV.B.5.b. The owner or operator of a lithographic and letterpress printing operations subject to Sections IV.B.1.d. and IV.B.2. through IV.B.4. must keep the following records for two (2) years and make them available for inspection by the Division upon request:
 - IV.B.5.b.(i) If applicable, records demonstrating that a listed exemption to this Section IV.B. applies.

- IV.B.5.b.(ii) If applicable, monthly records of the type, alcohol content or alcohol substitute content, and total volume of fountain solution used in printing operations.
- IV.B.5.b.(iii) If applicable, monthly records of the type, VOC content or composite vapor pressure, and total volume of the cleaning materials used in printing operations.
- IV.B.5.b.(iv) If applicable, monthly records of the type, VOC content, and total volume of inks (including varnishes) and coatings used in printing operations.
- IV.B.5.b.(v) If applicable, monthly records demonstrating compliance with the control requirements in Section IV.B.4.
- IV.B.5.b.(vi) Records of calendar year VOC emission estimates demonstrating whether the printing operation meets or exceed the applicability thresholds in Section IV.B.1.b.
- IV.B.5.c. Compliance with control requirements must be demonstrated using the following methods as applicable:
 - IV.B.5.c.(i) Safety data sheets or other analytical data from the ink, cleaning material, or fountain solution manufacturer to demonstrate compliance with VOC content limit for inks in Section IV.B.1.d., the cleaning material requirements in Section IV.B.2., and the fountain solution requirements in Section IV.B.3.;
 - IV.B.5.c.(ii) A manufacturer guarantee of the control equipment's emission control efficiency and operation and maintenance of control equipment according to manufacturer's specifications to demonstrate compliance with the control equipment requirements in Section IV.B.4.; or
 - IV.B.5.c.(iii) A performance test conducted during representative conditions using one of the following methods as applicable:
 - IV.B.5.c.(iii)(A) EPA Method 24 (40 CFR Part 60, Appendix A) (October 7, 2020) to determine VOC content for inks, fountain solutions and cleaning materials; or

IV.B.5.c.(iii)(B) EPA Method 18, 25, or 25A (40 CFR Part 60, Appendix A) (March 29, 2023) to determine control efficiency or outlet concentration of the emission control equipment.

IV.C. Repealed

V. Pharmaceutical Synthesis

V.A. General Provisions

V.A.1. Applicability

This section applies to all sources of volatile organic compounds associated with pharmaceutical manufacturing activities, including, but not limited to, reactors, distillation units, dryers, storage of VOCs, extraction equipment, filters, crystallizers, and centrifuges.

V.A.2. Exemptions

Extraction of organic substances from animal or vegetable material; fermentation and culturing; formulation and packaging of pharmaceutical or medicinal products.

V.A.3. Definitions

For the purpose of this section, the following definitions apply:

V.A.3.a. "Control System" means any number of control devices, including condensers, which are designed and operated to reduce the quantity of VOC emitted to the atmosphere.

V.A.3.b. "Pharmaceutical" means a medicine or drug which appears in the United States Pharmacopoeia National Formulary, or which is so designated by the National Drug Code of the United States FDA Bureau of Drugs.

V.A.3.c. "Production Equipment Exhaust System" means a device for collecting and directing out of the work area VOC fugitive emissions from reactor openings, centrifuge openings, and other vessel openings for the purpose of protecting workers from excessive VOC exposure.

V.A.3.d. "Reactor" means a vat or vessel, which may be jacketed to permit temperature control, designed to contain chemical reactions.

V.A.3.e. "Separation Operation" means a process that separates a mixture of compounds and solvents into two or more components. Specific mechanisms include, but are not limited to, extraction, centrifugation, filtration, distillation, and crystallization.

V.A.3.f. "Synthesized Pharmaceutical Manufacturing" means manufacture of pharmaceutical products by chemical synthesis. It includes the manufacture of chemical intermediates (of sufficient purity) which are typically used by the pharmaceutical industry as precursors to finished mixtures of chemicals. (Thus, it excludes those chemical processes which are not directed at creating finished pharmaceutical or chemical intermediates to finished pharmaceuticals.)

V.B. Provisions for Specific Processes

V.B.1. The owner or operator of a facility subject to this section shall control the volatile organic compound emissions from each vent which has the potential to emit 6.80 kg/day (15 lb./day) or more of VOC from reactors, distillation operations, crystallizers, centrifuge and vacuum dryers. Surface condensers or equivalent controls shall be used, provided that, if surface condensers are used, the condenser outlet gas temperature shall not exceed the following values:

VOCs True Vapor Pressure* at 20° in torr (and psia) from (minimum) up to ** (maximum)	Maximum temperature of Gas Stream immediately exiting the condenser
0-26(0-0.5)	35°C (95°F)
26-52(0.5-1.0)	25°C(77°F)
52-78(1.0-1.5)	10°C(50°F)
78-150(1.5-2.9)	0°C(32°F)
150-300(2.9-5.8)	-15°C(5°F)
Greater than 300(Greater than 5.8)	-25°C(-13°F)

*The calculation methods for gases containing more than one condensable component are complex. As a simplification, the temperature necessary for control by condensation can be roughly approximated by the weighted average of the temperatures necessary for condensation of each VOC considered separately but at concentrations equal to the total organic concentration.

**But not including the maximum value of the range.

V.B.2. Division approval shall be required for control equipment used to control VOCs of 570 torr (11 psia) and above.

V.B.3. The owner or operator of a facility subject to this section shall reduce the VOC emissions from each air dryer and production equipment exhaust system:

V.B.3.a. By at least 90 percent if emissions are 150 kg/day (330 lbs/day) or more of VOC, or,

V.B.3.b. To 15.0 kg/day (33 lb/day) or less if emissions are less than 150 kg/day (330 lb/day) of VOC.

V.B.4. The owner or operator of a facility subject to this section shall:

V.B.4.a. Provide a vapor balance system or equivalent control that is at least 90.0 percent effective in reducing emissions from truck or railcar deliveries to storage tanks with capacities greater than 7,570 liters (2,000 gallons) that store VOC with true vapor pressure greater than 210 torr (4.1 psia) at 20°C; and,

V.B.4.b. Install pressure/vacuum conservation vents set at plus or minus 0.2 kPa on all storage tanks that store VOC with true vapor pressures greater than 10.0 kPa (1.5 psi) at 20°C.

V.B.5. The owner or operator of a facility subject to this section shall enclose all centrifuges, rotary vacuum filters, and other filters having an exposed liquid surface, where the liquid contains VOC and exerts a total VOC true vapor pressure of 26 torr (0.5 psia) or more at 20°C.

V.B.6. The owner or operator of a synthesized pharmaceutical facility subject to this section shall install covers on all in-process tanks containing a volatile organic compound at any time. These covers shall remain closed unless sampling, maintenance, short-duration production procedures or inspection procedures require access.

V.B.7. The owner or operator of a facility subject to this section shall repair all leaks from which a liquid, containing VOC, can be observed running or dripping. The repair shall be completed the first time the equipment is off-line for a period of time long enough to complete the repair, except that no leak shall go unrepaired for more than 14 days after initial detection unless the Division issues written approval.

V.B.8. Each surface condenser shall have at least one temperature indicator with its sensor located in the outlet gas stream.

V.C. Testing and Monitoring

V.C.1. Sources subject to the requirements of this section are also subject to the requirements of Sections I.A.3., I.A.7., I.A.8., and I.A.9.

Appendix D Minimum Cooling Capacities for Refrigerated Freeboard Chillers on Vapor Degreasers

The specifications in this Appendix apply only to vapor degreasers that have both condenser coils and refrigerated freeboard chillers. (The coolant in the condenser coils is normally water.) The amount of refrigeration capacity is expressed in Calories/Hour per meter of perimeter. This perimeter is measured at the air/vapor interface. For refrigerated chillers operated below 0°C., the following requirements apply:

DEGREASER WIDTH	*CALORIES/HR METER OF	BTU/HR FOOT OF
Less than 1.1 meters (3.5	165	200
1.1 - 1.8 meters (3.5 - 6.0	250	300
1.8 - 2.4 meters (6.0 - 8.0	335	400
2.4 - 3.0 meters (8.0 - 10.0	145	500
Greater than 3.0 meters (10	500	600

* Kilocalories (1 Kilocalorie = 4184.0 joules)

For refrigerated chillers operating above 0°C., there shall be at least 415 Calories/Hr. - meter of perimeter (500 BTU/Hr-ft.), regardless of size.

Definition: "Air/Vapor Interface" - means the surface defined by the top of the solvent vapor layer within the confines of a vapor degreaser.

Appendix E Emission Limit Conversion Procedure

The following procedure shall be used to convert emission limits expressed as lb VOC/gallon coating less water and exempt solvents to limits expressed as lb VOC/gallon solids. This example uses the emission limit of 3.7 lb VOC/gallon coating.

Assume VOC density of the 'Presumptive' RACT coating is 7.36 pounds per gallon because this same value was used to determine the "Presumptive" recommended RACT emission limits from volume solids data.

$(3.7) \text{ LB VOC / GAL COATING LESS WATER} \times 1 \text{ GAL VOC} \times 100 / 7.36 \text{ LB VOC} = (50) \text{ VOL\% VOC}$
$100 - (50) \text{ VOL\% VOC} = (50) \text{ VOL\% SOLIDS}$
$(3.7) \text{ LB VOC / GAL COATING LESS H}_2\text{O} \times 100 \text{ GAL COATING} / (50) \text{ GAL SOLIDS} = (7.4) \text{ LB VOC / GAL SOLIDS}$

See "A Guideline for Surface Coating Calculations" EPA - 340/1-86-016 for additional examples.

The following table lists equivalent mass VOC/volume solids emission limits for various coating operations.

Equivalency Data for Surface Coating Processes (VOC Density = 7.36 lb/gal)			
Industrial Finishing Categories	Lb VOC per Gallon Coating less water	Lb VOC per Gallon of Solids	Kg VOC per Liter of Solids
<i>Can Industry</i>			
Sheet Basecoat (Exterior and Interior) and over-varnish; two-piece can exterior (base-coat and over-varnish)	2.8	4.5	0.55
Two- and three-piece can interior body spray, two-piece can exterior end spray or roll coat	4.2	9.8	1.19
Three-piece can side-seam spray	5.5	21.7	2.61
End sealing compound	3.7	7.4	0.88
Any additional coats	4.2	9.8	1.19
<i>Coil Coating</i>			
Any coat	2.6	4.0	0.48
<i>Fabric Coating</i>			
Fabric coating line	2.9	4.8	0.58
Vinyl coating line	3.8	7.9	0.93
<i>Paper Coating</i>			
Coating line	2.9	4.8	0.58
<i>Automotive and Light-Duty Truck Assembly Plant</i>			
Primer (electrodeposition) application, flashoff area and oven	1.9	2.6	0.31
Topcoat application, flashoff area and oven	2.8	4.5	0.55

Equivalency Data for Surface Coating Processes (VOC Density = 7.36 lb/gal)			
Industrial Finishing Categories	Lb VOC per Gallon Coating less water	Lb VOC per Gallon of Solids	Kg VOC per Liter of Solids
Final repair application, flashoff area and oven	4.8	13.8	1.67
<i>Metal Furniture</i>			
Coating line	3.0	5.1	0.61
<i>Magnet Wire</i>			
Wire coating operation	1.7	2.2	0.26
<i>Large Appliances</i>			
Prime, single, or topcoat application area, flashoff area and oven	2.8	4.5	0.55
<i>Miscellaneous Metal Parts and Products</i>			
Air-dried items	3.5	6.7	0.80
Clear-coated items	4.3	10.3	1.25
Extreme performance coatings	3.5	6.7	0.80
Other coatings and systems	3.0	5.1	0.61
<i>Plastic Film Coating</i>			
Plastic film coating line	2.9	4.8	0.58

PART C Statements of Basis, Specific Statutory Authority and Purpose

I. April 20, 2023

This Statement of Basis, Specific Statutory Authority, and Purpose complies with the requirements of the State Administrative Procedure Act, § 24-4-101, C.R.S., et seq., the Colorado Air Pollution Prevention and Control Act, § 25-7-101, C.R.S., et seq., and the Air Quality Control Commission's (Commission) Procedural Rules, 5 C.C.R. §1001-1.

Basis

To improve the readability and usability of Regulation Number 7 and Regulation Number 22, the Commission adopted revisions restructuring and reorganizing the parts and sections.

Specific Statutory Authority

The Colorado Air Pollution Prevention and Control Act, § 25-7-101, C.R.S., et seq. (the State Air Act or the Act), specifically § 25-7-103.3, directs rule-making agencies, such as the Commission, to review their rules and consider whether the rule is necessary; whether the rule overlaps or duplicates other rules of the agency or with other federal, state, or local government rules; whether the rule is written in plain language and is easy to understand; whether the rule has achieved the desired intent and whether more or less regulation is necessary; whether the rule can be amended to give more flexibility, reduce regulatory burdens, or reduce unnecessary paperwork or steps while maintaining its benefits; whether the rule is implemented in an efficient and effective manner, including the requirements for the issuance of permits and licenses; whether a cost-benefit analysis was performed by the applicable rule-making agency; and whether the rule is adequate for the protection of the safety, health, and welfare of the state or its residents. Based on this review, the rule-making agency will determine whether the existing rules should be continued in their current form, amended, or repealed.

Purpose

The following section sets forth the Commission's purpose in adopting the revisions to Regulation Number 25.

The Commission reorganized Regulation Number 7 into four regulations: Part B became Regulation Number 24; Part C became Regulation Number 25; Part D remained in Regulation Number 7; and Part E became Regulation Number 26. The upstream oil and gas intensity and midstream combustion program provisions currently in Regulation Number 22 moved to Regulation Number 7. The manufacturing sector greenhouse gas provisions in Regulation Number 22 became a new Regulation Number 27.

To assist in tracking the history of the regulatory revisions, associated statements of basis and purpose, and restructured location, the Commission provides the following tracking table.

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
1995	Dec. 21	Clarify substances that are negligibly reactive VOCs.	Section II.B.	Part A, Section II.B.	Regulations 7 and 24-26, Part A
1996	Mar. 21	Revisions related to the maintenance demonstration.	Sections I.A.1. through I.A.4.; II.D.; II.E.	Part A, Sections I.A.1. through I.A.4.; II.D.; II.E.	Regulations 7 and 24-26, Part A
1996	Nov. 21	Updated NRVOG list. Removed control of VOC emissions from dry cleaning facilities using perchloroethylene.	Section XII.	NA	NA
1998	Oct. 15	Revisions specific to Gates Rubber Company.	Section II.F.	NA	Regulation 24-25, Part A
2001	Jan. 11	Correct discrepancies in posted versus adopted provisions.	Sections III.C.; IX.L.2.c.(1); X.D.2. through XI.A.3.	Part B, Section I.; Part C, Section I.; Part C, Sections II. through III.	Regulation 24, Part B; Regulation 25, Part B (fkna Part C)
2003	Nov. 20	Repealed provisions establishing a procedure for granting exemptions for de minimis sources	Sections I.A.2. through I.A.4.; II.D.; II.E.	Part A, Sections I.A.1. through I.A.4.; II.D.; II.E.	Regulations 7 and 24-26, Part A

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
		and for approving alternative compliance plans.			
2004	Mar. 12	Revisions adopted in conjunction with the early action compact ozone action plan – control of emissions from condensate operation at oil and gas facilities, emissions from internal combustion engines, emissions from gas processing plants, and emissions from oil and gas operations dehydrators.	Sections I.A.; I.B.; XII.; XVI.	Part A, Section I.A.; Part A, Section I.B.; Part D, Section I.; Part E, Section I.	Regulations 7 and 24-26, Part A
2004	Dec. 16	Revisions adopted in response to EPA comments (re practical enforceability) on the ozone action plan adopted 3/2004.	Sections I.A.; II.A.; XII.; XVI.;	Part A, Section I.A.; Part A, Section II.A.; Part D, Section I.; Part E, Section I.	Regulations 7 and 24-26, Part A
2006	Dec. 17	Expanding oil and gas condensate tank emission controls.	Section XII.	Part D, Section I.	Regulation 7, Part B (fkna Part D)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2006	Dec. 17	Reduce emissions from oil and gas operations and natural gas fired engines.	Sections I.A.1.b.; XVII.	Part A, Section I.A.; Part D., Section II. & Part E. Section I. (for engines)	Regulation 7, Part A and Part B (fkna Part D); Regulation 26, Part A and Part B (fkna Part E)
2008	Dec. 12	Expand VOC RACT requirements for 100 tpy sources and clarify how RACT requirements in Regulation Numbers 3 and 7 interact in the ozone nonattainment area. Make typographical and formatting changes. Revise oil and gas condensate tank and pneumatic controller requirements.	Title; Sections I.; II.; VI. through XIII.; XVII.; XVIII.; and Appendices A through F	Part A, Section I.; Part A, Section II.; Part B, Sections IV. through VI. & Part C, Sections I. through IV. & Part D, Section I.; Part D, Section II. and Part E, Section I. (for engines); Part D, Section III.; Part A, Appendix A. & Part B, Appendices B and C & Part C,	Regulation 24, Part B; Regulation 25, Part B (fkna Part C); Regulation 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
				Appendices D and E (formerly Appendix F)	
2011	Jan. 7	Include engine requirements in the Regional Haze SIP.	Outline; Sections I.; XVII.	Part A, Section I.; Part E, Section I.	Regulation 26, Part B (fkna Part E)
2012	Dec. 20	Address EPA comments on the June 2009 submittal. Revise state-only requirements for consistency.	Sections II.; XII.; XVII.	Part A, Section II.; Part D, Section I.; Part D, Section II.	Regulation 7, Part B (fkna Part D)
2014	Feb. 23	Adopt additional oil and gas emission reduction requirements – auto-igniters, expand condensate tank controls, limit storage tank venting, expand dehydrator control, establish leak detection and repair program, limit venting during well maintenance and liquids unloading, expand pneumatic controller	Sections II.; XVII.; XVIII.	Part A, Section II.; Part D, Section II.; Part D, Section III.	Regulation 7, Part B (fkna Part D)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
		requirements.			
2016	Nov. 17	Adopt RACT requirements for industrial cleaning solvents, lithographic and letterpress printing, and specific major sources. Including existing combustion device auto-igniter and storage tank inspection requirements in the SIP. Adopting major source combustion equipment combustion process adjustment requirements and incorporate by reference NSPS and NESHAP for specific major sources.	Sections I.; X.; XII.; XIII.; XVI.; XIX.	Part A, Section I.; Part C, Section II.; Part D, Section I.; Part C, Section IV.; Part C, Section V.; Part E, Section III.	Regulation 25, Part B (fkna Part C); Regulation 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)
2017	Nov. 16	Adopt provisions based on recommendations in EPA's Oil and Gas Control Techniques Guideline. Revise state-only requirements for	Sections II.; XII.; XVII.; XVIII.	Part A, Section II.; Part D, Section I.; Part D, Section II.; Part D, Section III.	Regulation 7, Part B (fkna Part D)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
		consistency.			
2018	July 19	Adopt requirements for existing major source boilers, turbines, lightweight aggregate kilns, glass melting furnaces, engines.	Sections XVI.; XIX.	Part E, Section II., Part E, Section III.	Regulation 26, Part B (fkna Part E)
2018	Nov. 15	Adopt requirements for major source breweries and wood furniture manufacturing. Address EPA concerns with requirements for industrial cleaning solvents, metal furniture surface coating, and miscellaneous metal surface coating. Updated incorporation by reference dates.	Sections I.; II.; VI.; VIII.; IX.; X.; XII.; XIII.; XVI.; XVII.; XIX.; XX.; XXI.	Part A, Section I.; Part A, Section II.; Part B, Section IV.; Part B, Section VI.; Part C, Section I.; Part C, Section X.; Part D, Section I.; Part C, Section IV.; Part C, Section V.; Part D, Section II.; Part E, Section III.; Part E, Section IV. Part F	Regulation 24, Part B; Regulation 25, Part B (fkna Part C); Regulation 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E); Regulation 23 (fkna Part F)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2019	Dec. 19	Reorganized into Parts A through F. Replaced the SIP system-wide condensate tank control program with a fixed threshold storage tank control program. Increased state-only, state-wide storage tank controls. Adopted oil and gas storage tank measurement system, hydrocarbon liquids loadout, leak detection and repair, well plugging, and pneumatic controller requirements. Adopted an oil and gas transmission and storage segment methane intensity program. Adopted an annual oil and gas inventory program. Expanded SIP requirements to 50 tpy sources.	Sections I. through XX. and Appendices A through F	(see reorganization cross walk)	

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
		Aligned gasoline tank truck testing requirements with federal requirements as SIP clean-up.			
2020	Sept. 23	Adopted requirements for natural gas fired 1,000 horsepower engines. Adopted flowback vessel control requirements and pre- and early-production monitoring requirements. Expanded hydrocarbon liquids loadout requirements to class II disposal well facilities.		Part D, Sections II.; IV.; V.; VI.; Part E, Section I.	Regulation 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)
2020	Dec. 18	Adopted requirements for major source foam manufacturing, boilers, turbines, landfill and biogas fired engines, and wood surface coating.		Part D, Section II.; Part E, Sections II.; IV.; V.	Regulation 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
2021	Feb. 18	Adopted non-emitting pneumatic controller requirements for new facilities and existing pneumatic controller retrofit requirements for existing facilities.		Part D, Section III.	Regulation 7, Part B (fkna Part D)
2021	July 16	Adopted requirements for metal parts surface coating and major source process heaters.		Part C, Section I.; Part D, Section III.; Part E, Section II.	Regulation 25, Part B (fkna Part C); Regulation 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)
2021	Dec. 17	Adopted SIP revisions to address EPA concerns with the EPA Oil and Gas CTG. Adopted oil and gas combustion device performance testing requirements. Expanded reciprocating compressor rod packing, leak		Part D, Sections I., II., III., V., VI.	Regulation 7, Part B (fkna Part D)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
		detection and repair, and pneumatic controller requirements at natural gas processing plants. Expanded leak detection and repair, separator, and well maintenance requirements. Adopted pigging and blowdown requirements.			
2022	Dec. 15	Adopted requirements for major source combustion equipment, wood coating, solvent use, bakery operation, digital printing, poultry waste processing, oil stabilization facilities, class II injection well facilities, and industrial waste; included state only provisions as SIP strengthening		Part E, Sections I., II., III., VI., VII., and VIII., Part C., Sections I., II., and IV., and Part D, Section II.; Part D, Sections II., ; Part A, Sections I. and II.; Part C, Section I.; and Part B, Section IV.	Regulation 24, Part B (fkna Part B); Regulation 25, Part B (fkna Part C); Regulation 7, Part B (fkna Part D); Regulation 26, Part B (fkna Part E)

Year of rule adoption	Date of rule adoption	Summary of rule(s) adopted	Regulation 7 Section (pre-2019 numbering)	Regulation 7 Section (numbering as of 12.2022)	Rule & Section (as of 4.2023)
		measures; clarified the applicability of requirements to newly classified ozone nonattainment areas; included requirements for motor vehicle materials and automotive coatings; expanded gasoline tank truck testing requirements.			

The Commission also made typographical, grammatical, and formatting corrections throughout the regulations.

Incorporation by Reference

The Commission will update regulatory references as needed as opportunities arrive.

Additional Considerations

These revisions are administrative in nature and, therefore, do not exceed or differ from the requirement of the federal act or rules. Therefore, § 25-7-110.5(5)(a) does not apply.

Findings of Fact

To the extent that § 25-7-110.8, C.R.S., requirements apply to this rulemaking, and after considering all the information in the record, the Commission hereby makes the determination that:

(I) These rules are based upon reasonably available, validated, reviewed, and sound scientific methodologies, and the Commission has considered all information submitted by interested parties.

(II) Evidence in the record supports the finding that the rules shall result in a demonstrable reduction of greenhouse gas and VOC emissions.

(III) Evidence in the record supports the finding that the rules shall bring about reductions in risks to human health and the environment that justify the costs to implement and comply with the rules.

(IV) The rules are the most cost-effective alternative to achieve the necessary reduction in air pollution and provide the regulated entity flexibility.

(V) The selected regulatory alternative will maximize the air quality benefits of regulation in the most cost-effective manner.

II. December 18-20, 2024 (Revisions to Part A, Section II.C.2. and Part B, Sections I.L.6., I.O.5., and I.P.)

This Statement of Basis, Specific Statutory Authority, and Purpose complies with the requirements of the State Administrative Procedure Act, § 24-4-101, C.R.S., et seq., the Colorado Air Pollution Prevention and Control Act, § 25-7-101, C.R.S., et seq., and the Air Quality Control Commission's (Commission) Procedural Rules, 5 C.C.R. §1001-1.

Basis

The Commission adopted revisions in response to EPA actions on Colorado's state implementation plan (SIP) submissions. Specifically, a disapproval related to recordkeeping and reporting requirements and a disapproval of contingency measures for the serious SIP under the 2008 ozone National Ambient Air Quality Standard (NAAQS). See 88 Fed. Reg. 29827 (May 9, 2023); 88 Fed. Reg. 76676 (Nov. 7, 2023).

Specific Statutory Authority

The State Air Act, specifically § 25-7-105(1), directs the Commission to promulgate such rules and regulations as are consistent with the legislative declaration set forth in § 25-7-102 and that are necessary for the proper implementation and administration of Article 7. The Act broadly defines air pollutant to include essentially any gas emitted into the atmosphere (and, as such, includes VOC, NO_x, methane and other hydrocarbons) and provides the Commission broad authority to regulate air pollutants. § 105(1)(a)(I) directs the Commission to adopt a state implementation plan (SIP) to attain the NAAQS. § 25-7-106 provides the Commission maximum flexibility in developing an effective air quality program and promulgating such combination of regulations as may be necessary or desirable to carry out that program. § 25-7-106 also authorizes the Commission to promulgate emission control regulations applicable to the entire state, specified areas or zones, or a specified class of pollution. § 25-7-106(6) further authorizes the Commission to require owners and operators of any air pollution source to monitor, record, and report information. §§ 25-7-109(1)(a) and (2) of the Act authorize the Commission to promulgate regulations requiring effective and practical air pollution controls for significant sources and categories of sources and emission control regulations pertaining to nitrogen oxides and hydrocarbons.

Purpose

The following section sets forth the Commission's purpose in adopting the revisions to Regulation Number 25 and includes the technological and scientific rationale for the adoption of the revisions.

Periodic Reporting

To address EPA's concern with the lack of specified reporting and to avoid federal sanctions the Commission included requirements for major sources to include compliance demonstration information with the semi-annual reporting requirements in the operating permit, specifically wood coating operations. The Commission also included requirements for metal coating operations, which are a VOC source category subject to a CTG published by EPA and upon which the adopted provisions, including the recordkeeping without corresponding fixed frequency reporting recommendations, are based, to submit annual or semi-annual reports documenting compliance with coatings VOC content limits. See 88 Fed. Reg. 29827 (May 9, 2023) (EPA's limited disapproval, discussing EPA's perceived lack of periodic reporting sufficient to determine compliance by regulated entities). The semi-annual reports for major sources are intended to align with the sources' operating permit reporting requirements.

Contingency Measures

The Commission also updated the contingency measures adopted in 2022 in Section I.P. concerning automotive VOC content limits. These provisions were adopted as contingency measures for the moderate SIP under the 2015 ozone NAAQS. The revisions were adopted to include these contingency measures in the Serious SIP under the 2008 Ozone NAAQS, in response to EPA's disapproval. Under the federal Clean Air Act § 172(c)(9), nonattainment area SIPs must provide for the implementation of specific measures, termed contingency measures, if an area fails to attain the NAAQS or to demonstrate RFP by the required deadline. The Commission had adopted contingency measure for the Serious SIP in 2020 with anticipated future year emission reductions in VOC and NOx from on-road mobile sources. However, the U.S. Court of Appeals for the District of Columbia Circuit issued a decision in January 2021 interpreting the CAA requirements for contingency measures to require such measures be prospective and conditional, in contrast to previous EPA interpretation and corresponding state implementation. See 88 Fed. Reg. at 54978 discussing *Sierra Club v EPA*. Therefore, EPA disapproved Colorado's Serious SIP contingency measures, finding the measures did not satisfy the CAA requirements as interpreted by the DC Circuit. The Commission now includes the automotive VOC content limits and associated work practices in Regulation Number 25, Part B, Section I.P. as contingency measures for the 2008 Ozone NAAQS serious SIP. These requirements impact any person who supplies, sells, offers for sale, distributes for sale, or manufacturers for sale in Colorado certain products after a finding by EPA of failure to attain the 2008 ozone NAAQS by the serious ozone attainment date. The Commission adopted the contingency measures in order to ensure emission reductions within a reasonable timeframe after submission to EPA for approval of the measures.

The Commission also made typographical, grammatical, and formatting corrections throughout the regulations.

Incorporation by Reference

The Commission will update regulatory references as needed as opportunities arrive.

Additional Considerations

Colorado must revise Colorado's ozone SIP to address the ozone nonattainment area requirements. The CAA does not expressly address all of the provisions adopted by the Commission. Rather, federal law establishes the ozone NAAQS and requires Colorado to develop a SIP adequate to attain the NAAQS. Therefore, the Commission adopted certain revisions to Regulation Number 25 to satisfy Colorado's nonattainment area obligations and further achieve reductions of ozone precursor emissions. These revisions do not exceed or differ from the federal act due to state flexibility in determining what control strategies to implement to reduce emissions. However, where the proposal may differ from federal rules under the federal act, in accordance with § 25-7-110.5(5)(b), CRS, the Commission determines:

- (I) The revisions to Regulation Number 25 address wood coating, metal coating, and automotive coatings operations. NESHAP JJ, NESHAP DDDD, NESHAP MMMM, NESHAP QQQQ, NESHAP XXXXXX and EPA's National Volatile Organic Compound Emission Standards for Automobile Refinish Coatings at 40 CFR Part 59 Subpart B may also apply to the above described operations. However, the revisions to Regulation Number 25 apply on a broader basis.
- (II) The federal rules discussed in (I) are primarily technology-based in that they largely prescribe the use of specific technologies or work practices to comply.
- (III) The CAA establishes the 2008 and 2015 ozone NAAQS and requires Colorado to develop SIP revisions that will ensure attainment of the NAAQS. The ozone NAAQS was not determined taking into account concerns unique to Colorado. Similarly, EPA develops NSPS or NESHAP considering national information and data, not Colorado specific issues or concerns. In addition, Colorado cannot rely exclusively on a federally enforceable permit or federally enforceable NSPS or NESHAP to satisfy Colorado's ozone nonattainment area RACT obligations. Instead, Colorado can adopt applicable provisions into its SIP directly, as the Commission has done here.
- (IV) In addition to the 2008 NAAQS, Colorado must also comply with the lower 2015 ozone NAAQS. These current revisions may improve the ability of the regulated community to comply with new requirements needed to attain the lower NAAQS insofar as RACT analyses and efforts conducted to support the revisions adopted by the Commission may prevent or reduce the need to conduct additional RACT analyses for the more stringent NAAQS.
- (V) EPA has established Colorado's SIP RACT implementation deadlines. There is no timing issue that might justify changing the time frame for implementation of federal requirements.

- (VI) The revisions to Regulation Number 25 address EPA's May 9, 2023, disapproval. These sections currently address emissions from wood coating and metal coating operations, while allowing for continued growth of Colorado's industry.
- (VII) The revisions to Regulation Number 25 establish reasonable equity for owners and operators subject to these rules by providing the same standards for similarly situated and sized sources.
- (VIII) If EPA does not approve Colorado's SIP, EPA may promulgate a Federal Implementation Plan; thus potentially determining RACT for Colorado's sources. This outcome may subject others to increased costs.
- (IX) Where necessary, the revisions to Regulation Number 25 include minimal monitoring, recordkeeping, and reporting requirements that correlate, where possible, to similar federal or state requirements.
- (X) Demonstrated technology is available to comply with the revisions to Regulation Number 25. The revisions do not expand upon emission reduction requirements already applicable but minimally add additional reporting requirements.
- (XI) As set forth in past Economic Impact Analyses, the revisions to Regulation Number 25 will reduce emissions in a cost-effective manner.
- (XII) Alternative rules could also provide reductions in ozone, VOC, and NO_x to help to attain the NAAQS. However, a no action alternative would very likely result in sanctions.

Findings of Fact

To the extent that § 25-7-110.8, C.R.S., requirements apply to this rulemaking, and after considering all the information in the record, the Commission hereby makes the determination that:

- (I) These rules are based upon reasonably available, validated, reviewed, and sound scientific methodologies, and the Commission has considered all information submitted by interested parties.
- (II) Evidence in the record supports the finding that the rules shall result in a demonstrable reduction of greenhouse gas and VOC emissions.
- (III) Evidence in the record supports the finding that the rules shall bring about reductions in risks to human health and the environment that justify the costs to implement and comply with the rules.
- (IV) The rules are the most cost-effective alternative to achieve the necessary reduction in air pollution and provide the regulated entity flexibility.

- (V) The selected regulatory alternative will maximize the air quality benefits of regulation in the most cost-effective manner.

III. November 19-21, 2025 (Revisions to Part B, Section I.Q. and Repeal of Part B, Section IV.C.)

This Statement of Basis, Specific Statutory Authority, and Purpose complies with the requirements of the State Administrative Procedure Act, § 24-4-101, C.R.S., et seq., the Colorado Air Pollution Prevention and Control Act (State Air Act or the Act), § 25-7-101, C.R.S., et seq., and the Air Quality Control Commission's (Commission) Procedural Rules, 5 C.C.R. §1001-1.

Basis

On October 7, 2022, EPA reclassified the Denver Metro/North Front Range (DM/NFR) to severe for the 2008 8-hour Ozone National Ambient Air Quality Standard of 75 parts per billion (ppb) (2008 ozone NAAQS). See 86 Fed. Reg. 60926. Separately, on July 24, 2024, EPA reclassified the (DM/NFR) and northern Weld County ozone nonattainment area to serious for the 2015 8-hour Ozone National Ambient Air Quality Standard of 70 parts per billion (ppb) (2015 ozone NAAQS). See 89 Fed. Reg. 59832.

To ensure progress towards attainment of the ozone NAAQS and respond to Colorado's requirements under the Clean Air Act (CAA), the Commission adopted revisions to include reasonably available control technology (RACT) for major sources of volatile organic compounds (VOC) in the nonattainment areas, specifically adopting requirements for aerospace coatings.

Specific Statutory Authority

The State Air Act, specifically § 25-7-105(1), C.R.S., directs the Commission to promulgate such rules and regulations as are consistent with the legislative declaration set forth in § 25-7-102, C.R.S. and that are necessary for the proper implementation and administration of Article 7. The Act broadly defines air pollutant to include essentially any gas emitted into the atmosphere (and, as such, includes VOC, nitrogen oxides (NOx), methane and other hydrocarbons) and provides the Commission broad authority to regulate air pollutants. Section 105(1)(a)(I), C.R.S., directs the Commission to adopt a state implementation plan (SIP) to attain the NAAQS. Section 25-7-106, C.R.S., provides the Commission maximum flexibility in developing an effective air quality program and promulgating such combination of regulations as may be necessary or desirable to carry out that program. Section 25-7-106, C.R.S., also authorizes the Commission to promulgate emission control regulations applicable to the entire state, specified areas or zones, or a specified class of pollution. Section 25-7-106(6), C.R.S., further authorizes the Commission to require owners and operators of any air pollution source to monitor, record, and report information. Sections 25-7-109(1)(a) and (2) of the Act authorize the Commission to promulgate regulations requiring effective and practical air pollution controls for significant sources and categories of sources and emission control regulations pertaining to NOx and hydrocarbons.

Purpose

The following section sets forth the Commission's purpose in adopting the revisions to Regulation Number 25 and includes the technological and scientific rationale for the adoption of the revisions.

Due to the reclassifications to serious and severe, Colorado must submit revisions to its SIP to address the CAA's ozone nonattainment area requirements, as set forth in CAA §§ 172, 182(c), 182(d), and the final SIP Requirements Rules (83 Fed. Reg. 62998 (Dec. 6, 2018)). SIPs must include provisions that require the implementation of RACT for major sources of VOC and/or NOx (i.e., sources that emit or have the potential to emit 50 tons per year (tpy) or more in serious nonattainment areas and 25 tpy or more in severe nonattainment areas) and for each category of VOC sources covered by a Control Technique Guideline (CTG) for which Colorado has sources in the nonattainment area.

Therefore, to address nonattainment area requirements under CAA § 182(c), the Commission adopted revisions to Regulation Number 25 to include RACT requirements in Colorado's ozone SIP for 25 tpy VOC major stationary aerospace coating operations in a new Part A, Section I.Q. The Commission adopted aerospace coatings requirements that are based on EPA's Control of Volatile Organic Compound Emissions from Coating Operations at Aerospace Manufacturing and Rework Operations (1997), which also recommends a 25 tpy VOC applicability threshold. These requirements include VOC content limits on coatings, work practices, recordkeeping, and reporting.

The Commission also repealed requirements for major stationary digital printing operations in Part A, Section IV.C. These provisions were adopted in 2022 for major sources (25 tpy) located in the 8-hour ozone control area. The subject sources are no longer a major stationary source and, therefore, these SIP-RACT provisions are no longer required.

Lastly, the Commission adopted clean-up revisions in Part A, Section I.A.5.d. to correspond to the revisions made to update Regulation Number 3, Part A, Section V., Certification of Trading of Emission Reduction Credits Offset and Netting Transactions.

The Commission also made typographical, grammatical, and formatting corrections throughout the regulations.

Incorporation by Reference

The Commission will update regulatory references as needed as opportunities arrive.

Additional Considerations

Colorado must revise Colorado's ozone SIP to address the ozone nonattainment area requirements. The CAA does not expressly address all of the provisions adopted by the Commission. Rather, federal law establishes the ozone NAAQS and requires Colorado to develop a SIP adequate to attain the NAAQS. Therefore, the Commission adopted certain revisions to Regulation Number 25 to satisfy Colorado's nonattainment area obligations and further achieve reductions of ozone precursor emissions. These revisions do not exceed or differ from the federal act due to state flexibility in determining what control strategies to implement to reduce emissions. However, where the proposal may differ from federal rules under the federal act, in accordance with § 25-7-110.5(5)(b), C.R.S., the Commission determines:

- (I) The revisions to Regulation Number 25 address aerospace coatings operations. National Emissions Standards for Hazardous Air Pollutants (NESHAP) GG may also apply to these operations. However, the revisions to Regulation Number 25 apply on a broader basis.

- (II) The federal rules discussed in (I) are primarily technology-based in that they largely prescribe the use of specific technologies or work practices to comply.
- (III) The CAA establishes the ozone NAAQS and requires Colorado to develop SIP revisions that will ensure attainment of the NAAQS. The ozone NAAQS was not determined taking into account concerns unique to Colorado. Similarly, EPA develops New Source Performance Standards (NSPS) or NESHAP considering national information and data, not Colorado specific issues or concerns. In addition, Colorado cannot rely exclusively on a federally enforceable permit or federally enforceable NSPS or NESHAP to satisfy Colorado's ozone nonattainment area RACT obligations. Instead, Colorado can adopt applicable provisions into its SIP directly, as the Commission has done here.
- (IV) In addition to the 2008 NAAQS, Colorado must also comply with the lower 2015 ozone NAAQS. These current revisions may improve the ability of the regulated community to comply with new requirements needed to attain the lower NAAQS insofar as RACT analyses and efforts conducted to support the revisions adopted by the Commission may prevent or reduce the need to conduct additional RACT analyses for the more stringent NAAQS.
- (V) EPA has established Colorado's SIP RACT implementation deadlines. There is no timing issue that might justify changing the time frame for implementation of federal requirements.
- (VI) The revisions to Regulation Number 25 address emissions from major stationary source aerospace coating operations, while allowing for continued growth of Colorado's industry.
- (VII) The revisions to Regulation Number 25 establish reasonable equity for owners and operators subject to these rules by providing the same standards for similarly situated and sized sources.
- (VIII) If EPA does not approve Colorado's SIP, EPA may promulgate a Federal Implementation Plan; thus, potentially determining RACT for Colorado's sources. This outcome may subject others to increased costs.
- (IX) Where necessary, the revisions to Regulation Number 25 include minimal monitoring, recordkeeping, and reporting requirements that correlate, where possible, to similar federal or state requirements.
- (X) Demonstrated technology is available to comply with the revisions to Regulation Number 25. The revisions do not expand upon emission reduction requirements already applicable but minimally add additional reporting requirements.

- (XI) As set forth in past Economic Impact Analyses, the revisions to Regulation Number 25 will reduce emissions in a cost-effective manner.
- (XII) Alternative rules could also provide reductions in ozone, VOC, and NO_x to help to attain the NAAQS. However, a no action alternative would very likely result in sanctions.

Findings of Fact

To the extent that § 25-7-110.8, C.R.S., requirements apply to this rulemaking, and after considering all the information in the record, the Commission hereby makes the determination that:

- (I) These rules are based upon reasonably available, validated, reviewed, and sound scientific methodologies, and the Commission has considered all information submitted by interested parties.
- (II) Evidence in the record supports the finding that the rules shall result in a demonstrable reduction of NO_x and VOC emissions.
- (III) Evidence in the record supports the finding that the rules shall bring about reductions in risks to human health and the environment that justify the costs to implement and comply with the rules.
- (IV) The rules are the most cost-effective alternative to achieve the necessary reduction in air pollution and provide the regulated entity flexibility.
- (V) The selected regulatory alternative will maximize the air quality benefits of regulation in the most cost-effective manner.

Editor's Notes

History

New rule eff. 06/14/2023.

Rules Part A II.C.2, Part B I.L.6, I.O.6, I.P.1.b, I.P.3, I.P.4.b, I.P.7, I.P.8, Part C II eff.
02/14/2025.

Rules Part A II.A.18, Part B I.A.1, I.A.3.a, I.A.5.d, I.A.5.d(iv), I.A.11, I.A.12.a(iv), I.L.2.d,
I.N.7, I.Q, II.D.2.a, II.E.3.b(iii)(B), II.F.5.c(i)(B), IV.B.5.c(iii)(B), Part C III eff.
01/14/2026. Part B IV.C repealed eff. 01/14/2026.